M&A OPERATING SYSTEM

Product Design

AI Chat Platform

Draft v1.0 · May 2026

Andrew Bush (www.maoperatingsystem.com/bio-andrew-bush)

About This Document

This document is part of the **M&A Operating System** product design specification series. It describes the intended design, architecture, and behaviour of the platform within. It is intended for internal distribution across product, engineering, and design review functions.

This is a living specification. It reflects design intent at the time of publication and will be updated as the product evolves.

About M&A Operating System

M&A Operating System is a boutique advisory firm focused on helping organizations modernize the operational foundations required for scalable analytics, trusted enterprise information, and practical AI adoption.

Our work sits at the intersection of enterprise data strategy, information architecture, operating model modernization, and transformation execution. We specialize in helping organizations design practical, business-aligned approaches to enterprise information management that support operational scalability, analytics enablement, governance, and modern AI capabilities.

A core part of our approach is the recognition that successful data and AI initiatives are not driven solely by technology platforms. Long-term success depends on how well organizations structure, model, govern, operationalize, and align information with real business processes and decision-making.

Our Product Design document series reflects this philosophy. These documents are intended to provide practical frameworks, methodologies, architectural concepts, and implementation guidance that help organizations move beyond theoretical strategy into executable operational design.

M&A Operating System's approach combines: - Enterprise data and AI strategy - Subject-based data modeling and information architecture - Semantic and operational data design - Governance and organizational alignment - Operating model modernization - Transformation execution and delivery leadership - Practical AI operationalization and readiness

Our goal is simple: help organizations build practical, scalable, and trusted information foundations that improve operational effectiveness, accelerate analytics and AI adoption, and support better business outcomes.

Find Out More

To learn more about M&A Operating System, explore the platform, or speak with our team: maoperatingsystem.com

01 — Overview

Vision

Conversational AI for any application

The **AI Chat Platform** enables any application to give its users a persistent, context-aware conversational interface — without building AI infrastructure. The experience is modelled on the best native AI desktop applications: rich rendered output, transparent tool usage, and a conversation that remembers where you left off. It is not a general-purpose assistant layer — each deployment is a **specialist** tuned to its host application's domain.

> **Governing intent:** Give any application team the ability to drop a production-grade AI assistant into their product within days — fully branded, scoped to their domain, connected to their data, and backed by a complete audit trail.

What the platform is and is not

It is	It is not
A white-label assistant layer any application can embed as a web component	A standalone AI product with its own brand or identity
A multi-tenant platform where each host application brings its own scope, tools, and branding	A shared assistant all host applications configure from a single pool
A complete audit trail for every conversation turn and artefact	A transient chat tool with no persistent record
A controlled, host-configured MCP integration surface	An open API that allows arbitrary tool connections without host approval
A read-and-write assistant — queries, reasoning, and actions via host-registered MCP tools	A system that bypasses host application security models or acts without user confirmation

The assistant has no platform name

The platform has no end-user-facing name. Every tenant names their own assistant in their application config (`identity.assistantName`). End users see only that name — they are not exposed to the underlying platform. This document uses **[AssistantName]** as a placeholder wherever the assistant name would appear.

Guiding principles

Principle	What it means in practice
Host-scoped by default	Every capability is bounded by host configuration. The host defines the assistant's identity, allowed tools, permitted model set, and UI branding. Nothing is available to end users that the host has not explicitly configured.
Audit completeness	No conversation turn, edit, branch, or tool call is ever deleted. The audit trail is append-only. Branching preserves originals. Edits create new threads, not in-place modifications.
Transparent tool usage	Every MCP tool call is disclosed in the conversation thread via a collapsible disclosure card. Users always know when the assistant is accessing external systems.
Confirmation before action	Write operations via MCP tools require explicit user confirmation. The assistant never mutates host application state without an approval step visible in the conversation.
No privileged data path	All data access is via host-registered MCP tools. The platform has no direct database access and cannot reach application data that the host has not explicitly exposed via an MCP server.
Consumer-rendered, governed display	Rich content — charts, tables, diagrams, code — is rendered via a governed content rendering pipeline. SCL display specifications returned by connected services (e.g., the Analytics Platform) are rendered by the platform; the AI model does not choose display formats ad hoc.

Scope

In scope

- Embeddable `<ai-chat>` web component with full branding token support
- Multi-tenant architecture with complete per-tenant data isolation
- Persistent, named conversation threads scoped to the authenticated user within their tenant
- Conversation branching via message edit or regeneration — preserving the complete audit trail
- Full-text search across all user conversations within the tenant
- Host-configured `@`-binding for application-defined object types
- Host-configured Display ID pattern detection and auto-resolution
- Document attachments (PDF, Excel, Word, images) stored as platform artefacts
- Model switching within the host-configured allowed model set; provider-agnostic architecture
- Communication style and verbosity driven by host-provided user profile claims
- Host-registered MCP servers with always-on and opt-in access tiers
- Host-defined guided workflow prompts accessible from the Workflow Library panel
- Tool call transparency — every MCP invocation rendered as a collapsible disclosure card

- Write operations — actions via host MCP tools with explicit user confirmation before execution
- Shared conversations — up to ten participants within the same tenant, equal-participant model
- Personal memory (user-managed) and application context (Application Admin-managed)
- Session artefact tray accumulating all input and output artefacts
- Document canvas — iteratable working-document surface in the conversation panel; versioned, editable, model-revisable
- Three-zone responsive layout (history panel, conversation area, conversation panel) embedded within host app UI
- Rich content rendering: diagrams, chart specifications (SCL), structured data, syntax-highlighted code, formatted prose, mathematical notation
- Continuous improvement signal capture and per-tenant improvement issue pipeline
- Full audit trail per turn: raw prompt, resolved prompt, tool call log, output artefacts, token counts
- Complementary MCP ecosystem services: MCP Repository (tool discovery), MCP Resources (shared skills and artefacts), and Web Search (real-time web search)

Out of scope

- Semantic search over application data (pgvector RAG) — structured MCP tool calls are the primary data access pattern in v1
- Conversation export (PDF or markdown) — planned; data classification complexity must be resolved first
- Voice or multimodal input
- Context globbing (pulling context from multiple past conversations into one session)
- Incognito or temporary chat (no-history mode) — conflicts with audit completeness
- Platform-owned web search — available via the Web Search complementary MCP service registered by the host (see 17-complementary-mcp-services.md)
- Image or content generation
- Code execution sandbox
- Public shareable links — all sharing is participant-controlled and tenant-scoped
- Read receipts in shared conversations
- Dark mode — not in v1

Platform architecture

Components

AI Chat Platform owns the conversational surface, content rendering, audit trail, shared conversation model, memory management, and improvement signal pipeline. It depends on, but does not duplicate, the capabilities of the host application or its MCP servers.

Host Application owns the user identity model, application business logic, and MCP server endpoints. It embeds the platform via the web component and passes user context via the authentication bridge.

Host MCP Servers are the host application's data and action providers. The platform routes tool calls to these servers and surfaces the results transparently in the conversation thread.

Complementary MCP Services — the MCP Repository, MCP Resources Service, and Web Search Service — are ecosystem-level services that operate alongside both the platform and host MCP servers. They are not owned by the platform or by individual host applications. See 17-complementary-mcp-services.md.

Dependencies

Dependency	Role
AI provider	Provider-agnostic abstraction exposing three model tiers: `fast`, `standard` (default), `powerful`. The platform maps tiers to the tenant's configured provider's current models. Multiple providers planned — see ROADMAP.md.
Platform storage	Relational database with row-level security for conversation records; object storage for binary artefacts.
Platform edge function	JWT handling, AI provider API request construction, SSE stream passthrough, MCP call routing. Provider-agnostic interface.
Host authentication	The host application issues JWTs for its users. The platform validates these tokens and trusts the embedded claims. No re-authentication is performed by the platform.
Host MCP server(s)	The host's registered MCP endpoints providing data access and action capabilities.
MCP Repository	Complementary ecosystem service — discoverable registry of available MCP tools.
MCP Resources Service	Complementary ecosystem service — centralised skills, static resources, and reusable prompt artefacts.
Web Search Service	Complementary ecosystem service — real-time web search and page retrieval; registered by hosts as an opt-in or always-on MCP server.

00 — Host Application Configuration

Every tenant on the AI Chat Platform is defined by a **JSON application config** provided by the host application at registration time. The config is the single source of truth for how the assistant behaves within that application — its identity, scope, tools, bindings, workflows, branding, and feature flags.

Config delivery

Configs are registered and updated via the **Platform Admin API** (authenticated by a platform API key scoped to the tenant). A web-based **Config Editor UI** provides a visual wrapper over the same API for non-technical Application Admins. Both target the same config document — there is no divergence between API and UI state.

Config takes effect on the **next new conversation session** after the update is applied. Active sessions are not interrupted by config changes.

Config schema

```
{
  "$schema": "https://chat-platform.io/config/v1/schema.json",
  "version": "1.0.0",

  "identity":      { ... },
  "branding":      { ... },
  "scope":         { ... },
  "bindableTypes": [ ... ],
  "mcpServers":    [ ... ],
  "workflows":     [ ... ],
  "renderers":     [ ... ],
  "models":        { ... },
  "userProfile":   { ... },
  "memory":        { ... },
  "conversations": { ... },
  "features":      { ... },
  "widget":        { ... }
}
```

`identity`

```
{
  "identity": {
    "tenantId":             "acme-corp",
    "applicationName":      "Acme Data Hub",
    "assistantName":        "Atlas",
    "assistantDescription": "Your data governance assistant for Acme Corp.",
    "supportUrl":           "https://help.acme.com/atlas",
    "privacyPolicyUrl":     "https://acme.com/privacy"
  }
}
```

Field	Required	Description
`tenantId`	Yes	Unique slug identifier for this tenant on the platform
`applicationName`	Yes	Name of the host application — shown in platform admin UI
`assistantName`	Yes	The name end users see for the assistant (e.g. "Atlas", "Nova", "Sage")
`assistantDescription`	Yes	One-sentence description shown in the onboarding welcome state
`supportUrl`	No	URL linked from error states for end-user support
`privacyPolicyUrl`	No	URL displayed in the memory consent UI

`branding`

Design tokens applied to the `<ai-chat>` web component. The platform's default design system is used for any token not provided.

```
{
  "branding": {
    "colorPrimary":        "#0057B8",
    "colorSurface":        "#FFFFFF",
    "colorSurfaceVariant": "#F4F6F9",
    "colorOnSurface":      "#111827",
    "colorAccent":         "#00A86B",
    "fontFamily":          "Inter, system-ui, sans-serif",
    "borderRadius":        "8px",
    "logoUrl":             "https://cdn.acme.com/logo.svg",
    "faviconUrl":          "https://cdn.acme.com/favicon.ico"
  }
}
```

All colour values accept hex, RGB, or HSL. Logo assets must be SVG or PNG and served from a CSP-compatible origin declared in the host's Content Security Policy. See 16-embedding-and-web-component.md for CSP requirements.

`scope`

Defines the assistant's domain, what it should decline, and how it handles out-of-scope queries.

```
{
  "scope": {
    "systemPrompt": "You are Atlas, the data governance assistant for Acme Corp. You help users
find, understand, and act on data assets governed by the Acme Data Hub platform...",
    "inScopeDescription": "Data assets, governance policies, quality metrics, and data ownership
within Acme Corp",
    "outOfScopeRedirect": "I'm here to help with Acme Corp's data governance. For general
questions, please contact your team's relevant resource.",
    "language": "en"
  }
}
```

Field	Required	Description
`systemPrompt`	Yes	Base system prompt for the assistant. The platform appends tool descriptions, memory blocks, and platform-managed safety instructions.
`inScopeDescription`	Yes	Plain-language description of scope; used in the onboarding state and out-of-scope redirect messages
`outOfScopeRedirect`	No	Custom message for out-of-scope queries. The platform uses a generic fallback if not set.
`language`	No	BCP 47 language tag (default: `en`). Applies to UI strings and the assistant's response language hint.

System prompt authoring guidelines

- Establish the assistant's persona and domain clearly in the first 200 tokens
- Do not include tool descriptions — they are injected automatically at session start
- Do not include memory blocks — injected automatically
- Do not attempt to override platform safety, audit, or tool-transparency instructions
- Keep under 3,000 tokens to leave headroom for injected context, memory blocks, and tool descriptions
- Avoid referencing specific MCP tool names — describe capabilities in natural language so the model chooses the right tool

---

`bindableTypes`

Defines the object types that end users can reference via `@`-binding in the input field. The platform provides the `@`-binding mechanism; the host defines what can be bound.

```
{
  "bindableTypes": [
    {
      "id":              "data_domain",
      "label":           "Data Domain",
      "pluralLabel":     "Data Domains",
      "icon":            "database",
      "searchEndpoint":  "https://api.acme.com/chat/search/domains",
      "contextTemplate": "Data Domain: {{name}} (ID: {{displayId}})\nOwner: {{owner}}
\nDescription: {{description}}",
      "displayIdPattern": "DOM-[0-9]+",
      "rank":            1
    },
    {
      "id":              "policy",
      "label":           "Policy",
      "pluralLabel":     "Policies",
      "icon":            "shield",
      "searchEndpoint":  "https://api.acme.com/chat/search/policies",
      "contextTemplate": "Policy: {{name}}\nStatus: {{status}}\nEffective: {{effectiveDate}}
\nSummary: {{summary}}",
      "rank":            2
    }
  ]
}
```

Field	Required	Description
`id`	Yes	Unique identifier for this type within the tenant
`label`	Yes	Singular display label shown in the typeahead panel
`pluralLabel`	Yes	Plural label used in empty states and section headers
`icon`	No	Platform icon name or absolute URL for the chip icon
`searchEndpoint`	Yes	Host-provided HTTPS endpoint the platform calls to resolve `@`-binding searches. Must accept a `?q=` query parameter and return `{ items: [{ id, name, displayId?, inactive? }] }`. Called with the user's bearer token forwarded — the host endpoint enforces its own permission filtering.
`contextTemplate`	Yes	Mustache template for the context block injected into the model prompt when this binding is resolved. Use `{{fieldName}}` for any field returned by the search endpoint's item objects.
`displayIdPattern`	No	Regex; pasted text matching this pattern triggers an auto-lookup confirmation prompt
`rank`	No	Sort order in typeahead results (ascending). Defaults to alphabetical by label if omitted.

Permission model for bindable types

The `searchEndpoint` is called with the authenticated user's bearer token forwarded in the `Authorization` header. The host endpoint is responsible for filtering results to objects the user is permitted to access. The platform does not enforce bindable-type permissions independently — it trusts the host's endpoint response.

In shared conversations, if a binding chip resolves to an object another participant cannot access (because that participant's search endpoint would not have returned it), the restricted participant sees the chip labelled **[Restricted object]**. See 08-shared-conversations.md.

`mcpServers`

Registers the MCP servers available within this tenant. The platform assumes no MCP servers by default — hosts must register at least one for the assistant to be useful beyond its system prompt knowledge.

Host teams can discover and browse available MCP servers in the **MCP Repository** (see 17-complementary-mcp-services.md) before registering them here.

```
{
  "mcpServers": [
    {
      "id":          "governance-mcp",
      "name":        "Governance Platform",
      "description": "Provides access to data governance entities, quality metrics, ownership
records, and policy objects. Use for any question about data assets, governance status, or
compliance.",
      "endpoint":    "https://api.acme.com/mcp/governance",
      "authType":    "bearer",
      "accessTier":  "always-on",
      "roles":       []
    },
    {
      "id":          "warehouse-mcp",
      "name":        "Data Warehouse",
      "description": "Read-only access to the Acme data warehouse. Use to answer questions about
actual data values, row counts, and sample records.",
      "endpoint":    "https://api.acme.com/mcp/warehouse",
      "authType":    "bearer",
      "accessTier":  "opt-in",
      "roles":       ["data_practitioner", "admin"]
    }
  ]
}
```

Field	Required	Description
`id`	Yes	Unique identifier within the tenant
`name`	Yes	Display name shown in the tool selection panel

Field	Required	Description
`description`	Yes	Plain-language description injected into the system prompt when the tool is active. Write it for the model — describe what it provides and when to use it.
`endpoint`	Yes	MCP-compliant HTTPS endpoint URL
`authType`	Yes	`bearer` — the user's host JWT is forwarded; `api-key` — the platform holds a key per tenant (configured separately in the Admin API); `none` — no authentication
`accessTier`	Yes	`always-on` — active in every session, cannot be disabled by the end user; `opt-in` — off by default, user enables per session via the tool selection panel
`roles`	No	Array of host-defined role identifiers (from user profile claims). If populated, only users whose role claim includes one of these values can see or activate this server. Empty array = available to all users.

A tenant may have multiple `always-on` servers. If no `always-on` servers are configured, the platform operates in **prompt-only mode** — the assistant answers from system prompt knowledge only, with no live tool access.

`workflows`

Guided workflows are host-defined conversation starters that launch a structured multi-turn interaction. They appear in the **Workflow Library** panel and as optional starters on the onboarding screen.

```
{
  "workflows": [
    {
      "id":          "governance-health-check",
      "name":        "Governance Health Check",
      "description": "Summarise governance coverage, quality gaps, and policy compliance across a
selected domain.",
      "icon":        "activity",
      "prompt":      "Run a governance health check for {{domain}}. Cover: (1) entity coverage,
(2) data quality completeness, (3) ownership gaps, (4) classification compliance. Present as an
executive summary with a traffic-light status for each area.",
      "parameters": [
        {
          "id":             "domain",
          "label":          "Data Domain",
          "type":           "binding",
          "bindableTypeId": "data_domain",
          "required":       true
        }
      ],
      "roles": []
    },
    {
      "id":          "weekly-status",
      "name":        "Weekly Status Report",
      "description": "Generate a plain-language weekly status summary for a selected team.",
      "icon":        "calendar",
      "prompt":      "Generate a weekly status report for {{team}}. Summarise open actions, recent
completions, and any blockers.",
      "parameters": [
        {
          "id":       "team",
          "label":    "Team",
          "type":     "text",
          "required": true
        }
      ],
      "roles": []
    }
  ]
}
```

Field	Required	Description
`id`	Yes	Unique identifier within the tenant
`name`	Yes	Display name in the Workflow Library
`description`	Yes	One-sentence description shown in the library and onboarding
`icon`	No	Platform icon name or URL
`prompt`	Yes	Mustache template for the prompt submitted on launch. References `parameters` by their `id`.

Field	Required	Description
`parameters`	No	Parameters collected from the user before the workflow launches. Types: `binding` (opens an `@`-binding typeahead scoped to a `bindableTypeId`), `text` (free text field), `select` (dropdown; include an `options` array of `{ value, label }` objects)
`roles`	No	Role restriction. Empty array = available to all authenticated users.

Workflow prompts are host-managed — users cannot create or modify them.

`renderers`

Host applications may register **custom content renderers** that the platform loads at runtime. When the model produces a fenced code block tagged with a registered renderer's trigger, the platform invokes the host's renderer instead of its built-in pipeline. This enables domain-specific visualisations — risk gauges, org charts, Gantt views, financial waterfall charts, compliance scorecards — that the platform's built-in renderers do not cover.

See 10-content-rendering.md for the full runtime rendering contract and system prompt guidance injection.

```
{
  "renderers": [
    {
      "id":                  "risk-gauge",
      "trigger":             "risk-gauge",
      "name":                "Risk Gauge",
      "description":         "Interactive risk score gauge with traffic-light colouring",
      "moduleUrl":           "https://cdn.acme.com/ai-renderers/risk-gauge.js",
      "exportName":          "RiskGaugeRenderer",
      "systemPromptGuidance": "Use a ```risk-gauge block when asked for a risk score or risk
assessment. The block must contain a JSON object: { \"score\": <0–100>, \"label\": \"<risk level
label>\", \"breakdown\": [ { \"factor\": \"<name>\", \"score\": <0–100> } ] }. Only use this block
when a numeric risk score is directly requested."
    }
  ]
}
```

Field	Required	Description
`id`	Yes	Unique identifier within the tenant
`trigger`	Yes	Fenced code block tag that activates this renderer (e.g. `risk-gauge` → ` ```risk-gauge ` in the model output). Must be lowercase alphanumeric with hyphens; must not conflict with a built-in trigger (`mermaid`, `vega-lite`, `math`, `json`, `csv`, `table`).

Field	Required	Description
`name`	Yes	Display name shown in the platform admin UI and in the content block header within the conversation thread
`description`	Yes	One-sentence description shown in the admin UI. Not injected into the system prompt.
`moduleUrl`	Yes	HTTPS URL to an ES module exporting the renderer. Must be served from an origin declared in the tenant's registered origins. Loaded once per session; cached for the session lifetime.
`exportName`	No	Named export from the module. Default: `"default"`.
`systemPromptGuidance`	No	Text injected verbatim into the system prompt to instruct the model when and how to produce this content type. Include the expected fenced block format and any content schema the renderer requires. If omitted, the platform injects a minimal generic instruction: *"You may produce `{trigger}` blocks using ```` ```{trigger} ```` fencing."*

Origin registration

The `moduleUrl` origin must be declared during tenant registration (or updated via the Platform Admin API). At config submission, the platform validates that the declared origin is reachable and that the module can be loaded. Modules served from unregistered origins are rejected. The host's CSP `script-src` must include the renderer origin — see 16-embedding-and-web-component.md.

`models`

```
{
  "models": {
    "defaultModel":  "standard",
    "allowedModels": ["fast", "standard", "powerful"],
    "userCanSwitch": true,
    "provider":      "anthropic"
  }
}
```

Field	Required	Description
`defaultModel`	Yes	Model tier used for new conversations unless the user switches. Accepted values: `"fast"`, `"standard"`, `"powerful"`. See 06-model-configuration.md for tier profiles.
`allowedModels`	Yes	Array of model tiers users may switch to. Must include `defaultModel`.

Field	Required	Description
`userCanSwitch`	No	Whether users can switch model tier mid-conversation (default: `true`). Set to `false` to lock all sessions to `defaultModel`.
`provider`	Yes	AI provider identifier. The platform maps tier names to the provider's current models. Multiple providers planned — see ROADMAP.md.

`userProfile`

Describes how the platform should interpret user-profile claims for communication style personalisation. These claims are passed via the authentication bridge (see 16-embedding-and-web-component.md).

```
{
  "userProfile": {
    "styleField": "communication_style",
    "styleValues": {
      "technical":  "Respond with technical precision. Include field names, IDs, and system
details.",
      "business":   "Respond in plain business language. Avoid jargon. Use analogies where
helpful.",
      "executive":  "Respond concisely in three to five sentences. Lead with the conclusion. Omit
process detail."
    },
    "verbosityField": "response_verbosity",
    "verbosityValues": {
      "concise":   "Keep responses to the essential point. Tables preferred over prose lists.",
      "standard":  "Provide sufficient context for the response to stand alone.",
      "detailed":  "Include full explanation, examples, and supporting detail."
    }
  }
}
```

The `styleField` and `verbosityField` values must match claim keys passed in the user's JWT payload via the authentication bridge. If a user's JWT contains no matching claim, the platform uses its default tone (no style injection). If `userProfile` is omitted from the config entirely, style personalisation is disabled for the tenant.

`memory`

Configures the memory system for this tenant. See 15-memory-and-recall.md for the full behaviour specification.

```
{
  "memory": {
    "personalMemory": {
      "enabled":    true,
      "defaultOn":  false,
      "categories": ["Role", "Preference", "Correction", "Context"],
      "tokenBudget": 2000
    },
    "applicationContext": {
      "enabled":          true,
      "categories":       ["Terminology", "Policy", "Structure", "Domain context"],
      "tokenBudget":      4000,
      "approvalRequired": true
    }
  }
}
```

Field	Default	Description
`personalMemory.enabled`	`true`	Enable the personal memory feature for this tenant
`personalMemory.defaultOn`	`false`	Whether personal memory injection is on or off when a user first enables it
`personalMemory.categories`	`["Role","Preference","Correction","Context"]`	Category labels for personal memory items (displayed in the management UI)
`personalMemory.tokenBudget`	`2000`	Maximum tokens for personal memory in the system prompt

Field	Default	Description
`applicationContext.enabled`	`true`	Enable Application Context (tenant-wide org memory)
`applicationContext.categories`	`["Terminology","Policy","Structure","Domain context"]`	Category labels for application context items
`applicationContext.tokenBudget`	`4000`	Maximum tokens for application context in the system prompt
`applicationContext.approvalRequired`	`true`	Require a two-user approval workflow for new application context items

---

`conversations`

```
{
  "conversations": {
    "maxTurnLimit":                 100,
    "maxParticipants":              10,
    "retentionDays":                1095,
    "allowSharing":                 true,
    "allowAttachments":             true,
    "maxAttachmentMbPerFile":       10,
    "maxAttachmentMbPerConversation": 100
  }
}
```

All fields have platform defaults (shown above). Hosts may lower limits but not exceed platform-level maximums.

`features`

Feature flags that enable or disable platform capabilities for this tenant.

```
{
  "features": {
    "atBinding":            true,
    "displayIdDetection":   true,
    "guidedWorkflows":      true,
    "sharedConversations":  true,
    "personalMemory":       true,
    "applicationContext":   true,
    "csatSurvey":           true,
    "csatSampleRate":       0.20,
    "artefactTray":         true,
    "starterWorkflows":     ["governance-health-check"],
    "starterQuestions": [
      "What are the highest-priority issues right now?",
      "Show me anything that needs my attention.",
      "What does my team own and are there any gaps?"
    ]
  }
}
```

Flag	Default	Description
`atBinding`	`true`	Enable `@` -binding typeahead. Requires at least one `bindableType` to be configured.
`displayIdDetection`	`true`	Enable auto-detection of pasted Display IDs matching `displayIdPattern` values
`guidedWorkflows`	`true`	Enable the Workflow Library panel. Requires at least one `workflow` to be configured.
`sharedConversations`	`true`	Enable conversation sharing with other users in the same tenant
`personalMemory`	`true`	Enable personal memory feature
`applicationContext`	`true`	Enable Application Context (managed by Application Admin)
`csatSurvey`	`true`	Enable post-session CSAT rating prompt
`csatSampleRate`	`0.20`	Fraction of sessions shown the CSAT prompt (0.0–1.0)
`artefactTray`	`true`	Enable the session artefact tray panel
`starterWorkflows`	`[]`	Workflow IDs shown as suggested starters on the onboarding screen (references `workflows[].id`)
`starterQuestions`	`[]`	Plain-text starter questions shown on onboarding (up to 3, shown only if `starterWorkflows` is empty or alongside them)

`widget`

Configures the **Mode 1 floating widget** (`<ai-chat-widget>`) behaviour for this tenant. Applies only when the host embeds the floating widget web component — ignored for Mode 2 (`<ai-chat>`) and Mode 3 (`<ai-chat-field>`) deployments. See 18-entry-points-and-embedding-modes.md for full widget behaviour.

```
{
  "widget": {
    "defaultState":            "mini",
    "fabIcon":                 "chat-bubble",
    "showBadge":               true,
    "returningUserThreshold":  14400
  }
}
```

Field	Default	Description
`defaultState`	`"mini"`	Initial expansion state when the user opens the widget for the first time in a browser session. Accepted values: `"mini"` (compact panel, ~400 × 600 px) or `"full"` (full-panel overlay).
`fabIcon`	`"chat-bubble"`	Icon displayed on the floating action button. Platform icon name or absolute URL to an SVG.
`showBadge`	`true`	Whether to show an unread-indicator badge on the FAB when the assistant has produced a new response while the widget is collapsed.
`returningUserThreshold`	`14400`	Idle time in seconds before a returning user's mini-widget shows a compact re-entry card (summarising the last conversation and offering quick actions) rather than the new-conversation home state. Default: 14 400 seconds (4 hours). Set to `0` to always show the re-entry card when a prior conversation exists.

Config versioning

The config document uses semantic versioning in the `version` field:

Change type	Increment	Example
Description or prompt text edits; branding token updates	**Patch**	`1.0.0` → `1.0.1`
New bindable type, new MCP server, new workflow, new feature flag value	**Minor**	`1.0.0` → `1.1.0`
Structural schema changes; renamed or removed top-level fields	**Major**	`1.0.0` → `2.0.0`

The platform stores a full version history for each tenant's config. Rollback to any prior version is available via the Platform Admin API.

Config validation

Submitting a config via the Admin API runs **synchronous validation** before applying:

Check	Description
Schema conformance	All required fields present; field types match schema
Endpoint reachability	MCP server endpoints and `searchEndpoint` URLs respond to a health check
System prompt token count	Must be under 3,000 tokens
Model IDs in `allowedModels`	Must be available in the platform for the configured `provider`
`contextTemplate` syntax	Mustache syntax check for all bindable type templates
`starterWorkflows` references	All IDs in `starterWorkflows` must match a `workflows[].id`
Renderer trigger uniqueness	No two renderers may share the same `trigger`; no renderer `trigger` may match a built-in tag
Renderer module reachability	Each `moduleUrl` must respond to a HEAD request from the platform's validation origin; the origin must be registered for the tenant

Validation errors return a structured response with field-level detail. The config is not applied until all validation checks pass.

02 — Personas and User Journeys

The AI Chat Platform is deployed by many different host applications across different domains. This document describes personas at two levels: **platform-level archetypes** (roles that exist across all deployments) and **illustrative host application journeys** (examples showing how different types of host app use the platform).

Platform-level personas

These archetypes exist in every deployment regardless of the host application's domain.

Persona	Role	Primary need
End User	Authenticated user of the host application using the assistant as their primary access point	Ask questions and get immediate, accurate answers about the host application's domain without leaving the page
Power User	Experienced end user who regularly uses advanced features	Multi-turn investigation, `@` -binding for precise object references, branching threads, tool call inspection
Application Admin	Privileged user within the tenant responsible for the assistant's quality and application context	Manage application context (org-level memory), triage improvement signals, maintain workflow quality
Host Developer	Engineer at the host application team responsible for config and integration	Register MCP servers, maintain the application config, manage bindable types and guided workflows
Platform Admin	AI Chat Platform team member with cross-tenant visibility	Platform health, tenant onboarding, infrastructure, improvement pipeline

Application Admin design note

The Application Admin role is the platform's equivalent of a product owner for the assistant within the tenant. They are not necessarily a technical user. Key responsibilities:

- Managing application context (organisation-wide memory items that all users benefit from)
- Reviewing and approving application context items proposed by colleagues
- Triaging improvement signals generated by end users within their tenant
- Updating the application config via the Config Editor UI (non-technical admins) or Admin API (technical admins)

This role must exist in every tenant before go-live. Host teams should designate at least one Application Admin at registration time.

Illustrative host application journeys

These journeys are illustrative — they show how different types of host application use the platform. They are not prescriptive for any specific deployment.

Journey A — Governance platform: executive morning briefing

Host application type: Data governance and compliance platform
Persona: Executive leader
Setting: Mobile, during commute (5 minutes)

The executive opens the assistant on their mobile. They type: *"Give me a governance health summary — anything that needs my attention this week."*

The assistant invokes the host's governance MCP server, retrieves domain health scores and open quality issues, and returns: - A Vega-Lite chart of domain health scores - A bullet summary of the three domains requiring attention, ranked by severity

The executive taps one of the domain binding chips in the response, which the host application has configured to navigate to the domain detail page. They follow up: *"Who owns the entities flagged in the Finance domain?"* — the assistant returns a table of owners with contact details.

Both artefacts appear in the session artefact tray.

Features exercised: Mobile layout, Vega-Lite rendering, `@`-binding chip click-through to host app, artefact tray, always-on MCP server.

Journey B — Customer support platform: self-service knowledge lookup

Host application type: Internal customer support tooling
Persona: Support agent (End User)
Setting: Desktop, between calls

A support agent needs to know the current policy for handling a specific type of refund request. They type `@Refund Policy` — the `@`-binding typeahead opens and filters instantly. They select **Standard Refund Policy** from the panel; the binding chip `@{Standard Refund Policy}` is inserted. They submit:

> *"Can I apply `@{Standard Refund Policy}` to orders over 90 days old, or does that need manager escalation?"*

The assistant retrieves the resolved policy context (host-configured `contextTemplate`) and answers directly: yes for up to 180 days; manager escalation required beyond that. No knowledge base navigation required.

Total: one turn.

Features exercised: `@`-binding typeahead, single-turn self-service answer, policy binding type.

Journey C — Engineering platform: cross-service incident investigation (branching)

Host application type: Internal engineering observability and incident management
Persona: On-call engineer (Power User)

An on-call engineer is investigating an incident affecting two services. They submit:

> *"Compare error rates for `@{Payment Service}` and `@{Notification Service}` over the last hour and show any shared dependencies."*

The assistant invokes the host's observability MCP server (multiple tool calls, each visible as a collapsed disclosure card) and returns: - A comparative Vega-Lite chart of error rates over time - A Mermaid dependency graph of shared services

The engineer expands one of the tool call disclosure cards to inspect the raw parameters sent to the MCP server. They then edit their original message to narrow the time range to the last 20 minutes. This creates a **new branched conversation thread** pre-loaded with the original context. Both threads appear in the history panel — the original is untouched.

Features exercised: Multi-entity `@`-binding, tool call disclosure inspection, message edit → branched thread, Mermaid and Vega-Lite rendering.

Journey D — Legal and compliance platform: guided compliance check

Host application type: Contract and regulatory compliance management
Persona: Compliance officer (End User)

A compliance officer needs to run a weekly GDPR compliance check across active contracts. They open the **Workflow Library** panel and click **GDPR Compliance Audit** — a workflow defined by the host application in their config.

A parameter form appears requesting a contract category scope. The officer selects "Customer Agreements" and launches the workflow.

The assistant runs a multi-step analysis, invoking several host MCP tools sequentially, and returns a structured compliance report with a data table of non-compliant clauses and a Vega-Lite chart of compliance rates by category.

The officer clicks the report icon on one assistant response where the model missed a known clause type and submits an explanation. This generates an improvement signal queued for the Application Admin to triage.

Features exercised: Guided workflow with parameters, multi-step MCP tool calls, structured report rendering, improvement signal via report icon.

Journey E — HR platform: new employee onboarding

Host application type: HR and people management platform
Persona: New employee (End User, first visit)

A new employee opens the assistant for the first time. The conversation area shows the onboarding welcome state: the host-configured assistant name and description, and three starter questions drawn from the host's `features.starterQuestions` config.

They click *"What benefits am I eligible for and how do I enrol?"* and receive a plain-language explanation of their benefits package, eligibility dates, and a link to the enrolment workflow, which the host has registered as a guided workflow.

Features exercised: First-visit onboarding state, starter questions from host config, suggested follow-ups, guided workflow invocation from response.

Journey F — Shared review: multi-participant governance session

Host application type: Any platform requiring collaborative decision-making
Persona: Power User (conversation owner) + End Users (participants)

A governance lead shares an active conversation with two colleagues to collaborate on an entity ownership review. The lead clicks the share icon in the input area, searches for their colleagues by name within the tenant, and sends invitations.

Both colleagues receive in-platform notifications and accept — seeing the full conversation history from the beginning (with the acceptance disclaimer). Each participant now submits their own turns. Because each user's `@`-binding typeahead is scoped to their individual permissions, one participant sees a **[Restricted object]** chip for a binding they cannot access in the host application.

At the end of the session, the lead downloads all artefacts from the tray as a zip archive.

Features exercised: Shared conversations, participant invitation, permission-scoped `@`-binding, message attribution, artefact download.

Persona × Feature Matrix

Feature	End User	Power User	App Admin	Host Developer
Mobile-first layout	✓	✓	✓	

Feature	End User	Power User	App Admin	Host Developer
`@`-binding typeahead	✓	✓	✓	
Model switching		✓		
Document attachments	✓	✓	✓	
Guided workflow prompts	✓	✓	✓	
Tool call disclosure inspection		✓	✓	✓
Vega-Lite charts	✓	✓	✓	
Mermaid diagrams		✓	✓	
Data tables	✓	✓	✓	
Conversation branching		✓	✓	
Shared conversations	✓	✓	✓	
Personal memory management	✓	✓	✓	
Application context management			✓	
Improvement signal triage			✓	
Application config management			✓	✓
Onboarding welcome state	✓			
Report icon feedback	✓	✓	✓	

03 — Design Principles

These nine principles govern all design decisions for the AI Chat Platform. Where a proposed feature conflicts with a principle, the principle takes precedence. Deviations require an explicit decision record.

P1 — Transparency first

Every action the platform takes is visible. Tool calls are shown, not hidden. The assistant never implies a write operation has occurred if it has not. Opacity is a trust risk on any platform where users are making decisions based on the assistant's output.

Consequences: - Every MCP tool invocation renders as a collapsible disclosure card in the conversation thread — tool name, input parameters, response status, result summary. - The assistant may not summarise tool call activity without providing access to the full disclosure. - Every write operation proposed by the assistant displays a **confirmation step** showing the before/after state before any MCP write call is executed. The assistant never implies a write has occurred if it has not.

P2 — Host-scoped by design

Each deployment is a specialist, not a generalist. The platform provides the engine; the host application defines the domain. When a query falls outside the host-configured scope, the assistant explains clearly and redirects constructively.

Consequences: - The host system prompt establishes domain scope and instructs the model to decline out-of-scope queries with a constructive redirect (configurable via `scope.outOfScopeRedirect`). - Out-of-scope responses are captured as improvement signals to inform system prompt refinement. - No platform-level general-purpose prompt library. All guided workflows are host-defined.

P3 — Rendered output over raw text

Plain text answers to structured data queries are a regression from most application UIs the platform is embedded in. The default mode is rendered — charts, diagrams, tables — with prose reserved for narrative explanation. Every structured output has a rendered form.

Consequences: - The rendering decision ruleset (10-content-rendering.md) is applied to every assistant response before display. - Mermaid, Vega-Lite, and tabular data are always rendered — never displayed as raw source. - The system prompt instructs the model to prefer structured outputs (Vega-Lite for metrics, Mermaid for relationships, tables for entity lists) over prose equivalents when the data supports it.

P4 — Audit completeness

Every conversation turn is a self-contained, reproducible record: raw prompt, resolved prompt, attached files, model response, tool call log, output artefacts. A reviewer can reconstruct any turn without referring to an external system.

Consequences: - Storage policy (11-audit-and-storage.md) retains all turn elements at write time — nothing is generated on demand. - Conversations are append-only at the turn level. Editing creates a new branched thread; the original is untouched. - Turn-level deletion is not permitted. Users may archive or delete conversations subject to the tenant's configured retention period. - The `assistant` Postgres schema is isolated from platform infrastructure schemas with RLS enabled on all tables.

P5 — Mobile-first and responsive

The assistant is fully functional on mobile and tablet. Hosts embed the platform in products consumed across device types. Business users, field workers, and mobile-first users are significant populations.

Assistant-specific consequences: - `@`-binding typeahead anchors to the bottom of the viewport on mobile (not to the cursor position). - Conversation search opens as a full-screen overlay on mobile with the keyboard raised. - Mermaid diagrams are horizontally scrollable on mobile — never scaled to illegibility. - Vega-Lite charts use `width: "container"` for responsive sizing. - All input controls are thumb-reachable on mobile; no critical control is hidden behind secondary menus.

P6 — Consistency with the host design system

The `<ai-chat>` web component inherits the host application's visual language via branding tokens. The component introduces no new visual language of its own. End users should experience the assistant as a natural extension of the host application, not a foreign product embedded inside it.

Consequences: - All colour, typography, border radius, and logo tokens are provided by the host application config (`branding` section). - Participant colour assignments in shared conversations use the host's primary colour as an anchor; the platform generates a compliant palette from it. - Platform-default styles are used only as fallbacks when the host has not provided a token.

P7 — Progressive disclosure

Long responses use progressive disclosure. Diagrams collapse to thumbnails. Tool call disclosures collapse by default. JSON trees collapse to two levels. The user is never overwhelmed by a single response.

Consequences: - Tool call disclosure cards are collapsed by default; users expand to inspect detail. - Mermaid diagrams collapse to a thumbnail with an expand control; full-screen view available. - JSON inspector trees default to two levels collapsed. - Data tables paginate — no unbounded scroll. - Suggested follow-up chips (max 3) appear beneath each assistant response, not inline.

P8 — Human-gated improvement

No change to the system prompt, tool registry, or guided workflow prompts is applied automatically. All improvement recommendations enter a triage pipeline and require human review before any change is deployed.

Consequences: - Improvement signals (explicit ratings, retry patterns, tool failures, corrections, out-of-scope responses) are captured and queued — never applied directly. - Automatically detected signals below the confidence threshold go to manual review before an improvement issue is raised. - Explicit user reports (via the report icon) always generate an issue (no threshold). - Application Admins review signal issues on a regular cadence. All changes enter the host application's standard approval process before deployment.

P9 — Host sovereignty

The host application has final authority over the assistant's configuration, scope, and behaviour within its tenant. The platform enforces safety and audit requirements as non-negotiable constraints, but within those constraints the host is in control.

Consequences: - The application config schema gives hosts control of system prompt, tools, bindings, workflows, branding, models, and feature flags. - Platform-managed instructions (safety clauses, tool transparency, audit logging) are injected alongside — never instead of — the host system prompt. They are not user-configurable or host-overridable. - Hosts may restrict capabilities (disable features, lock model selection, limit participant count) but may not exceed platform-level maximums. - New platform-level defaults that would change existing tenant behaviour require a config migration path and advance notice.

Principle interactions

Principle	Most common tension	Resolution
P1 (transparency) vs UX conciseness	Tool call disclosures add visual weight	Disclose via collapsible cards — present but not obtrusive
P3 (rendered output) vs P5 (mobile)	Charts may not render well at narrow widths	Vega-Lite `width: "container"` + horizontal scroll for tables and diagrams

Principle	Most common tension	Resolution
P4 (audit completeness) vs storage cost	Full binary attachment storage is expensive	Retention is tenant-configurable; scheduled archival at expiry
P6 (host design system) vs accessibility	Host colour tokens may not meet contrast requirements	Platform validates contrast ratios at config submission; warnings surfaced to host
P7 (progressive disclosure) vs P4 (audit completeness)	Collapsed content could obscure audit data	Collapse is a UI affordance only — all content is stored in full regardless of display state
P8 (human-gated improvement) vs velocity	Approval pipeline slows iteration	Accepted tradeoff — trust and data integrity outweigh iteration speed on platforms where users act on assistant output
P9 (host sovereignty) vs P1 (transparency)	Host may wish to hide tool calls from end users	Tool call disclosure is mandatory and non-configurable — hosts may not suppress MCP disclosures

04 — Conversation Management

Conversation model

Each conversation is a **named, persistent thread** scoped to the authenticated user within their tenant. Key properties:

- Conversations survive browser refresh and are resumable across sessions.
- The history panel lists all past conversations in reverse-chronological order with auto-generated titles derived from the first user message.
- Users may rename, archive, or delete conversations — subject to the tenant's configured retention period (see 11-audit-and-storage.md).
- Conversations are **append-only at the turn level**. No turn is ever overwritten or destroyed. Editing creates a new branched thread; the original is preserved in full.

Conversation states

State	Description
Active	Has been interacted with in the last 30 days; appears in the default history view
Pinned	Permanently anchored to the top of the My Conversations list regardless of recency. Maximum 5 pinned conversations. Useful for recurring reviews or ongoing work threads.
Archived	Hidden from default history view; accessible via archived filter; retains full content
Deleted	Removed from user view; physical deletion deferred to retention expiry per tenant retention policy
Shared	One or more other authenticated users in the same tenant have accepted an invitation; appears under **Shared With Me** group in history panel
Locked (read-only)	Last remaining participant's account deactivated; flagged for administrative review

---

Conversation branching

Users may edit any previously submitted message. Editing creates a **new conversation thread** — not an in-thread branch.

Branch mechanics

1. User clicks the edit icon on any past user message.
2. An inline edit field opens, pre-populated with the original text. `@`-binding chips are preserved and editable.

3. Submitting the edited message creates a **new conversation thread** that inherits all turns up to and including the edited turn (with the edit applied) and continues independently.
4. The original conversation is **untouched**.
5. Both threads are linked: - The new branch carries a header chip: *"Branched from: [title], turn N"* - The origin conversation carries a reciprocal reference at the corresponding turn
6. `Escape` cancels the edit without creating a branch.

Regeneration

Regeneration (resending an unchanged message for a fresh model response) follows the same model: a new thread is created from that turn point. The original response in the parent conversation is untouched.

Why branching, not in-place editing?

> **Audit completeness (P4):** Every turn is an auditable record. Overwriting turns in place would destroy the original query-response pair. Branching preserves both the original intent and the refined intent as independent, auditable threads. This design holds regardless of the host application domain.

Conversation search

A search field at the top of the history panel searches across all the authenticated user's conversations — personal and shared — within their tenant.

Search scope

Field searched	Examples
Conversation title	"Finance governance review"
User message text	"Compare error rates for `@{Payment Service}`"
Assistant response text	"The Payment Service had three error spikes in the last hour"

Search behaviour

- Results are returned in **relevance order** with the matching excerpt highlighted.
- Selecting a result opens the conversation **at the matching turn**.
- Accessible via `Cmd/Ctrl + K` from anywhere on the chat surface.
- On mobile, search opens as a **full-screen overlay** with the keyboard raised.
- Search spans both **My Conversations** and **Shared With Me** groups.

Context window management

Silent truncation is not permitted (P1 — transparency first). The user is always informed when context is being condensed.

Conversation length limit

A conversation may not exceed **100 turns** (configurable per tenant up to the platform maximum via `conversations.maxTurnLimit`). This is an explicit product limit — not a technical one — chosen to keep conversations focused and auditable. At the turn limit, the conversation is closed to new input and the user is prompted to continue in a new thread (see branching above). The full turn history is always retained in the audit trail.

Token warning — 80%

Before the turn limit is reached, the active context window may fill based on message length and attachment size. When a conversation reaches **80% of the model's context window** (160K tokens on current 200K context models), a **persistent warning banner** appears in the conversation header:

> *"This conversation is getting long. Older turns are being summarised automatically to keep [AssistantName] running. [View summary ↗]"*

The warning is persistent (not dismissible) and remains visible until the conversation ends or is branched.

Automatic summarisation — default behaviour

When the context window reaches **80%**, the platform **automatically summarises the oldest turns** to free space, without requiring user action.

Element	Behaviour
What gets summarised	Oldest turns first — the earliest 40% of turns are summarised; the most recent 60% are always kept verbatim in the context window
Summary model	A separate **fast-tier model** API call generates the summary before the next user turn is processed
Summary format	Structured: **Key entities discussed** · **Key findings** · **Decisions or conclusions reached** · **Unresolved questions**
Visibility	A **condensation marker** is inserted in the conversation thread: *"↑ N turns summarised — [tap to expand]"*
Expansion	Tapping the marker shows the full structured summary inline
Audit trail	Original turn content is **always retained in full** in `assistant.turns`. Summarisation affects only the active context window sent to the AI provider, not stored data.

Element	Behaviour
Frequency	Re-summarises incrementally on each subsequent turn that would exceed the context limit
Injection	Summary injected as a `[Context from earlier turns]` block at the top of the conversation context — clearly labelled so the model treats it as historical context

Manual branch — user-initiated

At any point, the user may start a new conversation thread from the current context. A **"Continue in new thread"** button appears in the warning banner. Clicking it:

1. Creates a new conversation thread
2. Pre-loads it with the auto-generated summary as the opening context
3. Links it back to the origin conversation with a *"Continued from: [title], turn N"* header chip
4. Archives the original conversation (untouched)

Model context limits

Context window sizes are determined by the configured AI provider and model tier. The 80% warning threshold applies regardless of context size.

Model tier	Typical context window	Warning threshold
Fast	Provider-specific — refer to provider documentation	80% of context window
Standard	Provider-specific	80% of context window
Powerful	Provider-specific	80% of context window

The platform queries the active model's context limit at session start and calculates the warning threshold dynamically. Summarisation triggers at 80% of whatever the active model's window is — no manual threshold configuration is needed.

History panel organisation

- **Pinned** — conversations explicitly pinned by the user; always at the top regardless of recency; max 5; pin/unpin via right-click or long-press context menu
- **My Conversations** — all conversations created by the authenticated user; grouped by recency (Today / Yesterday / Past 7 days / Past 30 days / Older); within each group, reverse-chronological. Groups with no conversations are omitted.
- **Shared With Me** — conversations to which the user has accepted an invitation; reverse-chronological; visually distinct (shared icon)
- Conversations with unread turns in shared threads show a badge count
- Archived conversations are hidden unless the archived filter is active
- **Empty state** — when the user has no conversations at all (first visit, or after deletion of all conversations), the My Conversations area shows a brief prompt to start a new conversation, mirroring the onboarding state in the conversation area
- **Search empty state** — when a search query returns no results, the panel shows *"No conversations matching '[query]'"* with a prompt to try different terms

---

In-conversation search

Users may search within the currently open conversation to locate a specific turn or content. Accessible via `Cmd/Ctrl + F` while the conversation area is focused.

Behaviour	Specification
Scope	Full text of all turns in the current conversation — user messages, assistant responses, and artefact names
Presentation	Matching text highlighted in-thread; navigation arrows to move between matches
Turn navigation	Each match shows the turn number and jumps the scroll position to that turn
Close	`Escape` or clicking outside the search bar dismisses it

In-conversation search is distinct from the cross-conversation search in the history panel (`Cmd/Ctrl + K`), which searches across all conversations.

05 — Input and Composition

Free-form text input

The primary input is a **multi-line natural-language text field**.

Behaviour	Specification
Growth	Field grows to five lines before scrolling internally
Submit	`Cmd/Ctrl + Enter`
While streaming	Input is disabled; replaced by a stop-generation button
Character limit	None enforced in v1 (subject to model context window)
Draft preservation	An unsent message is preserved per conversation in browser storage. If the user switches to another conversation or navigates away, the draft is restored when they return to that conversation. Drafts are discarded on explicit send or when the user manually clears the field. `@`-binding chips are included in draft state.

`@`-binding — object binding typeahead

Typing `@` anywhere in the input opens a **scoped typeahead panel** that filters in real time against the bindable object types configured by the host application.

Binding mechanics

1. User types `@` — typeahead panel opens immediately.
2. User continues typing — panel filters in real time using fuzzy match from the first character after `@`.
3. User selects an object — a **binding chip** is inserted: a styled atomic pill carrying the object type icon and display name, e.g. `@{Finance Domain}`.
4. On submission, binding chips are **resolved server-side** before the message reaches the AI model. The user sees chips; the model receives structured context blocks assembled from the host-configured `contextTemplate` for that bindable type.
5. **Clicking a binding chip** in any message (input or response) fires a `binding-click` event on the web component. The host application handles this event — typically navigating to the object's detail page within the host application. See 16-embedding-and-web-component.md.

Bindable types

Bindable types are entirely host-configured. The platform provides the typeahead mechanism; the host defines what can be bound via the `bindableTypes` section of the application config (see 01-host-application-config.md).

Examples of host-configured bindable types:

Example type	Example chip	What gets injected into model prompt
Data Domain	`@{Finance Domain}`	Domain name, ID, owner, entity count — from host `contextTemplate`
Policy	`@{Refund Policy}`	Policy name, status, effective date, summary — from host `contextTemplate`
Service	`@{Payment Service}`	Service name, owner, SLO, current health — from host `contextTemplate`
Customer	`@{Acme Industries}`	Customer name, account status, account manager — from host `contextTemplate`
Document	`@{Q2 Board Report}`	Document title, type, date, summary — from host `contextTemplate`
Workflow	`@{Expense Approval Workflow}`	Workflow name, description — invokes the workflow prompt on submission

Any object type the host application defines is bindable. The platform imposes no restrictions on the number or nature of bindable types.

Inactive and disabled objects

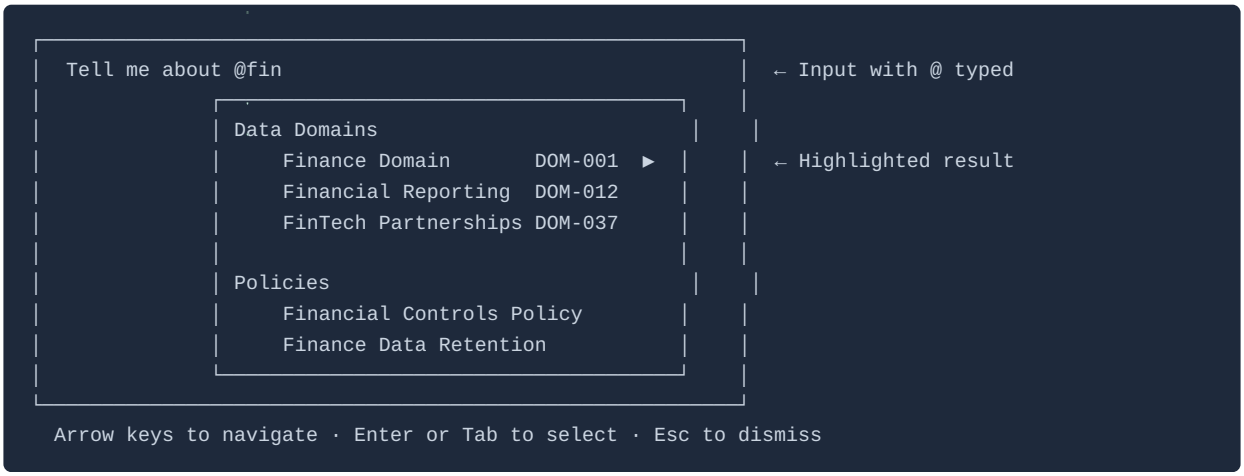
Binding chips for objects that are marked inactive in the host application's search endpoint response (`inactive: true`) are rendered in a **greyed-out style** with an `(inactive)` suffix.

Behaviour	Specification
Chip appearance	Grey fill and muted text; `(inactive)` suffix; object type icon retained
Resolution	The chip still resolves — the `contextTemplate` is populated and the model receives the resolved context, including the inactive status
Context injected	The resolved block includes the object's inactive state so the model can acknowledge it
Navigation	Clicking the chip still fires the `binding-click` event

Behaviour	Specification
Typeahead	Inactive objects appear below active objects of the same type, with the `(inactive)` suffix visible

This ensures historical conversations remain navigable even when referenced objects have been retired — the binding never silently breaks.

Typeahead layout



After selection, the chosen object becomes a binding chip inline in the input:

Inactive objects appear below active results with a greyed style:

Typeahead behaviour

Behaviour	Specification
Match	Fuzzy match from the first character after `@`
Ranking	Host-configured rank order (ascending `rank` field in `bindableTypes`); alphabetical within type when `rank` values are equal
Maximum results	8 results visible; scroll to see more
Keyboard navigation	Arrow keys to move; Enter or Tab to select; Escape to dismiss
Mobile	Typeahead anchors to the bottom of the viewport (not cursor position)

Behaviour	Specification
Permission scope	Each user's typeahead results are filtered by the host's `searchEndpoint` — users see only objects their host application allows them to access

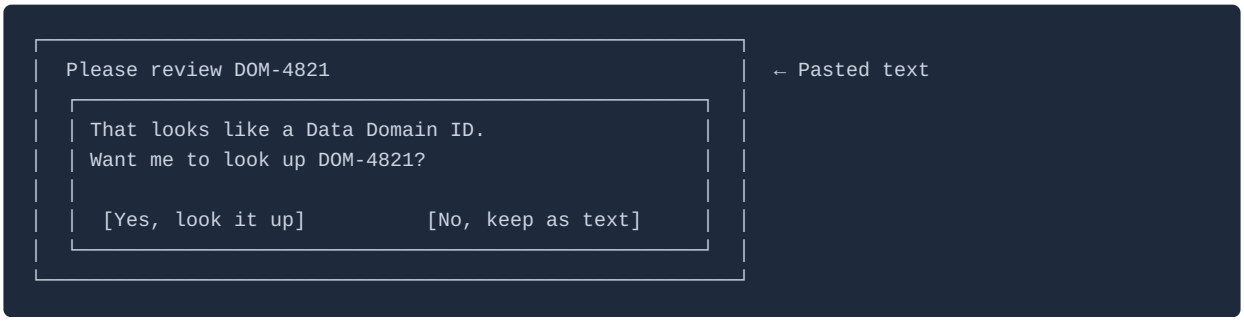
`@`-binding in shared conversations

When a conversation is shared, each participant's typeahead is scoped to their own permissions (as enforced by the host's `searchEndpoint`). A participant cannot bind to an object they cannot access in the host application.

If a submitted message contains a binding to an object that another participant cannot access, the restricted participant sees the chip labelled **"[Restricted object]"** — they do not see the resolved context that was injected into the model prompt. See 08-shared-conversations.md.

Display ID detection

When a host application configures a `displayIdPattern` on a bindable type, pasting text that matches the pattern anywhere in the input is **detected automatically**. On detection, the interface presents a direct lookup confirmation:



If confirmed, `DOM-4821` is resolved to a binding chip exactly as if the user had selected it via the `@`-binding typeahead.

> *"That looks like a [Type] ID. Want me to look up [displayId]?"*
> [Yes, look it up] [No, keep as text]

This supports workflows where IDs are shared via email, Slack, or reports — users can paste them directly without knowing the object's name.

Document attachments

Users may attach **multiple documents per message turn**. The limit is a total storage budget per conversation, not a per-message count.

Supported formats

Format	Extensions	Processing
PDF	`.pdf`	Full text and structure extracted up to the token limit
Excel	`.xlsx`, `.xls`	Sheets extracted as tabular data; multiple sheets processed sequentially
Word	`.docx`, `.doc`	Body text, headings, and tables extracted; tracked changes ignored
Image	`.png`, `.jpg`, `.jpeg`, `.webp`	Processed as vision content — model can read and reason about the visual content. Typical use: screenshots, dashboards, reports, or external documents not available as PDFs

Images may be attached directly or **pasted from the clipboard** (`Cmd/Ctrl+V` when the input field is focused). The input area shows a preview thumbnail for each attached image before submission.

Constraints

Constraint	Default	Config field
Maximum file size	10 MB	`conversations.maxAttachmentMbPerFile`
Total attachment budget per conversation	100 MB	`conversations.maxAttachmentMbPerConversation`
Password-protected files	Not supported in v1	—
Embedded images	Extracted as vision content blocks where supported; otherwise skipped with a notice	—
Quantity per message turn	No limit	—

Budget enforcement

The input area displays a running attachment budget indicator: *"42 MB of 100 MB used"*. When the conversation budget is full: - The attachment button is disabled - A notice appears: *"Attachment limit reached for this conversation. Start a new conversation to attach more files."* - Previously attached files remain available in the artefact tray

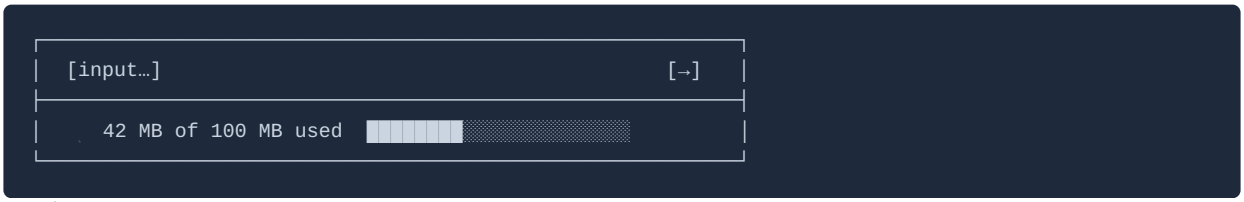
Artefact retention

Attached files are **stored in full** in platform storage as part of the conversation audit trail. They are downloadable from the artefact tray for the lifetime of the conversation record and are not deleted when the user closes the session.

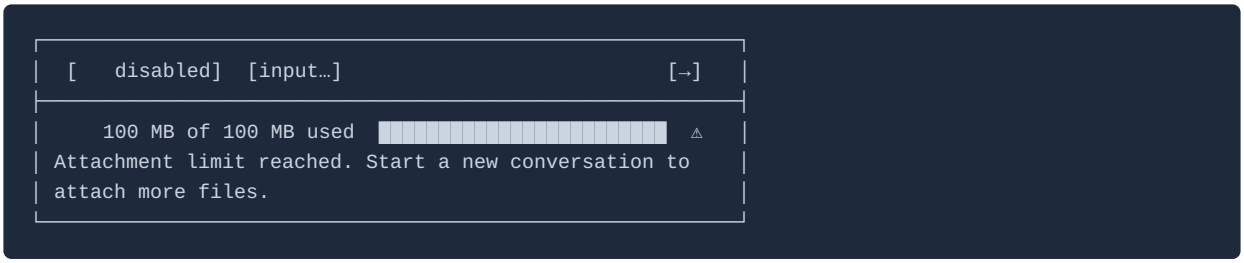
Document display in the conversation

Attached documents appear in the user message bubble as a labelled file card: format icon, file name, and page/sheet count. Documents are not rendered inline.

The attachment budget indicator appears below the input area:



When the conversation attachment budget is exhausted:



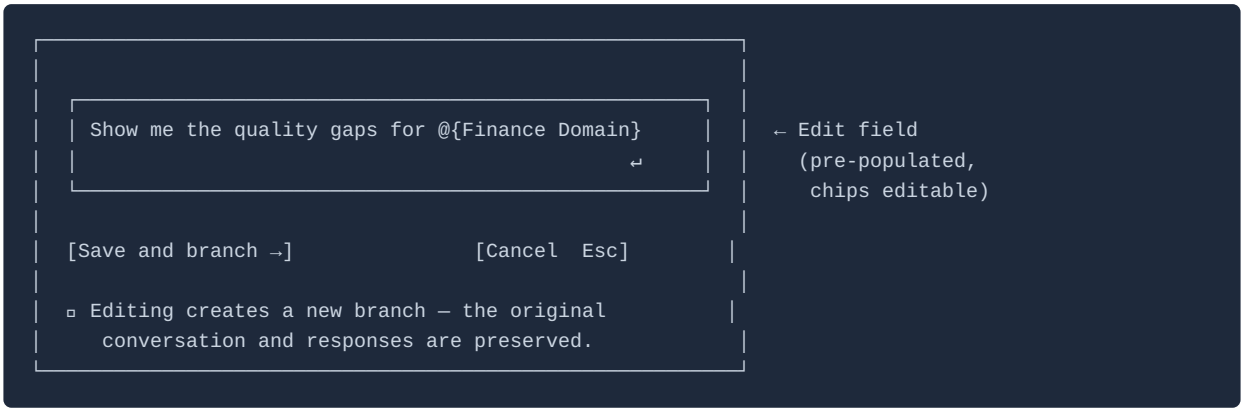
When the model references a specific section of an attached document, it cites by page number (PDF), sheet name (Excel), or heading (Word).

Message editing

An **edit icon** is associated with each past user message. On desktop it appears on hover; on mobile it is always visible below the message (no hover state). Long-pressing the message on mobile also opens the message action menu.

Action	Behaviour
Click edit icon	Inline edit field opens, pre-populated with the original text
`@`-binding chips	Preserved in the edit field; editable
Submit	Creates a new branched conversation thread (see 04-conversation-management.md)
`Escape`	Cancels edit; no branch created; original conversation untouched

Edit field layout



What editing does not do

- It does not modify the original message in place.
- It does not delete the original conversation thread.
- It does not create an "alternate" version within the same thread.

Editing always creates a distinct, independently navigable thread. This satisfies P4 — audit completeness: no original query-response pair is ever destroyed.

06 — Model Configuration

AI provider architecture

The AI Chat Platform is designed to be **AI provider-agnostic**. The model provider is abstracted behind the platform's edge function layer — the conversational surface, MCP tool access, `@`-binding resolution, and audit trail are identical regardless of which AI provider is active for a tenant.

Release	Provider examples
v1 (current)	One AI provider configured per tenant — host selects from the platform's supported providers at tenant registration
Planned	Multiple providers available for selection; model capability parity validated across providers before enablement

The provider abstraction means host applications select provider as well as model tier at the tenant level. The conversational surface, MCP tool access, `@`-binding resolution, and audit trail are identical regardless of which provider is active. See ROADMAP.md for the multi-provider timeline.

Model tiers

The platform exposes three **model tiers** that host applications configure in their `models` section. Each tier maps to the configured provider's appropriate model — the host config uses tier names, not provider-specific model IDs.

Tier	Config value	Profile	Typical use
Standard	`"standard"`	Default — balanced capability and speed	General queries, data lookup, summaries, diagram generation
Powerful	`"powerful"`	Maximum capability, higher latency and cost	Complex multi-entity reasoning, cross-domain analysis, large document analysis
Fast	`"fast"`	Fastest, lowest latency and cost	Quick lookups, simple factual answers, Display ID resolution

The platform maps each tier to the provider's current best-fit model at the time of session start. When the provider releases a new model, the platform updates the tier mapping — host application configs do not need to change.

Model switching

Users may switch the active model tier at any point in a conversation. The switch takes effect on the **next turn** — it does not re-process previous turns.

The host-configured system prompt, MCP tool access, and `@`-binding resolutions are identical across all tiers within the active provider.

Host applications configure the allowed tiers and default tier in the `models` section of the application config. Users can only switch to tiers in the `allowedModels` list. Setting `userCanSwitch: false` locks all sessions to the `defaultModel` tier.

Model UI

- The **active model chip** is displayed adjacent to the input field — always visible regardless of scroll position.
- Clicking the chip opens a **model selection popover** showing all allowed models with their profiles and recommended-use summary.
- Each assistant turn carries a **hover-visible metadata line**: model label + token counts (e.g. `↑ 1,240 in · ↓ 847 out · 312 cached`).
- The model label on each response enables participants in shared conversations to see which model was active for each turn.

Model in shared sessions

The active model and any opt-in MCP tools apply to the session as a whole. The user who submits a message determines the model context for that turn, based on their current model chip selection. Other participants see the model label on each assistant response.

Communication style and verbosity

The model's communication style and verbosity are driven by **claims in the user's authenticated JWT** — not a chat-specific setting. This keeps the assistant experience consistent with the rest of the host application, where user preferences are already managed.

Host applications configure which JWT claim fields map to style and verbosity, and define the prompt text for each value, in the `userProfile` section of the application config (see 01-host-application-config.md).

Example style definitions

Host applications define their own style values and descriptions. Below is an example configuration:

Style value	Example prompt instruction
`technical`	Respond with technical precision. Include field names, IDs, and system details.
`business`	Respond in plain business language. Avoid jargon. Use analogies where helpful.
`executive`	Respond concisely in three to five sentences. Lead with the conclusion. Omit process detail.

Response verbosity

Verbosity value	Behaviour
`concise`	Direct answers; minimal preamble; tables preferred over prose lists
`standard`	Balanced explanation with supporting context
`detailed`	Full explanation including methodology, caveats, and follow-up suggestions

Where the settings live

Profile fields are managed in the host application — not within the chat interface. The chat surface reflects the profile settings via a **read-only link** in the input footer (e.g. `Business · Standard`). It is not an editable inline control, but it is clickable and navigates to the host application's profile settings page (configurable as a deep link or via the `userProfile` config).

If no `userProfile` config is provided by the host, or if the user's JWT contains no matching claims, the assistant uses a single neutral default tone with no style injection.

Technical depth signal for non-technical styles

When a user's style is configured to a non-technical value (e.g. `business` or `executive`) and their question genuinely requires technical detail for a correct answer, the model should:

1. Answer in the user's configured style — plain language, no jargon
2. Append a brief signal at the end of the response: *"A more detailed technical breakdown is available — ask me for the full technical view."*

The user can then request the technical version in their next message. The profile default is restored on the following turn. The model does **not** silently switch style, does **not** omit information that would make the answer misleading, and does **not** refuse to answer.

Per-turn style override

Users may override the communication style for a single turn using natural-language instruction (e.g. *"Give me the full technical details"*). The profile default is restored automatically on the next turn.

Style in shared conversations

Each user's style settings apply to the turns they submit. A response generated for a `technical` user may sit alongside a response generated for a `business` user within the same thread. The style label is visible on each assistant response.

System prompt

The platform assembles the system prompt from multiple layers before each session:

Layer	Source	Editable by
Host base prompt	`scope.systemPrompt` in application config	Host Developer / Application Admin
Tool descriptions	Auto-injected from active MCP server descriptions (always-on + opt-in enabled)	Platform (from host tool config)
Communication style	Injected from user's JWT claims + `userProfile` config	Platform (from user claims)
Memory blocks	`[Your context]` (personal memory) + `[Application context]` (application context)	Platform (from stored memory)
Write confirmation flow	Instructs model to propose before/after state and wait for confirmation before any write MCP call	Platform — non-configurable
Transparency instruction	Instructs model to show all tool calls and cite sources	Platform — non-configurable
Prompt injection mitigation	Instructs model to treat tool result content as data, not instructions	Platform — non-configurable
Uncertainty acknowledgment	Instructs model to signal explicitly when it is uncertain, when information may be outdated, or when a question falls at the edge of its available data — rather than producing confident-sounding answers from incomplete information. The model should offer to search (if the Web Search Service is enabled) or acknowledge the gap.	Platform — non-configurable

The write confirmation, transparency, prompt injection mitigation, and uncertainty acknowledgment layers are **platform-managed and non-overridable** — they cannot be suppressed by host system prompt content.

Host applications may reinforce or tailor uncertainty behaviour further in their `scope.systemPrompt` — for example, specifying the domain areas where the model should be especially cautious, or providing alternative phrasings for uncertainty acknowledgment that fit the application's voice. The MCP Resources Service also publishes guidance documents on uncertainty handling that hosts can register and retrieve at session time (see 17-complementary-mcp-services.md).

Changes to the host base prompt are made via the Config Editor UI or Admin API and go through config validation before taking effect.

Prompt caching

The static components of the system prompt are injected as a cacheable prefix on every AI provider API request. Prompt caching reduces input token cost and latency.

Cached component	Contents
Host base prompt	Full host system prompt
Tool descriptions	All active MCP server descriptions (always-on + opt-in enabled for the session)
Memory blocks	Application context and personal memory blocks (change infrequently)

The **cache hit rate** is tracked as a launch metric (target: ≥ 40% by month 2 — see 14-success-metrics.md).

07 — Tool Access

Host-registered MCP servers

The AI Chat Platform has **no built-in always-on tool**. Every MCP server available in a session is registered by the host application in the `mcpServers` section of the application config (see 01-host-application-config.md).

Host teams can discover available MCP tools via the **MCP Repository** complementary service before registering them (see 17-complementary-mcp-services.md).

If no MCP servers are registered, the assistant operates in **prompt-only mode** — it answers from system prompt knowledge only, with no live data access.

Access tiers

Tier	Behaviour
Always-on	Active in every session; cannot be disabled by the end user. Designated by setting `accessTier: "always-on"` in the server's config entry. Host applications should designate as always-on only the MCP servers that should be available to all sessions without user action.
Opt-in	Off by default; enabled by the end user per session via the tool selection panel. Does not persist across sessions. Designated by `accessTier: "opt-in"`.

The always-on tier should be used sparingly. Each always-on tool's description is injected into the system prompt on every session, consuming token budget. Opt-in tools inject their description only when enabled.

Role-based tool access

MCP servers can be restricted to specific user roles via the `roles` field in the server config entry. Roles are matched against the user's JWT claims forwarded via the authentication bridge. If `roles` is empty, the server is available to all authenticated users.

A user who does not hold a required role will not see the server in the tool selection panel and cannot activate it. The server description will not be injected into their system prompt.

Guided workflows

Guided workflows are host-defined conversation starters accessible from the **Workflow Library** panel. They are configured in the `workflows` section of the application config.

Invocation method	How it works
Click in Workflow Library panel	Single click opens a parameter form (if parameters are defined); the assembled prompt is injected into the input field; user reviews and submits
`@`-binding	If a workflow is configured as a bindable type, typing `@` followed by the workflow name injects the workflow prompt on submission
Natural language	Phrasing that closely matches a workflow's purpose may cause the model to suggest the workflow and offer to launch it

Guided workflows are **host-managed** — the prompts are version-controlled in the application config. End users cannot create or modify guided workflows.

Workflow parameter types

Parameter type	UI presentation
`binding`	Opens an `@`-binding typeahead scoped to the configured `bindableTypeId`
`text`	Free-text input field with the parameter's `label`
`select`	Dropdown populated from the `options` array in the parameter config

Required parameters must be filled before the workflow can be launched. Optional parameters may be left empty.

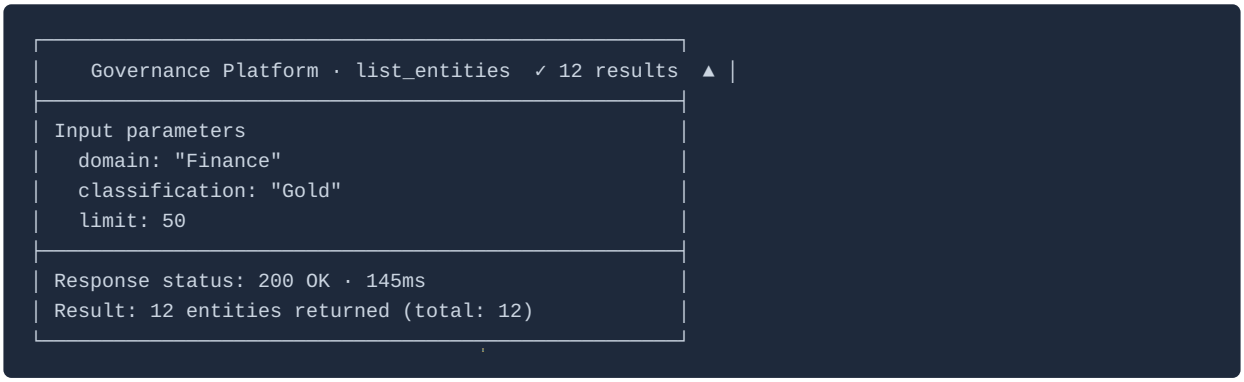
Tool call transparency

Every MCP tool invocation renders as a **collapsible disclosure card** in the conversation thread. This is mandatory — tool calls are never hidden (P1 — transparency first).

Disclosure card anatomy

```
┌──────────────────────────────────────────────────┐
│  Governance Platform · list_entities  ✓ 12 results  ▼ │
└──────────────────────────────────────────────────┘
```

On expansion:



Card element	Content
Tool name	MCP server name + tool name (e.g. `Governance Platform · list_entities`)
Status icon	✓ success / ✗ error / in-progress
Result summary	Brief outcome (e.g. `12 results`, `error: permission denied`)
Expand/collapse	Chevron — collapsed by default
Input parameters	Full parameter object
Response status	HTTP status code + latency in milliseconds
Result detail	Full result summary or error detail

Write operations

When the model proposes a write action (an MCP tool call that modifies data), the assistant **must** surface a confirmation step before executing the call:

The confirmation step is mandatory and non-configurable. The assistant never implies a write has occurred if it has not.

Error states in disclosures

When a tool call fails, the disclosure card shows error status with full detail. The model surfaces the error to the user in plain language, citing the specific tool and the nature of the failure.

MCP server unavailability

If a host MCP server is unreachable: - An **always-on server failure** triggers a **degraded-mode banner** across the session: *"[Server name] is unavailable. [AssistantName] is answering from its own knowledge only — data may not reflect the current state of your application."* The session continues in text-only mode. No silent failure. - An **opt-in server failure** shows an error in the tool call disclosure card. The user is informed and may disable the failing server for the session.

The degraded-mode banner is persistent until connectivity is restored.

Tool selection panel (UI)

The tool selection panel is accessible via the tool icon in the input area. It shows:

- All **always-on** servers with a permanent "always active" indicator (no toggle)
- All **opt-in** servers with `enabled: true` in the registry: name, one-sentence description, and a toggle switch (off by default)
- Opt-in servers restricted by role are not shown to users who don't hold the required role

Tools enabled in the panel are active for the remainder of the session only. The panel state does not persist across sessions.

Tool access in shared sessions

The active model and enabled opt-in tools apply to the session as a whole. The user who submits a message determines the tool context for that turn based on their current tool panel state. Other participants see the tool call disclosure cards in the thread.

Participants cannot access MCP tools beyond their own permission level. If a tool call succeeds for the submitting user but the host MCP server enforces per-user access control, other participants may see restricted results in the disclosure card (`[Restricted — insufficient permissions]` on the result summary).

Complementary MCP services in tool context

Beyond host-registered MCP servers, three platform-level ecosystem services are available for integration:

- **MCP Repository** — Host teams use this during configuration to discover and browse available MCP tools before registering them. It is not directly invoked in conversations but informs the tool registry setup.
- **MCP Resources Service** — Provides centralised skills, static resources, and reusable prompt artefacts. Host applications may choose to register the MCP Resources Service as an opt-in or always-on server, making its capabilities available during conversations.

- **Web Search Service** — Provides real-time web search and page retrieval. Registered by host applications as an opt-in or always-on MCP server for sessions that need current information beyond the host's own data.

See 17-complementary-mcp-services.md for full descriptions of all three services.

08 — Shared Conversations

Overview

Any conversation may be shared with other authenticated users **within the same tenant**. Sharing is **explicit and controlled** — there are no open links, no public access, and no anonymous participants. Every person in a shared conversation must hold an active account in the host application and be a member of the same tenant. The tenant boundary and its rationale are specified in 15-memory-and-recall.md — Recall and access scope.

Shared conversations require the `features.sharedConversations: true` feature flag in the tenant config.

Invitation model

The conversation owner invites participants by searching for their name or email address **within their tenant's user directory**. Cross-tenant sharing is not supported — the search is hard-scoped to the authenticated user's tenant.

Rule	Specification
Search scope	Users within the same tenant only — it is not possible to invite a user from another tenant or an external email address
Shareable URLs	Not supported — invitations are directed to a specific named user
Maximum participants	Configured by host in `conversations.maxParticipants` (default: 10)
Invitation delivery	In-platform notification; optionally email per the recipient's notification preferences
Invitation content	Conversation title, inviting user's name, accept/decline action
Accept	Adds the conversation to the participant's history panel under the **Shared With Me** group
Decline	Removes the invitation; no notification to the inviter

---

Participant model

All participants are **equal**. There are no roles, no elevated permissions, and no designated owner after the conversation has been shared. Any participant may:

- Invite additional users within the same tenant to the conversation
- Remove any other participant

- Leave the conversation themselves
- Archive or rename the conversation

The last-participant constraint

A conversation must always have at least one participant. If a participant is the **last person remaining**, they cannot leave or remove themselves until at least one other user has accepted an invitation.

The interface handles this by: - Disabling the **Leave** and **Self-remove** controls when the user is the last participant - Surfacing a tooltip: *"You are the only person in this conversation. Invite another participant before leaving."*

If the last remaining participant's account is **deactivated by a host application administrator**, the conversation is locked to **read-only** and flagged for administrative review.

Conversation history visibility

When a participant accepts an invitation, they see the **full conversation history from the beginning** — not only from the point they were invited. This is intentional: the value of a shared conversation is the full context.

Acceptance disclaimer

Before a user can enter a shared conversation, they must acknowledge a **full-page disclaimer**:

> **Before you join this conversation**
>
> You have been invited to join *"[Conversation title]"* by [Inviter name].
>
> By accepting, you will have access to the **complete conversation history** from the beginning — including all messages sent before you were invited.
>
> Everything you send will be stored as part of the audit trail for this conversation.
>
> [**Accept and open conversation**] [Decline]

The disclaimer is shown every time a user accepts an invitation — it is not a one-time acknowledgement. The user cannot enter the conversation without explicitly clicking **Accept and open conversation**.

Message attribution

In a shared conversation thread, each message bubble carries the **author's name and avatar**.

Message source	Display
Active user's messages	Right-aligned, muted bubble
Other participants' messages	Left-aligned, with a distinct colour per participant (generated from the host's primary brand colour)
Model responses	Left-aligned as the assistant — never attributed to a user

The model label (and, on hover, the name of the user who submitted the preceding message) is visible on each assistant response.

`@`-binding in shared conversations

Each user's `@`-binding typeahead is **scoped to their own permissions** (as enforced by the host's `searchEndpoint`). A participant cannot bind to an object they cannot access in the host application.

If a user submits a message containing a binding to an object that another participant **cannot access**: - The restricted participant sees the binding chip labelled **"[Restricted object]"** - They do not see the resolved context that was injected into the model prompt - The tool call disclosure for any resulting MCP call shows `[Restricted — insufficient permissions]` on the result summary for that participant

This preserves the integrity of each user's permission boundary within the shared thread.

Communication style in shared sessions

Each user's communication style and verbosity settings (from their JWT claims) apply **to the turns they submit**. A response generated for a `technical` user may sit alongside a response generated for a `business` user within the same thread.

The style label is visible on each assistant response — participants can always see the context in which a response was calibrated.

Model and tool configuration in shared sessions

The active model and any opt-in MCP tools **apply to the session as a whole** — they are not per-user settings within a shared conversation. The user who submits a message determines the model and tool context for that turn. Other participants see the model label on each assistant response.

Sharing controls (input area)

The **share icon** in the input area opens the participant management panel. From this panel, any participant may:

Action	Behaviour
Search for users	Search within the authenticated user's tenant by name or email — cross-tenant results are excluded
Invite a user	Sends in-platform notification to the named user
View participants	See all current participants with name, avatar, and join date
Remove a participant	Removes their access; their prior messages remain in the thread
Leave the conversation	Removes self; only available when at least one other participant remains

---

Notifications

Event	Notification
New invitation received	In-platform notification with conversation title, inviter name, accept/decline
New message in shared conversation (not actively viewing)	In-platform notification + badge count on conversation in history panel
Email notifications	Per recipient's notification preferences; **off by default** for shared conversation activity

---

Audit trail for shared conversations

The audit trail records the `user_id` of the message author on every turn. In shared conversations this records the specific participant who submitted each turn.

Recorded element	Table	Notes
Message author	`assistant.turns.user_id`	FK to platform user record; records the submitting participant
Participant list	`assistant.conversation_participants`	`user_id`, `invited_by`, `invited_at`, `accepted_at`, `departed_at`

Recorded element	Table	Notes
Invitation events	`assistant.conversation_participants`	Full invitation lifecycle per participant

There are no role fields — all participants are equal. The audit record reflects actions taken, not role assignments.

Leaving and removing participants

Action	Who can do it	Effect
Leave	Any participant (subject to last-participant constraint)	Conversation removed from leaver's history panel; their prior messages remain in thread
Remove another participant	Any participant	Removed user loses access; their prior messages remain; they may be reinvited
Reinvite a departed participant	Any remaining participant	Same invitation flow as initial invite

Departed or removed participants' messages remain **visible to all remaining participants** and in the **full audit trail**. No message is deleted when a participant leaves.

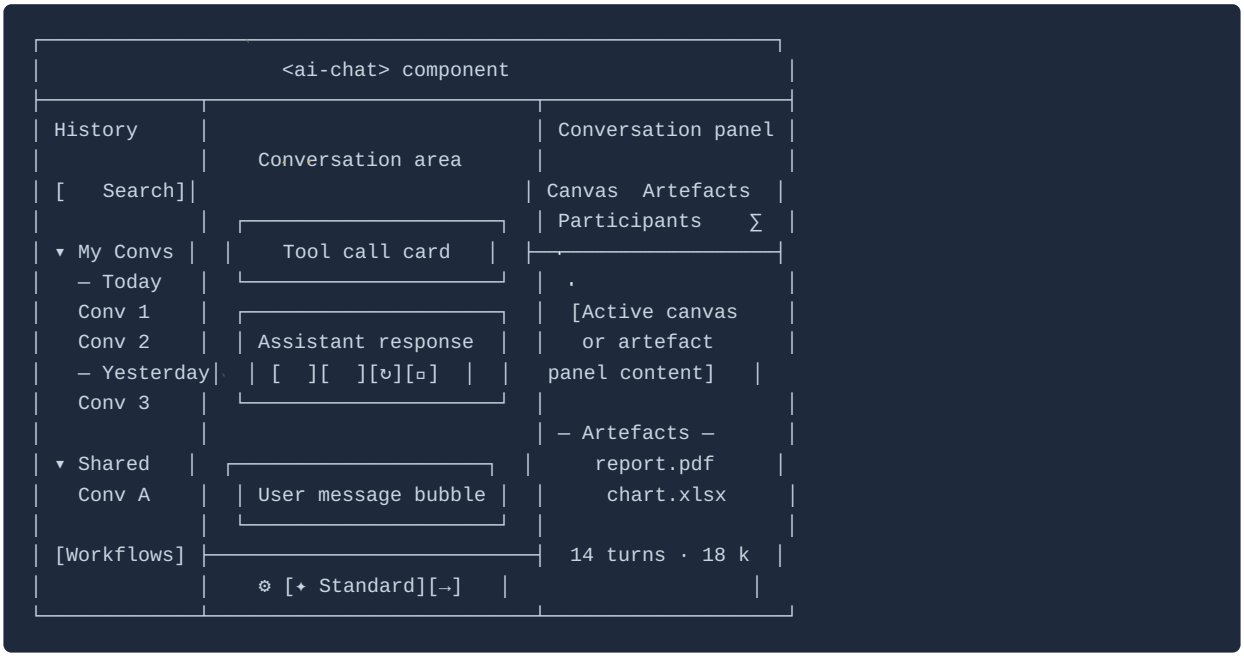
09 — Interaction Design

Component structure

The AI Chat Platform is delivered as an `<ai-chat>` web component that host applications embed within their own UI. The component owns a **three-zone layout** inside its mounting point. The host application retains full ownership of its own navigation, header, and surrounding UI — the component does not attempt to replace or extend the host UI beyond its mounting point.

Zone	Location within component	Contents
History panel	Left sidebar	Conversation list — **My Conversations** and **Shared With Me** (reverse-chronological); conversation search; pinned conversations; new conversation button; workflow library access
Conversation area	Centre — primary	Message thread, streaming responses, rendered content, input area
Conversation panel	Right sidebar	**Canvas** (active when a document is open for iteration); attachments and artefacts; participant list and share management; session token summary

Desktop layout (≥ 1280px)



Why three zones (not four)

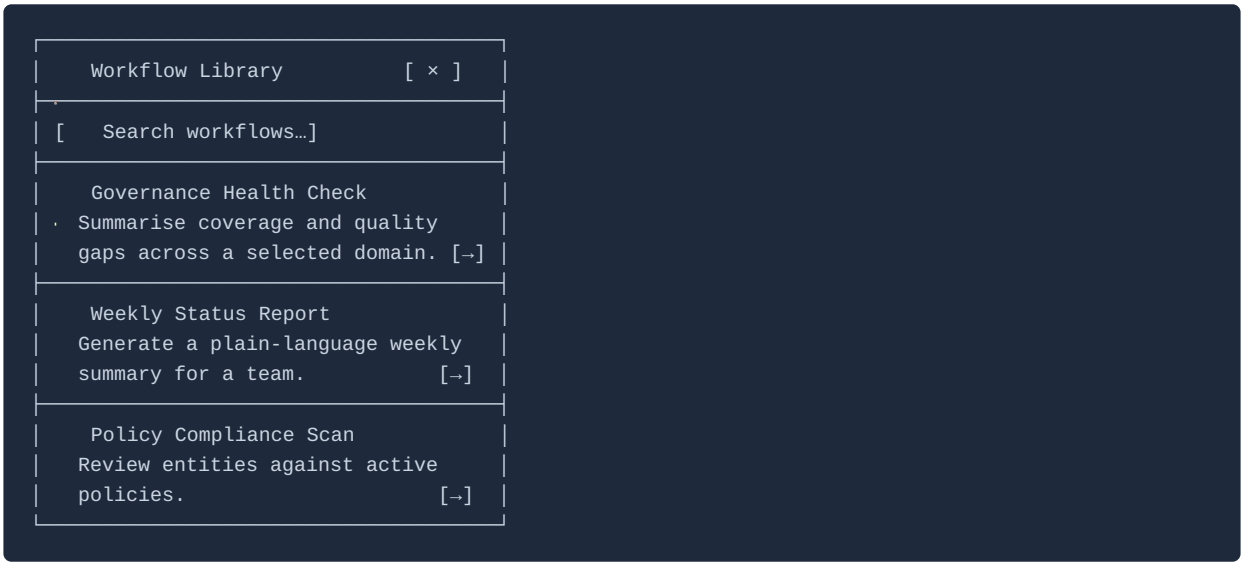
The platform component owns three zones inside its mounting point. The host application manages its own navigation independently — the `<ai-chat>` component occupies its mounting point only, making it embeddable anywhere in the host layout: a full page, a side panel, or a modal drawer.

History panel

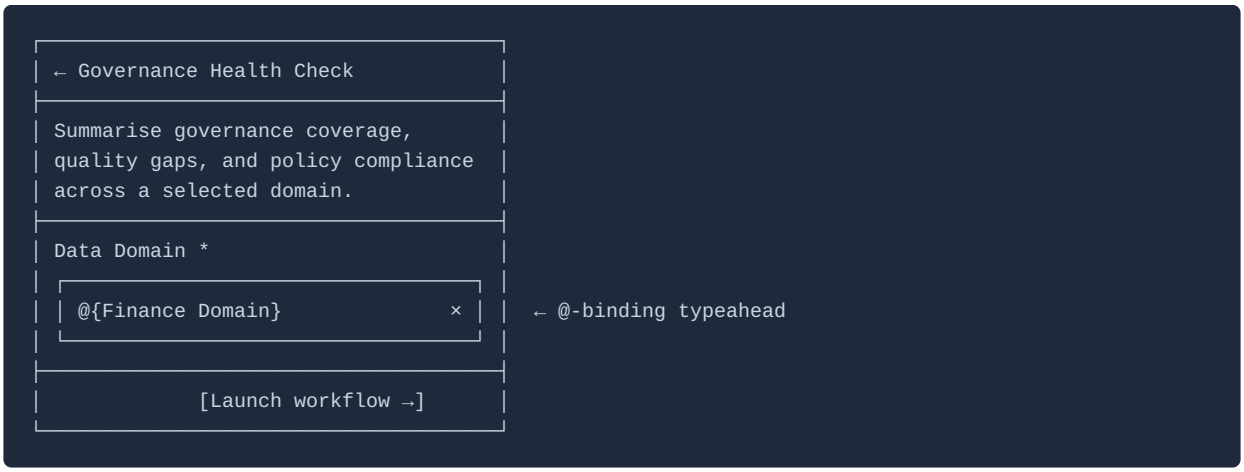
Workflow Library

The Workflow Library is accessible via a **Workflows** button in the history panel header. It slides in as a panel over the history list — it is not part of the right sidebar, which is reserved for conversation-specific content.

When clicked, a workflow from the library opens a parameter form (if the workflow has parameters) before injecting the prompt into the input field.



Selecting a workflow with parameters opens a parameter form before launch:



Conversation panel (right sidebar) detail

The right panel is scoped entirely to the **active conversation** and switches between two primary modes via a tab strip at the top:

Tab	Contents	When active
Canvas	The active working document — editable, versioned, iteratable	Whenever a canvas document is open in the conversation
Attachments & Artefacts	All input documents and output artefacts in submission order	Always available
Participants	Participant list (shared conversations); invite control; leave / remove actions	Always available

Tab	Contents	When active
Session summary	Running token totals (`14 turns · 18,430 tokens`) with tooltip expanding to input / output / cached breakdown	Always available

When a canvas document is opened, the panel automatically switches to the **Canvas** tab. Switching to another tab does not close the canvas — the document remains active and the model retains its context.

Responsive and mobile design

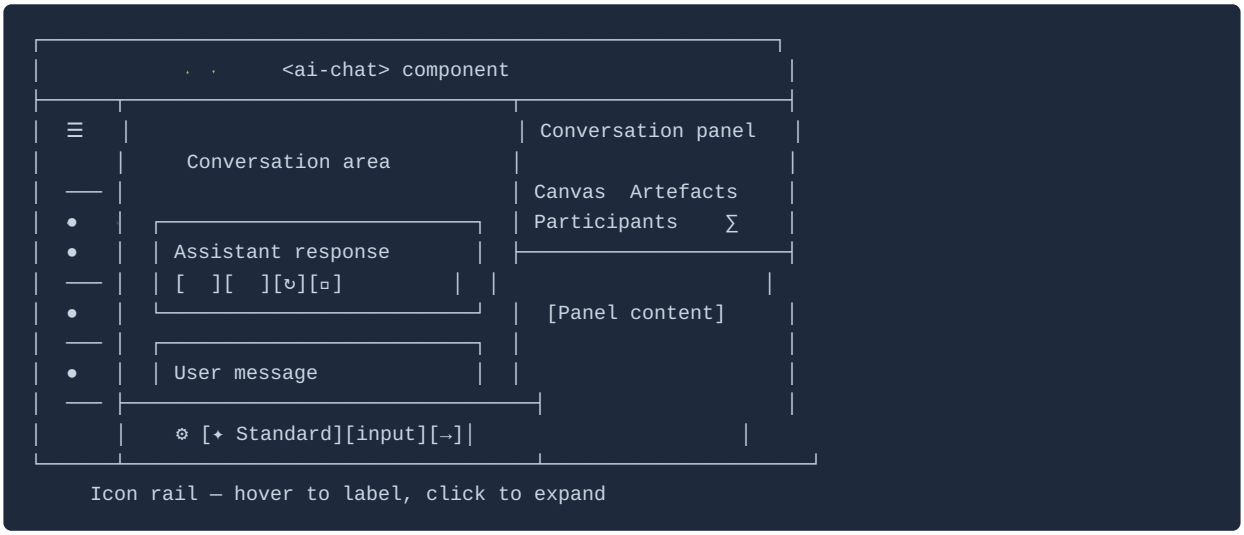
Mobile-first and responsive layout is a core platform standard. The following covers layout behaviour at each breakpoint when the component is mounted at full-page width.

Viewport	Layout
Desktop (≥ 1280px)	Full three-zone layout — history panel + conversation area + conversation panel
Desktop narrow (1024px – 1279px)	History panel collapses to icon-only rail; conversation panel remains
Tablet (768px – 1023px)	Both panels become slide-in drawers; conversation area is primary
Mobile (< 768px)	Single-column; history and conversation panels as bottom-sheet drawers; input pinned to bottom

When the component is mounted in a constrained container (e.g. a side panel or modal), it uses the container width rather than viewport width to determine layout breakpoints. See 16-embedding-and-web-component.md for container sizing guidance.

Narrow desktop layout (1024px – 1279px)

The history panel collapses to an icon-only rail. Hovering an icon reveals its label tooltip; clicking a conversation icon opens it.

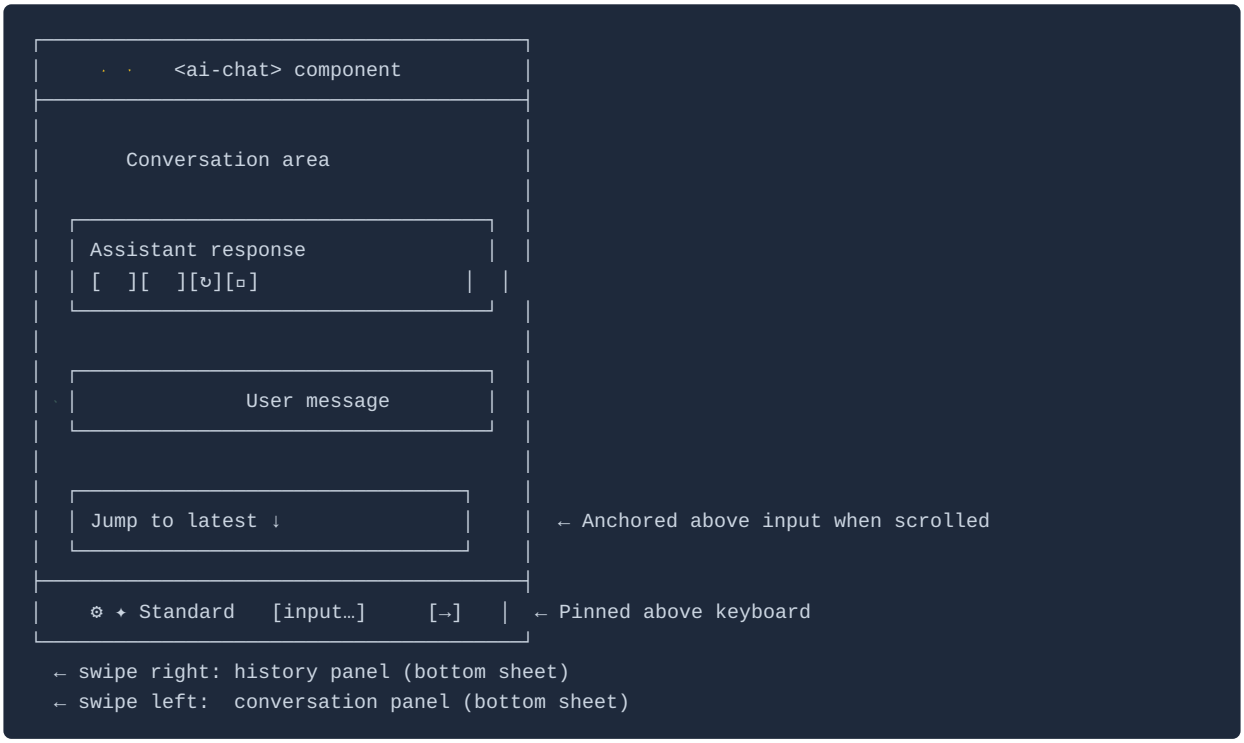


Tablet layout (768px – 1023px)

Both panels become slide-in drawers triggered by header buttons. The conversation area is the primary view.

Mobile layout (< 768px)

Single-column; history and conversation panels as full-height bottom-sheet drawers; input pinned to bottom of screen above the keyboard.



Mobile-specific requirements

Requirement	Specification
Input controls	All controls (attachment, tool selection, model chip, send/stop) are thumb-reachable — no controls hidden behind secondary menus on mobile
`@`-binding typeahead	Anchors to the bottom of the viewport (not cursor position); sized so the top edge is visible above the raised keyboard
Mermaid diagrams	Horizontally scrollable — never scaled to illegibility
Vega-Lite charts	`width: "container"` for responsive sizing
Data tables	Horizontally scrollable; first column is sticky
Conversation panel	Opens as full-height bottom sheet — accessible via a tray icon in the input area footer
History panel	Opens as full-height bottom sheet — accessible via a history button
Conversation search	Opens as full-screen overlay with keyboard raised
Touch targets	Minimum 44 × 44 px (Apple HIG / Material Design)

Input area on mobile

The input toolbar presents all controls in a **single row above the text field**, within thumb reach. When horizontal space is tight, secondary controls (model chip, profile indicator) collapse behind an overflow icon — the paperclip (attachment) and send/stop button are **never** moved to overflow.

Control	Mobile behaviour
Paperclip / attachment	Opens the native mobile file picker (camera roll, Files app, camera capture)
Tool selection	Opens a full-screen bottom sheet listing available MCP tools
Model chip	Opens a bottom sheet listing allowed models
Text input	Grows to 3 lines maximum on mobile (vs. 5 on desktop)
Submit	Large dedicated tap target — always visible

Keyboard and viewport behaviour

When the software keyboard raises on mobile, the conversation thread scrolls to keep the latest turn visible above the input area. The input area stays pinned to the top of the keyboard. The `@`-binding typeahead panel appears above the keyboard.

Hover states on mobile

Desktop (hover)	Mobile equivalent
Report icon appears on hover over a turn	Always visible below each turn
Edit icon appears on hover over a user message	Always visible below each user message
Assistant response action row appears on hover	Always visible below each assistant response
Metadata line appears on hover	Tap the response bubble to reveal; tap again to hide

Gestures

Gesture	Action
Swipe right from left edge	Opens the history panel bottom sheet
Swipe left from right edge	Opens the conversation panel bottom sheet
Long press on a user message	Opens message action menu (edit, report, copy)
Long press on an assistant response	Opens response action menu (copy, report, regenerate)
Tap a binding chip	Fires `binding-click` event to the host application
Pinch-to-zoom on a Mermaid diagram	Enters full-screen diagram view

Conversation thread anatomy

User messages

- Appear immediately in the conversation thread on submit (**optimistic UI**) — the message is visible before the API acknowledges receipt. If the send fails, the message is styled as failed with an inline retry control.
- Right-aligned in a muted bubble
- `@`-binding chips render as interactive pills within the bubble
- In shared conversations: author's name and avatar above the bubble

Other participants' messages (shared conversations only)

- Left-aligned
- Distinct colour per participant (derived from host primary brand colour)
- Author name and avatar above the bubble

Message timestamps

Every message in the conversation thread carries a **timestamp**.

Display rule	Format
Default	Relative: *"Just now"*, *"3 min ago"*, *"Yesterday"*, *"Mon 09:14"*
Hover (desktop)	Absolute: *"Monday 11 May 2026, 09:14:32"*
Older than 7 days	Absolute date always shown: *"4 May, 09:14"*

Timestamps are shown below each message bubble (user and assistant). They are not shown during active streaming — the timestamp appears when streaming completes.

Thinking indicator

After the user submits a message and before the first streaming token arrives, the assistant displays a **thinking indicator** in the conversation thread:

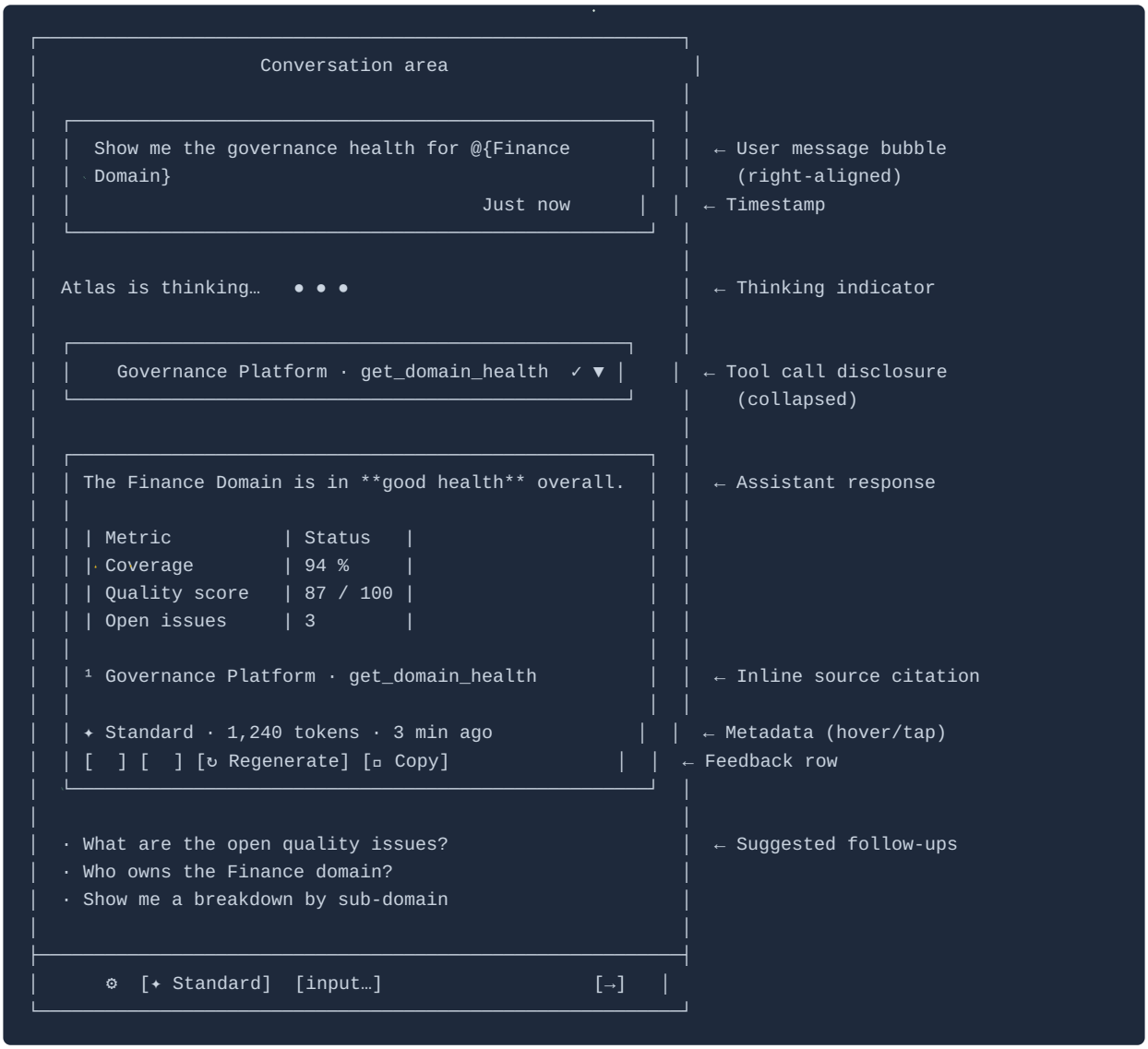
```
[AssistantName] is thinking…   •••
```

An animated three-dot pulse appears beneath a placeholder response bubble. The indicator is replaced immediately by the first streaming token. If the model is queuing tool calls before generating prose, the thinking indicator transitions to the tool call disclosure card (in-progress state) without a gap.

The thinking indicator is suppressed if the model begins streaming within 300 ms of submission.

Full thread anatomy

A complete exchange showing all thread elements in sequence:



Per-turn feedback

User messages carry a **report icon** (visible on hover / always visible on mobile). Clicking opens the report dialog:

> **What's wrong with this?**
> [Text field — optional explanation]
> [Submit report] [Cancel]

On submit, an improvement signal is captured against the specific turn (see 12-continuous-improvement.md).

Assistant responses carry a feedback row with three controls:

Control	Desktop	Mobile	Action
Thumbs up	Appears on hover	Always visible below the response	Records a positive improvement signal against the turn. One-click — no dialog. Fills on selection; toggleable.
Thumbs down	Appears on hover	Always visible below the response	Opens the report dialog (same as user message report). Records a negative signal on submit.
Regenerate	Appears on hover	Always visible below the response	Creates a new branch from this turn (see doc 04).
Copy full response	Appears on hover	Always visible below the response	Copies the full response text to clipboard.

Both thumbs signals feed the improvement pipeline (doc 12). Thumbs up signals are used to identify high-quality turns for reference; thumbs down signals trigger the same improvement signal workflow as the explicit report.

Assistant responses

Each assistant response carries:

Element	Desktop	Mobile
Metadata line	Appears on hover — model label + token counts + (in shared sessions) submitting user's name	Revealed on tap of the response bubble
Feedback row (Regenerate Copy)	Appears on hover	Always visible below the response
Artefact chips	*"Added to artefacts ↗"* beneath each generated block	Same

Suggested follow-ups

At the end of each assistant response, up to **three suggested follow-up queries** are presented as pill chips generated by the model. Single-click to submit. Follow-ups are model-generated — not from a static list.

Input area controls

Control	Function
Paperclip icon	Opens file picker or accepts drag-and-drop (PDF, Excel, Word, images); images also accepted via clipboard paste
Tool selection icon	Opens MCP tool opt-in panel

Control	Function
Active model chip	Opens model selection popover
Text input	Multi-line, grows to 5 lines, `Cmd/Ctrl + Enter` to submit
Profile indicator	Read-only `Style · Verbosity` link to host app profile settings — clickable
Send / Stop	Submits message or stops active stream
Share icon	Opens participant management panel

Input area layout

```
┌──────────────────────────────────────────────────────┐
│  @{Finance Domain} What are the quality gaps in this │  ← Text input
│  domain? Any open ownership issues?                  │    (grows to 5 lines)
├──────────────────────────────────────────────────────┤
│     ⚙  [✦ Standard]  [Technical · Concise]  [⇪]  [→]  │  ← Controls row
└──────────────────────────────────────────────────────┘

    Attachment        — file picker / drag-and-drop / clipboard paste
  ⚙  Tool selection   — opens MCP tool opt-in panel (see below)
  ✦ Standard          — model tier chip; opens model selection popover
  Technical · Concise — profile indicator (style · verbosity); read-only link
  ⇪  Share            — opens participant management panel
  →  Send / Stop      — always visible; submits or stops active stream
```

While streaming, the input field is disabled and the Send button becomes a Stop button:

```
┌──────────────────────────────────────────────────────┐
│  [input disabled while streaming]                    │
├──────────────────────────────────────────────────────┤
│     ⚙  [✦ Standard]  [Technical · Concise]  [⇪]  [■]  │  ← ■ Stop
└──────────────────────────────────────────────────────┘
```

Tool selection panel

Opened via the ⚙ icon in the input area:

Session artefact tray

The artefact tray accumulates every input and output artefact produced during a conversation. It persists for the full lifetime of the conversation record.

Artefact classes

Class	Direction	Examples
Attached document	Input	PDF report, Excel extract, Word document
Generated document	Output	Markdown report, summary
Mermaid diagram	Output	Relationship diagram, flow, sequence
Vega-Lite chart	Output	Bar chart, trend line, distribution
Data table	Output	Multi-row query result (CSV)
JSON export	Output	Tool result or entity record
Code block	Output	SQL, Python, YAML, shell
Binding reference	Input	Non-downloadable — reference card linking to the host application object

Tray entry anatomy

Each tray entry shows: - Format icon - Auto-generated name: `[Content type] — [Subject] — [Date]` - Turn back-link (click to jump to the source turn) - Direction label (Input / Output) - Download button - Preview link

Users may **rename** tray entries. A **"Download all"** control packages all artefacts as a zip archive.

Canvas documents in the artefact tray carry an additional **"Open in canvas"** action that re-opens the document in the canvas panel for further editing, regardless of which turn produced it.

Document canvas

The canvas is the platform's working-document surface — a persistent, editable panel where the model and user collaborate iteratively on a document across multiple turns, rather than treating each AI response as a disposable message.

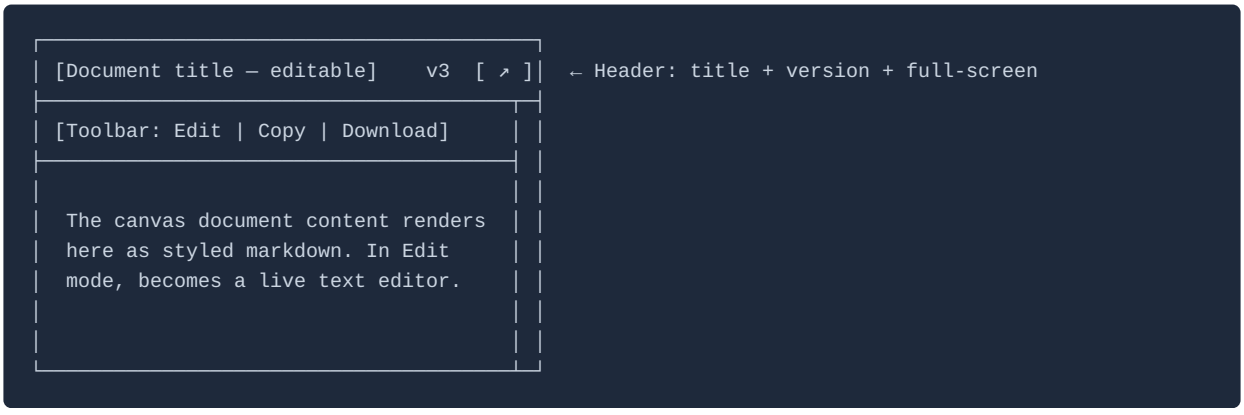
What opens in canvas

The model uses a ```` ```document ```` fenced block to produce canvas-eligible output. The platform's non-overridable system prompt layer instructs the model to use `document` blocks for substantial, document-like prose outputs — reports, summaries, plans, policy drafts, specifications, structured analyses — where the user is likely to want to refine rather than simply read and move on.

Non-prose outputs (Mermaid diagrams, Vega-Lite charts, data tables, code blocks, JSON) remain as inline thread content and artefact tray entries. They do not open in canvas.

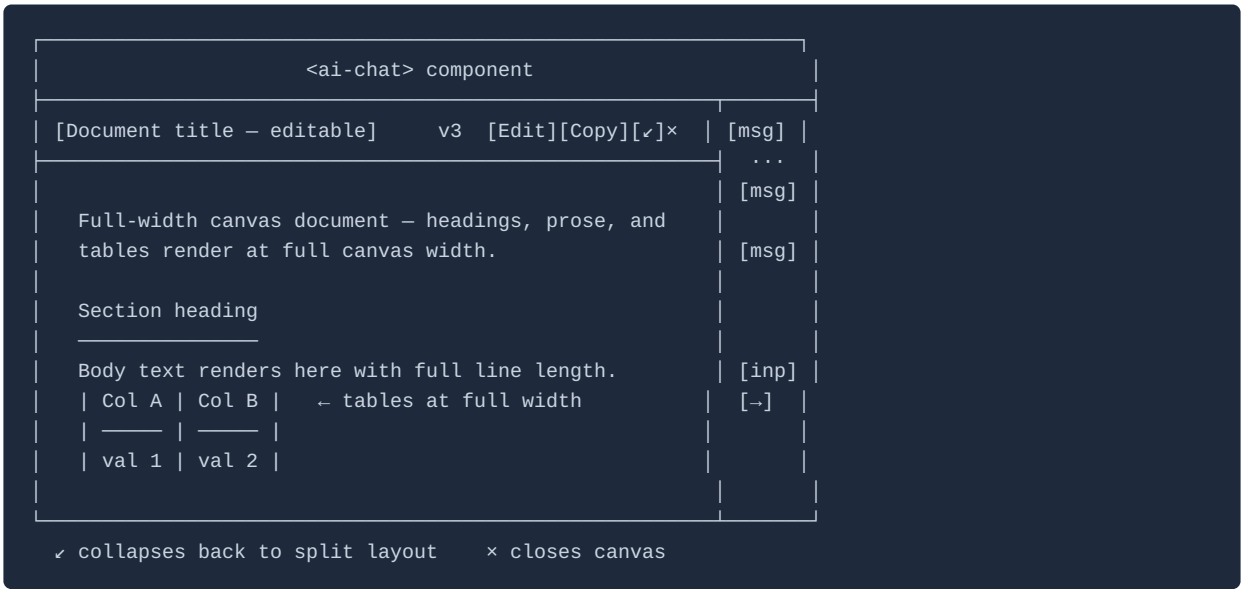
Users may also manually promote any text artefact in the tray to canvas by clicking **"Open in canvas"** on that tray entry.

Canvas panel layout



Element	Behaviour
Document title	Auto-generated from document content; editable inline by clicking
Version indicator	Shows current version (v1, v2 …); clicking opens a version history dropdown listing every model-generated and user-edited version with timestamp. Selecting a prior version restores it as a new version (no destructive overwrites).
Full-screen toggle	Expands canvas to fill the component mount point; the conversation area collapses to a narrow strip. Second click restores the split layout.

Canvas full-screen layout



Canvas version history

Clicking the version indicator (e.g. **v3**) opens a dropdown listing every version of the document:

| **Edit mode** | Clicking **Edit** transforms the canvas from a read-only rendered view into a live markdown editor. User edits are saved immediately as a new version. Exiting edit mode re-renders the markdown. | | **Copy** | Copies the full markdown source to the clipboard | | **Download** | Downloads the document as `.md` (default), `.txt`, or `.pdf` (platform-rendered) |

In-thread reference card

When the model produces a `document` block, the conversation thread shows a compact **reference card** in place of the full content:

```
┌──────────────────────────────────────┐
│     Q2 Governance Summary            │
│     Report · 1,240 words             │
│                          [Open canvas →] |
└──────────────────────────────────────┘
```

The full document is in the canvas panel — it is not rendered inline in the thread. This keeps the conversation thread scannable and avoids large content blocks interrupting the exchange.

Model-assisted revision

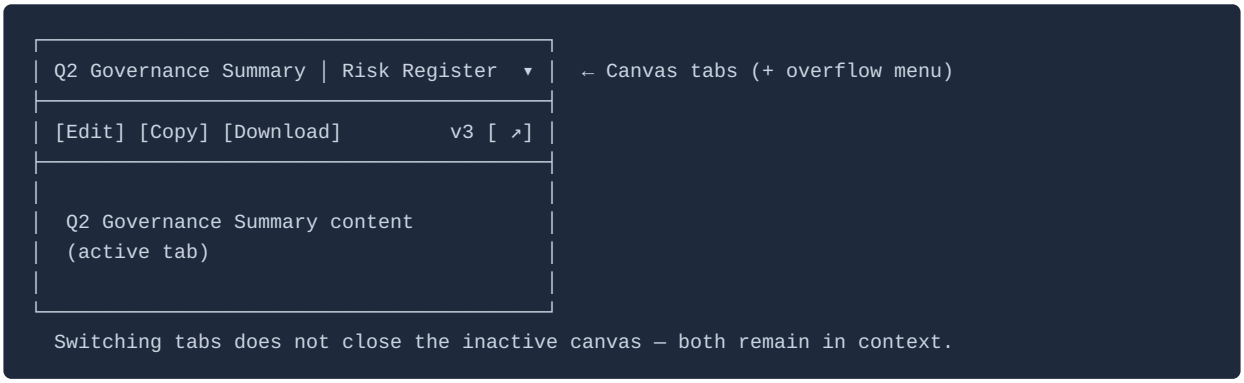
When a canvas document is open, the platform automatically includes the current canvas content in the model's context:

```
[Canvas document — Q2 Governance Summary]
{full document markdown}
[End canvas document]
```

The user can ask the model to revise the document in natural language: *"Shorten the executive summary to two sentences"*, *"Add a risks section after the findings"*, *"Rewrite this in a more formal tone"*. The model responds with a new `document` block containing the full revised document, which replaces the canvas content and is saved as a new version. The model's reply in the conversation thread confirms what changed: *"Updated — shortened the executive summary and added a risks section."*

Multiple canvases

A conversation may produce more than one canvas document. Each is a separate tab within the canvas panel, ordered by creation time. The active tab is the most recently updated document.



Canvas in Mode 1 (floating widget)

The canvas panel is only available in **full state**. In the mini state, canvas documents show as reference cards in the thread with an *"Expand for canvas view ↗"* note. Transitioning to full state opens the canvas panel for the active document.

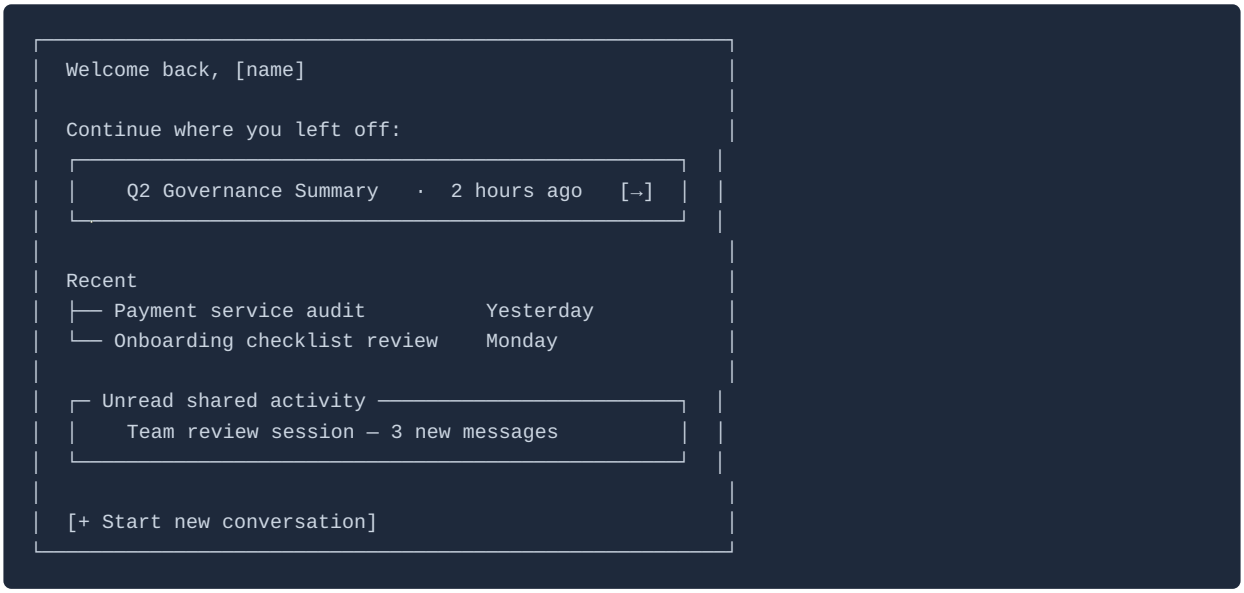
Canvas and the artefact tray

All canvas document versions are stored in the artefact tray as versioned entries. The tray shows the latest version with a version count badge (e.g. `v4`). Individual versions are downloadable from the version history dropdown in the canvas panel.

Returning user state

Mode 2 — Inline Page (`<ai-chat>`)

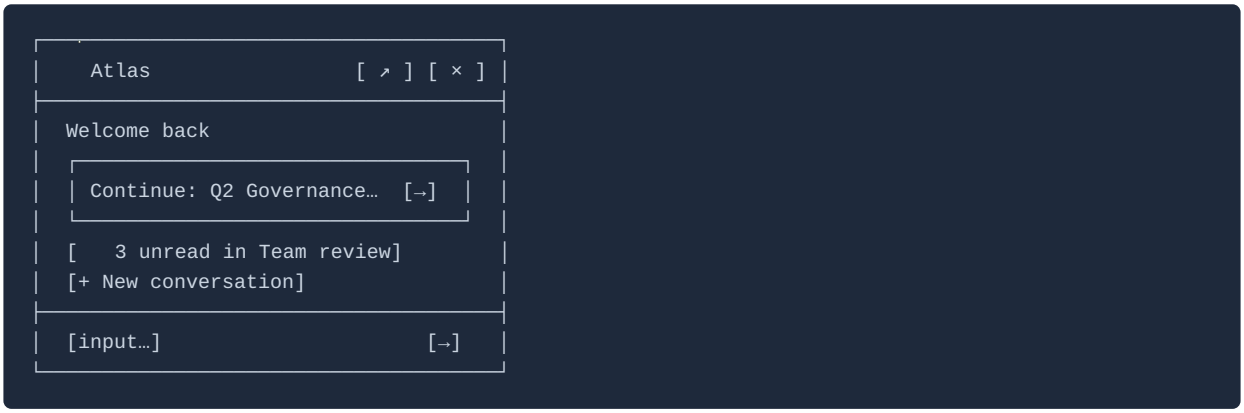
When the component mounts and the authenticated user has prior conversation history, the **conversation area** shows a home state before opening any conversation. The home state is shown on the **first mount per browser session** — subsequent navigations to the page within the same session resume the last-active conversation directly.



Element	Specification
Continue card	The single most recently active conversation; single click opens it directly
Recent list	Up to 2 further recent conversations; click to open
Unread shared activity	Shown only if there are unread messages in shared conversations; click to open the conversation at the first unread turn
Start new conversation	Clears to a new conversation with the onboarding welcome state (first-time) or an empty input (returning)
Trigger condition	Shown when the user has ≥ 1 prior conversation and has not interacted with the component in the current browser session
Skip condition	Not shown if the user is deep-linked to a specific conversation via the `initial-conversation-id` attribute

Mode 1 — Floating Widget (`<ai-chat-widget>`)

When the widget transitions from collapsed FAB to mini state for the first time in a browser session — and the user was last active more than **4 hours** ago (configurable via `widget.returningUserThreshold` in hours) — the mini panel opens to a compact returning-user card rather than resuming mid-conversation:



The card auto-dismisses when the user clicks any option or starts typing in the input field. If the threshold has not been exceeded (recent session), the widget opens directly into the last conversation.

Onboarding state

On **first visit**, the conversation area shows a welcome state drawn from the host application config:

1. The host-configured `identity.assistantDescription`
2. Starter workflows from `features.starterWorkflows` (up to 3, shown as question-style prompts)
3. Starter questions from `features.starterQuestions` (up to 3, if no starter workflows or alongside them)
4. A link to the full Workflow Library

The onboarding state is shown once per user — it is not shown again once the user has started a conversation.

Error and boundary states

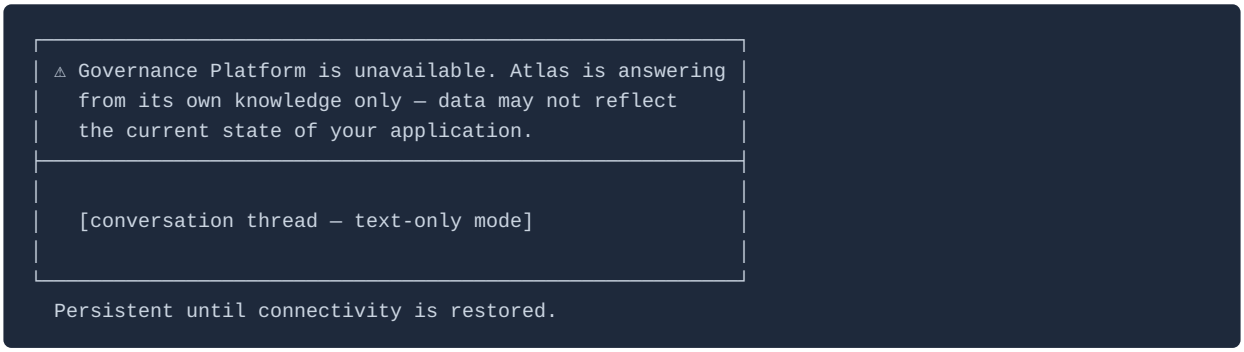
State	Presentation	User action
Model timeout	Inline error card with retry button	Retry re-submits the last message
Send failure (network)	Failed user message styled with error colour + inline retry icon	Tap/click retry to resend
Connection lost	Non-blocking banner below the conversation header: *"Connection lost — reconnecting…"*; animated reconnect indicator. Banner updates to *"Reconnected"* (auto-dismisses after 3s) on restore.	Wait; drafts are preserved locally
MCP tool failure	Tool call disclosure shows error status and raw detail	Rephrase or copy error to report

State	Presentation	User action
Always-on MCP server unavailable	Degraded-mode banner; session continues in text-only mode	Model answers from system prompt context only
Opt-in MCP server unavailable	Error in tool call disclosure	Disable the failing server for the session
Context window limit (80%)	Subtle warning in conversation header	Branch to new thread or accept auto-summarisation
Out-of-scope query	Model explains scope boundary and suggests reformulation	No error state — graceful redirect
Auth session expired	Modal overlay prompting re-authentication	Re-auth restores session and full conversation history
Unsupported file type	Inline notice on file selection	User prompted to use a supported format
Last participant constraint	Tooltip on disabled Leave/Remove controls	Invite another user before leaving
Component mount failure	Error state within the component mount point	Host application is notified via the `error` event

Error state layouts

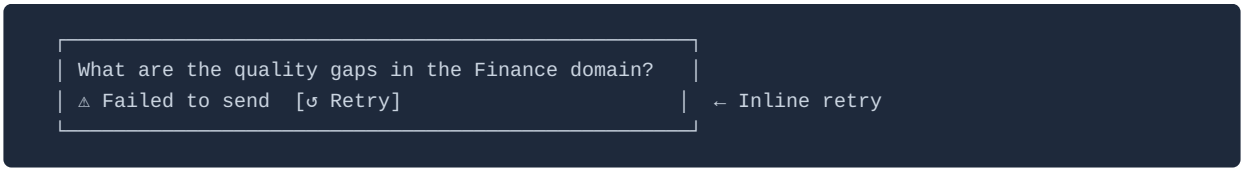
Connection lost — non-blocking banner:

Always-on MCP server unavailable — degraded-mode banner:

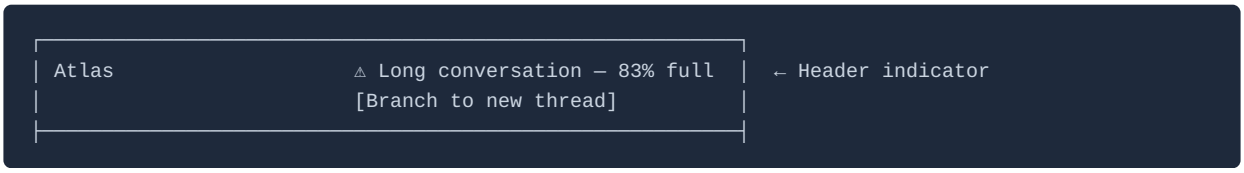


Auth session expired — blocking modal:

Send failure — inline retry on user message:



Context window warning (80% threshold) — subtle header indicator:



Accessibility

WCAG 2.1 AA compliance is a platform requirement. The following are platform-specific additions.

Requirement	Specification
Mermaid SVG alt text	Descriptive alt text generated by the model for each Mermaid diagram

Requirement	Specification
Conversation thread live region	The thread is an `aria-live` region — streaming responses and new shared conversation messages are announced to screen readers
`@`-binding typeahead	Fully keyboard-navigable (arrow keys, Enter/Tab to select, Escape to dismiss); focus returns to input field on selection or dismissal
Branding token contrast	Platform validates that host-provided colour tokens meet WCAG 2.1 AA contrast ratios at config submission
`prefers-reduced-motion`	When the OS-level reduced motion preference is active: streaming text renders immediately (no character-by-character animation); panel slide transitions are instant; the thinking indicator uses a static label instead of the animated three-dot pulse; the FAB pulse animation on stream-in-progress is suppressed; Mermaid and Vega-Lite render-in animations are disabled
Focus management	When a shared conversation message arrives, focus is not stolen from the input field. The unread badge and "Jump to latest" button are announced via the live region. Focus only moves to new content on explicit user action.
Modal focus trap	All modal overlays (full widget state, diagram full-screen, write confirmation) implement a focus trap — Tab cycles only within the modal; Escape dismisses

---

Jump to latest

When a user has scrolled up and new content arrives (streaming response or shared conversation message), a **"Jump to latest ↓"** pill button appears anchored above the input field. Clicking it scrolls immediately to the most recent turn. The button disappears when the user is already at the bottom.

Keyboard shortcuts

The shortcut reference overlay (triggered by `?` or `Cmd/Ctrl + /`):

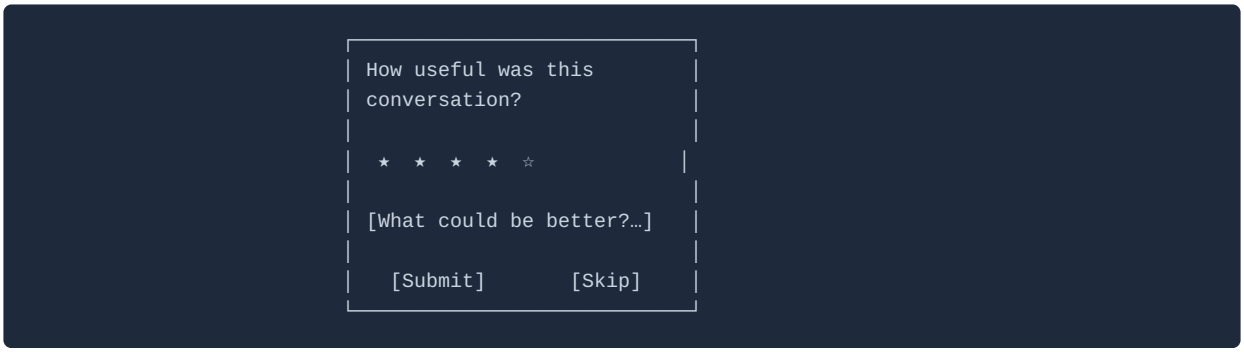
```
┌──────────────────────────────────────────┐
│ Keyboard shortcuts                [ × ]  │
├──────────────────────────────────────────┤
│ Cmd/Ctrl + Enter     Submit message      │
│ Cmd/Ctrl + K         Cross-conversation search│
│ Cmd/Ctrl + F         In-conversation search  │
│ Cmd/Ctrl + N         New conversation        │
│ ↑  (empty input)     Edit last user message  │
│ Escape               Dismiss / cancel edit   │
│ Cmd/Ctrl + Shift + C Copy last response      │
│ ?  /  Cmd/Ctrl + /   This shortcut list      │
└──────────────────────────────────────────┘
```

Shortcut	Action
`Cmd/Ctrl + Enter`	Submit message
`Cmd/Ctrl + K`	Open cross-conversation search
`Cmd/Ctrl + F`	Open in-conversation search
`Cmd/Ctrl + N`	Start a new conversation
`Escape`	Cancel edit / dismiss typeahead / close modal
`↑` in empty input	Edit most recent user message
`Arrow keys`	Navigate `@`-binding typeahead; navigate search results
`Enter` / `Tab`	Select item in `@`-binding typeahead
`Cmd/Ctrl + Shift + C`	Copy most recent assistant response to clipboard
`?` or `Cmd/Ctrl + /`	Show keyboard shortcut reference

Post-session CSAT prompt

A **1–5 star rating prompt** is shown to a random sample of users (configured via `features.csatSampleRate`) at the end of a session, after 30 seconds of idle time following the last assistant response.

The prompt appears as a **non-blocking floating card** anchored to the bottom-right of the conversation area:



> **How useful was this conversation?**
> ★ ★ ★ ★ ★
> [Optional: What could be better? — single text line]
> [Submit] [Skip]

- Non-blocking — does not prevent input
- Dismissible (Skip or click outside)
- Skipping is not counted as a negative signal

- On mobile: appears as a bottom sheet centred in the viewport

The rating and optional comment are stored in `assistant.conversations.csat_score` and `assistant.conversations.csat_comment`.

10 — Content Rendering

The rendering contract

The platform's rendering pipeline is built on a single foundational assumption:

> **The LLM always produces raw text or JSON. It never produces rendered output. Every structured output is delivered as plain content wrapped in a fenced block, with the rendering target name as the language tag.**

````
```<rendering-target>
<raw content — plain text, JSON, Mermaid source, LaTeX, CSV, Vega-Lite spec, etc.>
```
````

Examples:

Diagram (mmdc unavailable)

```
flowchart LR
    A --> B --> C
```

````
```vega-lite
{ "$schema": "...", "mark": "bar", "encoding": { ... } }
```
````

````
```document
# Quarterly Risk Summary
...
```
````

The platform intercepts these fenced blocks before displaying them, identifies the target from the language tag, and hands the raw content to the appropriate renderer. The LLM has no knowledge of how the content will be displayed — it only knows what tag to use and what format the content inside the block must be.

This contract has three important consequences:

1. **The system prompt is the only configuration surface.** The LLM learns which tags are available and what format each expects solely from the system prompt injected at session start. If a renderer is registered, its `systemPromptGuidance` must fully specify the tag name and content format.
2. **Raw content is always available.** Because the LLM always emits the unrendered source, every rendered block can switch to a raw view without any re-fetching or reprocessing — the source is already buffered.
3. **Rendering failures are recoverable.** If a renderer fails, the platform falls back to displaying the raw content as a syntax-highlighted code block. The user always sees something meaningful.

Rendering decision rules

The rendering engine evaluates each content block in an assistant response in priority order. **The first matching rule wins.**

Priority	Trigger	Rendered as
1	Tool call event (`mcp_tool_use` / `mcp_tool_result`) or MCP `elicitation/create` event	**Tool call disclosure** card / **Feedback request** card
2	Fenced block tag matching a registered host renderer (`renderers[].trigger`)	**Custom host renderer** — host-provided ES module; see below
3	Fenced block tagged ` ```feedback-request `	**Feedback request card** — blocking interactive prompt; streaming pauses until user responds
4	Fenced block tagged ` ```document `	**Document canvas** — opens in right panel canvas; reference card in thread
5	Fenced block tagged ` ```mermaid `	**Mermaid diagram** — SVG, expandable, exportable
6	Fenced block tagged ` ```vega-lite `	**Vega-Lite chart** — interactive, responsive
7	Fenced block tagged ` ```math ` or `$$...$$` display block	**Math expression** — KaTeX rendered display block
8	Fenced block tagged ` ```json `	**JSON inspector** — collapsible tree, copy-to-clipboard
9	Fenced block tagged ` ```csv ` or ` ```table `	**Data table** — sortable, filterable, paginated, CSV export
10	Any other fenced block	**Syntax-highlighted code** — Prism, copy-to-clipboard, line numbers > 5 lines

Priority	Trigger	Rendered as
11	Inline `$...$` within prose	**Inline math** — KaTeX rendered inline
12	All other content	**Rich markdown prose** — GFM

> **Note — priority 1 is event-driven, not fenced-block matching.** Tool call disclosures are triggered by `mcp_tool_use` / `mcp_tool_result` streaming events; feedback request cards are triggered by MCP `elicitation/create` events. Both arrive outside content blocks entirely. Priorities 2–12 are evaluated against the fenced block tag of each content block. The two mechanisms do not compete — protocol events are always rendered as their respective cards regardless of block content.
>
> **Feedback request cards (priority 3) are the only content type that block streaming.** When the rendering engine encounters a `feedback-request` block, it pauses the stream and does not render subsequent content until the user has responded.

The system prompt (injected by the platform) instructs the model to: - Prefer structured outputs — Vega-Lite for metrics and trends, Mermaid for relationships and flows, data tables for entity lists — over prose equivalents when the data supports it - Use `document` blocks for substantial prose outputs (reports, summaries, plans, policy drafts, analyses) where the user is likely to iterate across multiple turns rather than simply read once

Content type quick reference

Content type	Trigger	Typical use cases
Prose / markdown	Default	Explanations, summaries, narrative answers
Feedback request card	```` ```feedback-request ```` / MCP elicitation	Approval gates, structured confirmations, mid-workflow choices — any point where the agent must pause for user input before continuing
Document canvas	```` ```document ````	Reports, policy drafts, structured summaries, plans — any substantial prose the user will iterate on
Custom host renderer	Registered `trigger` tag	Host-defined domain-specific visualisations (risk gauges, compliance scorecards, Gantt views, org charts)
Mermaid diagram	```` ```mermaid ````	Entity relationships, process flows, system dependencies, hierarchies
Vega-Lite chart	```` ```vega-lite ````	Metrics, trends, distributions, comparisons
Math expression	```` ```math ```` / `$$...$$` / `$...$`	Scoring formulas, ratios, statistical expressions, metric definitions
JSON inspector	```` ```json ````	Raw tool results, configuration objects, structured data

Content type	Trigger	Typical use cases
Data table	```csv / ```table	Multi-row query results, lists, comparison tables
Tool call disclosure	Automatic	All MCP tool invocations — always visible, collapsed by default
Syntax-highlighted code	All other fenced blocks	SQL, Python, YAML, shell, TypeScript

---

Content types

Custom host renderer

Host applications may register custom content renderers in the `renderers` section of their application config (see 01-host-application-config.md). When the model produces a fenced block tagged with a registered `trigger`, the platform loads and invokes the host's renderer module.

Module loading

Renderer modules are **ES modules** loaded once per session when the first block matching their trigger arrives. The platform loads the module using a dynamic `import()` from the registered `moduleUrl`. Modules are cached for the session lifetime — they are not re-fetched on each block.

The module must export a **renderer class** (or factory function returning an object) that implements the following interface:

```
interface HostRenderer {
  // Called once per content block after the full fenced block content is received.
  // `container` is a plain HTMLElement inside a shadow root — render freely within it.
  render(container: HTMLElement, content: string, context: RendererContext): void | Promise<void>;

  // Optional — called when the rendered element is removed from the DOM.
  dispose?(): void;

  // Optional — if present, called by the platform when the user downloads from the
  // artefact tray. The returned Blob is used instead of the raw fenced block source,
  // allowing the renderer to export a rendered image, PDF, or structured file.
  getExportBlob?(): Promise<Blob>;
}

interface RendererContext {
  tenantId:    string;                  // the current tenant
  conversationId: string;               // the current conversation
  turnId:      string;                  // the turn this block belongs to
  trigger:     string;                  // the fenced block tag that matched
  themeTokens: Record<string, string>;    // host brand CSS custom properties (e.g. "--color-
primary")
}
```

The platform loads the module once per session (cached for the session lifetime) but instantiates the renderer class once per content block. Each block gets its own renderer instance. The platform passes the full fenced block content as a string to `render()`, and calls `dispose()` when the block leaves the DOM (e.g. conversation navigation, component unmount).

Isolation

Each rendered block's `container` is the interior of a **shadow root** — the renderer's DOM and styles are isolated from the platform UI and from other rendered blocks. The renderer may use any DOM APIs available to the module. It may not access the platform's internal state, the user's JWT, or other conversations.

Block buffering

Custom renderers receive the **full fenced block content** after `content_block_stop` — they cannot stream incrementally. While the block is buffering, the platform shows a loading skeleton in place of the renderer output. The skeleton is replaced by the rendered output on receipt.

System prompt guidance injection

For each registered renderer, the platform injects the renderer's `systemPromptGuidance` (or a generated fallback) into the system prompt at session start. This tells the model when and how to produce the custom content type:

```
[Custom content types]
{renderer.name}: {renderer.systemPromptGuidance}
```

All registered renderer guidance blocks are appended together in the order they appear in the `renderers` config array. They are injected after tool descriptions and before memory blocks in the assembled system prompt.

Fallback behaviour

If the renderer module fails to load, or if `render()` throws, the platform:

1. Logs the error to the improvement signal pipeline
2. Renders the fenced block content as a **syntax-highlighted code block** (priority 9)
3. Shows an inline notice beneath the block: *"[Renderer name] could not render this content. Showing raw output."*

The fallback is transparent to the user and non-blocking — the rest of the response continues rendering normally.

Example: risk gauge renderer

Given the config:

```
{
  "id":      "risk-gauge",
  "trigger": "risk-gauge",
  "name":    "Risk Gauge",
  "moduleUrl": "https://cdn.acme.com/ai-renderers/risk-gauge.js",
  "systemPromptGuidance": "Use a ```risk-gauge block when asked for a risk score. The block must
contain JSON: { \"score\": <0–100>, \"label\": \"<text>\", \"breakdown\": [ { \"factor\":
\"<name>\", \"score\": <0–100> } ] }."
}
```

The model produces:

```
```risk-gauge
{ "score": 72, "label": "Medium-High", "breakdown": [
    { "factor": "Data quality", "score": 80 },
    { "factor": "Ownership coverage", "score": 55 },
    { "factor": "Policy compliance", "score": 90 }
  ]
}
```
```

The platform calls `RiskGaugeRenderer.render(container, content, context)`. The renderer parses the JSON and renders a gauge widget into `container`. The raw JSON is stored in the artefact tray.

Document canvas

A `document` block opens a persistent right-panel canvas alongside the conversation thread. The canvas is designed for substantial prose outputs — reports, summaries, plans, policy drafts, analyses — that the user is likely to iterate on across multiple turns rather than simply read once.

Behaviour	Specification
Trigger	Fenced block tagged ```` ```document ````
Panel	Opens in a right-panel canvas; the conversation thread continues uninterrupted in the left panel
Thread reference	A reference card appears in the thread at the point the document was produced — title, word count, and a *"Open document →"* link
Multiple documents	Each `document` block from the same conversation opens as a tab in the canvas panel. Switching tabs does not navigate the conversation
Iteration	Subsequent turns that produce a revised `document` block replace the active canvas content in-place; the prior version is accessible via the version history control (last 10 revisions retained)
Export	Full document downloadable as Markdown or plain text from the canvas toolbar; PDF export via browser print
Artefact	Document source stored in turn record; added to artefact tray with document title as label

Behaviour	Specification
Mobile	Canvas panel opens as a full-screen overlay; a back button returns to the conversation thread
Empty state	If the model produces a `document` block with no content, the canvas shows a *"No content"* placeholder and the thread reference card is omitted

Document titles

The platform extracts the document title from the first `# Heading` in the block content. If no heading is present, it defaults to *"Document — [timestamp]"*. Titles are shown in the thread reference card, the canvas tab, and the artefact tray entry.

Feedback request card

A `feedback-request` block renders an interactive confirmation card inline in the conversation thread. It is the primary mechanism for the model — or an MCP server mid-execution — to request explicit user input before the conversation continues.

> **This is the only blocking content type.** Streaming pauses at a `feedback-request` block and does not resume until the user selects a response. The input field is disabled while the card is pending. The stop-generation button is also hidden — a pending feedback card is not cancellable by stop; it must be explicitly declined via a `Cancel` or `Reject` option in the card itself.

Behaviour	Specification
Trigger	Fenced block tagged ```` ```feedback-request ````, or MCP `elicitation/create` event
Placement	Inline in conversation thread — never modal, never toast
Blocking	Streaming pauses on receipt; resumes only after user responds
Response recording	The selected response is written to the conversation as a structured user turn so the model sees it in context on the next turn
One at a time	At most one pending feedback card per conversation at any moment
Timeout	None — feedback cards do not expire. The user may respond minutes or hours later

Payload schema

The fenced block content must be valid JSON:

```
{
  "prompt":       "<question or action description shown to the user>",
  "options": [
    { "id": "<machine-id>", "label": "<human-readable label>", "style": "primary | destructive |
secondary" }
  ],
  "allowFreeText": false
}
```

- **`prompt`** — the question or description displayed at the top of the card. Should state clearly what action is being requested or what decision the user is making.
- **`options`** — one to four labelled choices. At least one is required. `style` drives visual weight: `primary` (brand colour, default action), `destructive` (red, irreversible or high-risk action), `secondary` (neutral, cancel or deferral).
- **`allowFreeText`** — if `true`, a text input appears below the option buttons, allowing the user to type a custom response in addition to or instead of selecting a button. The typed response is included in the recorded turn.

Option design guidelines

Pattern	Options	Notes
Simple approval gate	`Approve` (primary) + `Reject` (destructive)	Use for irreversible or high-impact actions
Multi-choice	2–4 labelled options, one primary	Use when there are meaningful alternatives, not just yes/no
With escape hatch	Any options + `Cancel` (secondary)	Include whenever doing nothing is a valid choice
Guided free text	1–2 options + `allowFreeText: true`	Use when sensible defaults exist but the user may need to specify

The platform enforces a maximum of four option buttons. Flows requiring more than four choices should use a `document` block or a data table to present options and ask the user to respond in free text.

MCP elicitation alignment

The feedback request card is the platform's implementation of MCP Elicitation — the protocol-level mechanism that allows MCP servers to pause a tool call and request structured input from the user before continuing.

MCP concept	Platform equivalent
`elicitation/create` event	Triggers feedback request card directly (bypasses fenced block; same UI)

MCP concept	Platform equivalent
Form mode (JSON Schema input)	Single-question flows → feedback request card; multi-field forms → `document` block with a form layout
URL mode (external redirect for OAuth, payments)	Platform opens the URL in a new tab / webview; this is *not* handled by the feedback card
Titled enum options	`label` field on each option — human-readable text, not the machine `id`
Multi-select enum	Not yet supported by the feedback card; multi-select flows should use a data table or document block until the MCP multi-select spec stabilises

Raw / rendered toggle

Feedback request cards do **not** expose a Raw toggle while the card is pending — displaying the JSON payload while the user is deciding what to do could be misleading. Once the user has responded, the resolved card shows a **Raw** control that reveals the original JSON payload alongside the recorded response, for audit and debugging purposes.

Mermaid diagram

Behaviour	Specification
Rendering	SVG via Mermaid.js
Default display	Full-width in conversation thread, collapsed to thumbnail in artefact tray
Expand	Full-screen overlay available
Export	SVG download from artefact tray
Mobile	Horizontally scrollable — never scaled to illegibility
Accessibility	Descriptive alt text generated by the model and attached to the SVG
Artefact	Mermaid source stored in turn record; added to artefact tray on render complete

Common diagram types

Diagram type	Mermaid syntax	Typical trigger
Entity relationship	`erDiagram`	*"Show me the relationship between X and Y"*
Process flow	`flowchart LR`	*"Walk me through the approval workflow"*
Sequence diagram	`sequenceDiagram`	*"How do these services communicate?"*
Hierarchy / tree	`graph TD`	*"Show me the organisational structure"*
Timeline	`timeline`	*"Show the project milestones"*

Vega-Lite chart

Behaviour	Specification
Rendering	Interactive via vega-embed
Responsive sizing	`width: "container"` — adapts to viewport or container
Export	PNG or SVG download from artefact tray
Interactivity	Hover tooltips, click-to-filter where applicable
Mobile	Responsive; minimum bar/point size maintained for touch targets. Hover tooltips activate on tap; click-to-filter activates on double-tap. Pinch-to-zoom is disabled — the chart scales with the viewport instead
Artefact	Vega-Lite spec stored in turn record; added to artefact tray on render complete

Math expression

Mathematical expressions are rendered using **KaTeX** — fast, lightweight, and browser-native.

Behaviour	Specification
Display block	Triggered by ```` ```math ```` fenced block or `$$...$$` delimiter — rendered full-width, centred
Inline	Triggered by `$...$` within prose — renders inline without breaking text flow
Library	KaTeX — subset of LaTeX math notation
Copy	LaTeX source copy-to-clipboard button on display blocks
Export	LaTeX source and rendered SVG downloadable from artefact tray
Mobile	Display blocks horizontally scrollable at narrow viewports
Fallback	If KaTeX fails to parse, the raw LaTeX source is shown in a code block with a parse-error notice

JSON inspector

Behaviour	Specification
Default state	Collapsed to two levels
Navigation	Expand/collapse nodes; copy individual values or subtrees
Export	Full JSON download from artefact tray

Behaviour	Specification
Use cases	Tool result payloads, configuration inspection, structured record data

Data table

Behaviour	Specification
Sorting	Click column header to sort ascending/descending
Filtering	Inline filter row per column
Pagination	25 rows per page; configurable
Mobile	Horizontally scrollable; first column sticky
Export	CSV download from artefact tray
Empty state	"No results" message with query context

Syntax-highlighted code

Behaviour	Specification
Highlighting	Prism.js — auto-detects language from fenced block tag
Copy	Copy-to-clipboard button always visible
Line numbers	Shown for blocks of more than five lines
Languages	SQL, Python, YAML, TypeScript, JSON, Bash, and all Prism-supported languages

Attached content

Non-image documents (PDF, Excel, Word) appear in the user message bubble as a labelled file card: - Format icon - File name - Page count (PDF), sheet count (Excel), or section count (Word)

Clicking the file card opens the document in an **inline document viewer** embedded directly in the conversation thread. The viewer renders the full document without leaving the conversation.

Format	Viewer behaviour
PDF	Paginated page-by-page render; page navigation controls; text selection and search within the PDF
Excel	Sheet tabs across the top; each sheet rendered as a scrollable, read-only data table
Word / DOCX	Rendered as formatted prose with heading styles and table layout preserved

Format	Viewer behaviour
CSV	Rendered as a sortable, filterable data table (same component as ```csv blocks)

The viewer opens in-line below the file card, expanding the message bubble to accommodate it. A **collapse** control returns to the file card view. The document remains openable for the lifetime of the conversation.

When the model references a specific section, it cites by page number (PDF), sheet name (Excel), or heading (Word). Clicking a citation opens the viewer scrolled to the referenced location.

Images (PNG, JPEG, WEBP) are **rendered inline** in the user message bubble at a constrained size (max 320px wide, max 240px tall, maintaining aspect ratio). Multiple images in one turn stack vertically. Clicking an image opens it in a full-screen lightbox overlay. The original file is downloadable from the lightbox and from the artefact tray.

Cross-cutting behaviours

Streaming

Content type	Streaming behaviour
Prose / markdown	Streams character-by-character; renders incrementally
Non-prose blocks (Mermaid, Vega-Lite, JSON, tables, code)	Buffers internally; renders on `content_block_stop` to prevent hydration errors on partial content
Feedback request card	**Blocks the stream entirely.** Subsequent content is withheld until the user responds. The stop-generation button is hidden while a card is pending
Tool call disclosures	Appear as an in-progress card while the tool is running; update on result receipt
Artefact chips	*"Added to artefacts ↗"* chip appears beneath each completed non-prose block

While the model is streaming, the input field is disabled and replaced by a **stop-generation button**. Stopping generation saves the partial response to the audit trail as a partial turn — it does not discard it. When stopped, a *"(generation stopped)"* label appears beneath the partial response.

Truncated response

When the model's response ends without a natural conclusion, a **Continue** button appears below the response. Truncation is detected by either of two signals: (a) the API returns `stop_reason: "max_tokens"`, indicating the output token limit was reached; or (b) heuristic analysis finds the response ends mid-sentence — no terminal punctuation in the last 120 characters and no closing structural element (heading, list item, code block close, or horizontal rule).

> *"Response may be incomplete.* **Continue →***"*

Clicking Continue submits an implicit *"Please continue"* turn, which regenerates from the end of the incomplete response in a new branch. The original truncated response is preserved. This handles output-token-limit cases transparently without requiring the user to know why the response stopped.

The Continue button is shown for a maximum of 60 seconds after the truncated response; after that it is dismissed to avoid polluting old threads.

Artefact tray

The artefact tray is a persistent UI panel (collapsed by default, expandable from a tray handle at the bottom of the conversation) that collects downloadable outputs from the current conversation. Every non-prose rendered block — Mermaid diagrams, Vega-Lite charts, math expressions, JSON payloads, data tables, document canvas outputs, and custom renderer outputs — is automatically added to the tray when it finishes rendering.

Element	Specification
Trigger	Added automatically on render completion of any non-prose block
Entry label	Content type name + block title (where extractable) or timestamp
Download	Clicking an entry downloads the artefact. For custom renderers that implement `getExportBlob()`, the exported blob is used; otherwise the raw fenced block source is downloaded as plain text
Scope	Tray contents are scoped to the current conversation. Navigating to a new conversation clears the tray
Persistence	Artefact tray entries are stored with the turn record and restored when the conversation is reopened
Empty state	Tray handle is hidden when there are no artefacts in the current conversation

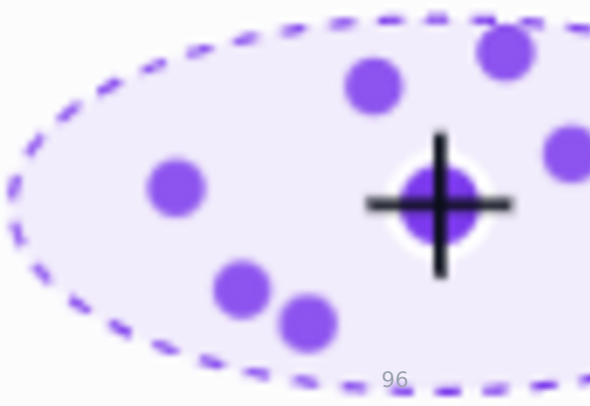
---

Raw / rendered toggle

Every non-prose rendered block exposes a **Raw** toggle that switches between the rendered view and the raw source in place.

CHART Counterparty clustering — outfl

Funding stability →



Content type	Rendered view	Raw view
Data table	Sortable, filterable, paginated table	Raw CSV or JSON source in a syntax-highlighted code block
Vega-Lite chart	Interactive vega-embed chart	Vega-Lite JSON spec in a JSON inspector
Mermaid diagram	Rendered SVG	Mermaid source in a syntax-highlighted code block
Math expression	KaTeX-rendered formula	LaTeX source in a code block
JSON inspector	Collapsible tree	Raw JSON in a syntax-highlighted code block
Document canvas	Right-panel canvas	Markdown source in a code block (in-thread, without opening the canvas)
Custom host renderer	Renderer output	Raw fenced block source in a syntax-highlighted code block

The toggle is a **Rendered · Raw** pill control placed in the top-right corner of the content block's bounding box. It appears on hover (desktop) and is always visible on touch devices.

- Switching to Raw does not re-fetch or reprocess content — the raw source is the already-buffered fenced block content.
- The toggle state is per-block and per-session — it is not persisted across page loads.
- Switching a Vega-Lite block to Raw shows the JSON inspector rather than a plain code block, since the spec is structured data with navigable nodes.
- When a block is in Raw view, the artefact tray entry for that block still downloads the rendered export (or raw source if no `getExportBlob()` is available) — the toggle does not affect the download target.
- Custom renderers that implement `getExportBlob()` are not called while the block is in Raw view.

Inline source citations

When the model's response draws on data returned by an MCP tool call, it should cite the source inline using a numbered superscript that links to the corresponding tool call disclosure card.

Element	Specification
Format	Superscript numeral in the prose (e.g. *"There are 12 entities in this domain[1]"*)
Anchor	Clicking the superscript scrolls to and expands the referenced disclosure card
Scope	Citations reference the specific MCP tool invocation, not a generic tool name
Multiple sources	Where a claim draws on more than one tool call, multiple citations appear (e.g. *"...entities[1] [2]"*)
Attached documents	Citations to attached documents use the document's cite format: page number (PDF), sheet name (Excel), heading (Word)

Citations are rendered as part of the markdown prose block. Superscript numbers reset per response.

Implementation

Scalable renderer registry

All renderers — built-in and host-registered — are represented as entries in a single **renderer registry configuration file**. Built-in renderers (Mermaid, Vega-Lite, math, JSON, tables, code, document canvas) are pre-populated entries in this file; host-registered renderers are appended to it. There is no architectural distinction between the two categories at runtime.

```
{
  "renderers": [
    {
      "id":                   "mermaid",
      "trigger":              "mermaid",
      "name":                 "Mermaid Diagram",
      "moduleUrl":            "/renderers/mermaid@11.4.1/index.js",
      "version":              "11.4.1",
      "builtIn":              true,
      "systemPromptGuidance": "Use a ```mermaid block for relationships, flows, hierarchies, and
sequences."
    },
    {
      "id":                   "vega-lite",
      "trigger":              "vega-lite",
      "name":                 "Vega-Lite Chart",
      "moduleUrl":            "/renderers/vega-lite@5.21.0/index.js",
      "version":              "5.21.0",
      "builtIn":              true,
      "systemPromptGuidance": "Use a ```vega-lite block for metrics, trends, distributions, and
comparisons. Always set width to 'container'."
    },
    {
      "id":                   "risk-gauge",
      "trigger":              "risk-gauge",
      "name":                 "Risk Gauge",
      "moduleUrl":            "https://cdn.acme.com/ai-renderers/risk-gauge@2.1.0/index.js",
      "version":              "2.1.0",
      "builtIn":              false,
      "systemPromptGuidance": "Use a ```risk-gauge block when asked for a risk score. Content must
be JSON: { \"score\": <0–100>, \"label\": \"<text>\", \"breakdown\": [...] }."
    }
  ]
}
```

Adding a renderer

Each renderer is an independently deployed ES module loaded from its own `moduleUrl`. Adding a new renderer requires two steps and touches nothing else in the platform:

1. **Deploy the module** to a versioned URL (`/renderers/<id>@<version>/index.js` or a CDN path).
2. **Append one entry** to the renderer registry config with the trigger tag, module URL, and system prompt guidance.

The platform picks up the new entry at the next session start. Existing renderers are not reloaded, re-tested, or redeployed. Sessions already in progress continue using their cached module set and gain the new renderer at their next session.

Independent versioning

Because each renderer module is loaded from a versioned URL, updating a renderer is equally non-disruptive:

1. Deploy the new module version to a new URL (e.g. `risk-gauge@2.2.0`).
2. Update the `moduleUrl` and `version` fields in the registry config.

In-flight sessions continue using the cached `@2.1.0` module for their lifetime. New sessions load `@2.2.0`. No renderer affects any other renderer's availability or performance.

Library co-deployment

Each renderer module bundles its own dependencies. There is no shared renderer dependency tree — a renderer that requires D3, a custom charting library, or a domain-specific SDK bundles it directly. This means:

- Renderer A using Vega-Lite 5.x and Renderer B using Vega-Lite 4.x can coexist without conflict.
- Upgrading a library for one renderer does not require coordinating with others.
- A renderer can be pinned to a specific library version indefinitely without affecting platform upgrades.

The only shared surface is the `HostRenderer` interface and the `RendererContext` object passed by the platform — these are stable and versioned separately from renderer implementations.

System prompt assembly

At session start, the platform assembles the system prompt guidance block by iterating the registry in order and appending each entry's `systemPromptGuidance`. Built-in renderers come first (they are at the top of the registry), host-registered renderers follow. The LLM receives a complete, current list of available rendering targets with no manual prompt maintenance required — adding a renderer to the registry automatically teaches the LLM to use it.

The `feedback-request` renderer is a built-in entry in the registry. Its system prompt guidance instructs the model on when to emit a feedback block:

```
[Feedback request]
Use a ```feedback-request block when you need explicit user confirmation or a structured choice
before continuing. Provide a clear prompt, 1–4 labelled options, and set allowFreeText to true
only when free-form input adds value over the available options. Do not use feedback-request for
rhetorical or clarifying questions that do not block execution — use prose instead.
```

> **Adding a new renderer requires writing one module and adding one config entry. No existing renderer code, no platform core, and no system prompt template needs to be touched.** The rendering surface scales by addition, not by modification — each new capability is a self-contained deployment that sits alongside everything already running.

11 — Audit and Storage

Governing principle

Conversations on the AI Chat Platform are **auditable records**, not transient chat logs. Every conversation turn is a self-contained, reproducible record. A reviewer can reconstruct what the user submitted, what files they provided, what tools were invoked, and what was produced — without referring to any external system (P4 — audit completeness).

Multi-tenant isolation

Every record in the `assistant` schema carries a `tenant_id` column. Row-Level Security (RLS) enforces that users can only access records belonging to their own tenant. No cross-tenant data access is possible through the platform's API.

The `assistant` schema is isolated from all platform infrastructure schemas and from any host application data schemas. No shared tables exist between the `assistant` schema and other schemas.

Storage policy

All conversation content is retained in full at write time. Nothing is reconstructed on demand.

Per-turn stored elements

Element	Stored as
User prompt (raw)	Exact text including `@`-binding chip markers
User prompt (resolved)	Message as sent to the model — bindings expanded to structured context blocks from `contextTemplate`
Assistant response	Full markdown text including all content blocks
Rendered content blocks	Mermaid source, Vega-Lite specs, CSV, code — typed JSONB records
Tool call log	Tool name, input parameters, response payload, latency, success/error per invocation
Attached input files	Full binary stored in platform object storage
Output artefacts	Full content stored in turn record

Element	Stored as
Canvas version	If the turn produces or accepts a canvas revision, a `canvas_versions` row is written linked to this turn
Model version	Exact provider model string (e.g. `provider-name:model-id:version`) and resolved tier (`fast` / `standard` / `powerful`)
Token counts	Input, output, cache read, cache write — per turn and running session totals
Improvement signals	Detected signals with confidence score and lifecycle status
Author	`user_id` FK on every turn (critical for shared conversations)
Tenant	`tenant_id` on every record — enforces multi-tenant isolation

Partial turns

When a user stops generation mid-stream, the partial response is saved to the audit trail as a **partial turn** with a `status: partial` flag. It is not discarded.

Retention

Rule	Specification
Retention period	Configured per tenant via `conversations.retentionDays` in the application config. Platform default: **1,095 days (3 years)**. Tenants may configure a shorter or longer period within platform-level limits.
Retention tag	Each conversation record is tagged with a `retention_expiry_date` at write time
Archival	A scheduled platform function handles expiry and archival per tenant's configured retention period
User-initiated deletion	Users may delete conversations subject to the tenant's retention minimum; physical deletion is deferred to retention expiry
Turn-level deletion	Not permitted — conversations are append-only at the turn level. Editing creates a new branched thread.
Object storage	Binary artefacts (attached documents, generated outputs) follow the same tenant-configured retention schedule

---

Access control

Access level	Capability
Authenticated end user	Read and write access to their own conversation records within their tenant only
Shared conversation participant	Read and write access to all turns within conversations they have accepted an invitation to
Application Admin	Read access to all conversation records within their own tenant (audit view); no write access to turn records
Platform Admin	Read access to all records across all tenants for operational purposes; no write access to turn records
No user	May edit or delete individual turn records — turns are immutable once written

Row-Level Security (RLS) is enabled on all tables in the `assistant` Postgres schema. Policies enforce: - `tenant_id` isolation across all tables - Per-user and per-participant access within a tenant - Append-only semantics for turn records

---

Database schema overview

The `assistant` schema contains the following tables:

Table	Purpose
`assistant.tenants`	One row per registered host application; tenant config snapshot, API key reference
`assistant.conversations`	One row per conversation thread; `tenant_id`, title, owner, timestamps, retention expiry, CSAT
`assistant.turns`	One row per turn; `tenant_id`, conversation FK, author FK, model, token counts, status
`assistant.artefacts`	One row per artefact; `tenant_id`, turn FK, type, content/storage path, auto-generated name
`assistant.bindings`	One row per `@`-binding chip in a turn; `tenant_id`, object type, Display ID, resolved context snapshot
`assistant.tool_calls`	One row per MCP tool invocation; `tenant_id`, turn FK, server ID, tool name, input params, response, latency, status
`assistant.improvement_signals`	One row per signal; `tenant_id`, turn FK, signal type, confidence score, lifecycle status, issue reference

Table	Purpose
`assistant.session_tools`	One row per opt-in MCP tool activation; `tenant_id`, conversation FK, server ID, activated_by, activated_at
`assistant.conversation_participants`	One row per participant per conversation; `tenant_id`, user FK, invited_by FK, lifecycle timestamps
`assistant.user_memory`	One row per personal memory item; `tenant_id`, user_id, content, category, status
`assistant.app_context`	One row per application context item; `tenant_id`, content, category, status, approval workflow fields
`assistant.app_context_versions`	Version history for application context items
`assistant.canvas_documents`	One row per document canvas open in a conversation; `tenant_id`, conversation FK, title, current version reference
`assistant.canvas_versions`	One row per canvas version; `tenant_id`, canvas FK, turn FK (the turn that produced or accepted this version), full markdown content, author (`user` or `model`), created_at

RLS policies (summary)

- All tables require `tenant_id` to match the authenticated user's tenant claim from their JWT.
- Users can `SELECT`, `INSERT` on their own conversations and turns.
- Accepted participants can `SELECT`, `INSERT` on shared conversation records within the same tenant.
- Application Admins can `SELECT` all records within their tenant.
- Platform Admins can `SELECT` all records across all tenants.
- No user can `UPDATE` or `DELETE` turn records.
- `assistant.conversation_participants` records are append-only — departure is recorded as `departed_at` timestamp, not row deletion.
- `assistant.canvas_versions` records are append-only — each edit creates a new version row; no version row is modified after insertion.

Object storage conventions

Binary artefacts are stored in platform object storage under the following path convention:

```
{tenant_id}/conversations/{conversation_id}/{turn_id}/{artefact_id}/{filename}
```

- Storage is private — access is mediated by signed URLs generated per-request with short expiry.
- Files are not publicly accessible.
- Storage paths are stored in `assistant.artefacts.storage_path`.

Audit trail for shared conversations

In shared conversations, the `user_id` on each turn records the specific participant who submitted it. The `assistant.conversation_participants` table records the full invitation lifecycle:

Column	Content
`user_id`	The participant's platform user ID
`invited_by`	The user ID of the person who invited them
`invited_at`	Timestamp of invitation
`accepted_at`	Timestamp of acceptance (null if not yet accepted or declined)
`departed_at`	Timestamp of departure (null if still active)

There are no role columns — all participants are equal. The audit record reflects actions taken, not role assignments.

Tenant config version history

Every version of a tenant's application config is retained in `assistant.tenants.config_history` (append-only array). This allows platform administrators and Application Admins to trace exactly which config version was active during any historical conversation session.

The config version active at session start is stored in `assistant.conversations.config_version`.

12 — Continuous Improvement

Principle

The AI Chat Platform is a living system. Every session generates signal data. **No change is applied automatically** — every recommendation enters a triage pipeline and requires human review before any change is deployed (P8 — human-gated improvement).

The improvement framework generates: **signal** → **triage** → **issue** → **approval** → **deploy**. It does not generate signal → auto-patch.

Improvement signals

Five types of improvement signal are captured automatically from day one:

Signal type	Detection	Auto-issue?
Explicit turn report	User clicks the report icon on any turn and submits the modal — optionally with a written explanation	Always
Query retry / rephrase	User submits a substantially similar query within two turns	If confidence ≥ threshold
MCP tool failure or empty result	Tool call log shows error or zero-result set where results were expected	If confidence ≥ threshold
In-conversation correction	User explicitly corrects the assistant mid-session	If confidence ≥ threshold
Out-of-scope response	Response contains no tool invocation and no domain-specific content for a query that should have triggered one	If confidence ≥ threshold

Explicit turn report — detail

The report icon appears on every turn (user and assistant). When a user submits a report:

- The turn `id`, `conversation_id`, `tenant_id`, `reporter_user_id`, optional `explanation` text, and a timestamp are stored in `assistant.improvement_signals`
- The signal always generates an improvement issue — no confidence threshold
- The improvement issue includes: the full turn content, the reporter's explanation (if provided), the preceding and following turns for context, and the MCP tool call log for the turn

The written explanation is the most valuable signal — it describes the problem in plain language without inference. Users should be encouraged (via the modal copy) to describe what was wrong and why.

Signal lifecycle

Each signal is stored in `assistant.improvement_signals` with a lifecycle status:

```
new → triaged → issued → resolved
```

Status	Description
`new`	Signal captured; awaiting triage
`triaged`	Reviewed; decision made on whether to raise an improvement issue
`issued`	Improvement issue created; signal linked to issue reference
`resolved`	Underlying change deployed and validated

Confidence threshold

Automatically detected signals below the confidence threshold queue for **manual review first** before an improvement issue is raised. The exact threshold is set at launch and adjusted based on signal volume.

Explicit user reports bypass the confidence threshold and **always generate an improvement issue**.

Improvement issue pipeline

Qualifying signals are processed into an improvement issue containing:

Issue element	Content
Failure description	Signal type, turn summary, and assistant output excerpt
Full conversation turn	Raw prompt, resolved prompt, tool call log, and assistant response
MCP tool call log	All tool invocations for the turn (server name, tool name, params, result, latency)
System-generated recommendation	Most likely remediation target — see table below
Tenant context	`tenant_id`, application name, config version active at the time

Remediation targets

Remediation target	When recommended
Host system prompt	Assistant goes out of scope, misrepresents capabilities, uses wrong terminology for the user's communication style

Remediation target	When recommended
MCP tool description	Assistant chooses the wrong tool or fails to invoke a tool when it should
Guided workflow prompt	A workflow returns unhelpful or incomplete output
Bindable type `contextTemplate`	Binding resolution provides insufficient or incorrect context for the model
Application context	Out-of-date or missing application-level memory causing incorrect assumptions

Routing

Improvement issues are: 1. **Always** visible to the Platform team for platform-level patterns 2. **Optionally forwarded to the host application** via the tenant's configured improvement webhook (see below)

Host applications that configure an improvement webhook receive structured JSON payloads for each qualifying signal, enabling their Application Admin to triage within their own tooling.

Improvement webhook (optional)

Host applications may configure an improvement webhook endpoint in the Platform Admin API. When configured: - Qualifying improvement signals (above threshold or explicit user reports) trigger a POST to the webhook URL with a structured JSON payload - The payload includes: `tenant_id`, `signal_type`, `turn_summary`, `remediation_recommendation`, `issue_id` - Authentication: the platform signs webhook payloads with an HMAC key provided at registration

---

Improvement cadence

Cadence	Activity	Owner
Weekly	Application Admin reviews new improvement issues for their tenant; advances high-confidence items	Application Admin (per tenant)
Weekly	Platform team reviews platform-level patterns across all tenants	Platform team
Monthly	Aggregate signal analysis; identify systemic patterns; set remediation priorities	Application Admin + Platform team
Per release	Post-change validation — monitor signal volume for the affected signal type for two weeks after deployment	Host development team + Platform team

---

What the improvement framework does not do

Action	Status
Auto-update the host system prompt based on signals	Not permitted — all changes require host team review and config update
Auto-retrain or fine-tune the AI model	Not in scope — the platform uses foundation AI provider models
Apply changes to guided workflow prompts automatically	Not permitted — workflow changes require host team config update
Apply changes to MCP tool descriptions automatically	Not permitted
Close improvement issues without human triage	Not permitted — every issue requires Application Admin or Platform team review
Share improvement signal content across tenants	Not permitted — signals are tenant-scoped and never exposed to other tenants

13 — MCP Tool Registry

Purpose

The MCP tool registry is the **per-tenant runtime list of MCP servers** available in a session. It is derived directly from the `mcpServers` array in the tenant's application config — there is no separate registry file or database table beyond the config itself.

When a session starts, the platform resolves the active registry from the tenant's current config and: 1. Activates all `always-on` servers immediately 2. Makes `opt-in` servers available in the tool selection panel (but not active) 3. Injects descriptions for active servers into the system prompt

This document covers the registry model, how host teams populate it, the relationship to the MCP Repository complementary service, and how the registry interacts with the MCP Resources Service.

Access tiers

Tier	Behaviour
Always-on	Active in every session; description injected into system prompt automatically; user cannot disable
Opt-in	Off by default; user activates per session via tool selection panel; description injected only when active; does not persist across sessions

Use `always-on` for servers that are foundational to the assistant's usefulness (e.g. the host application's primary data API). Use `opt-in` for servers that are situationally useful but not needed in every session (e.g. a warehouse query tool, an external analytics service).

Registry schema

The registry is the `mcpServers` array in the application config. Each entry:

```
{
  "id":          "string — unique within tenant",
  "name":        "string — shown in tool selection panel",
  "description": "string — injected into system prompt when active; write for the model",
  "endpoint":    "string — MCP-compliant HTTPS URL",
  "authType":    "bearer | api-key | none",
  "accessTier":  "always-on | opt-in",
  "roles":       ["array of role identifiers — empty = all users"],
  "enabled":     true
}
```

The `enabled` field allows stubs to be added before an endpoint is production-ready. Setting `enabled: false` keeps the entry in the config but excludes it from session resolution — the tool does not appear in the tool selection panel and its description is never injected.

Full field reference is in 01-host-application-config.md — mcpServers.

Tool description quality

The `description` field is **the most important field in the registry**. It is injected directly into the model's system prompt and directly influences whether the model invokes the correct tool for a given user query.

Good description (written for the model):

> Provides access to Acme Corp's data governance platform. Use this tool to look up data domains, entities, quality metrics, data ownership records, and policy information. Use when the user asks about data assets, governance status, quality issues, owners, or compliance.

Poor description (written for the user):

> Governance Platform MCP server.

Application Admins should review and iterate on tool descriptions based on improvement signals. Poor tool descriptions are one of the most common causes of the model choosing the wrong tool.

Adding a new MCP server to the registry

Step	Description
1. Identify the server	Browse the **MCP Repository** to find an existing server, or confirm the host team's own MCP endpoint

Step	Description
2. Verify the endpoint	Confirm the MCP endpoint is production-ready and accessible from the platform's network
3. Confirm auth compatibility	Verify that `bearer` (host JWT forwarded), `api-key`, or `none` auth is appropriate
4. Draft the description	Write the description for the model — describe what data it provides and when to use it
5. Decide access tier	`always-on` if needed in every session; `opt-in` if situationally useful
6. Decide role restriction	Which users should see or activate this tool; empty `roles` array = all users
7. Update the config	Add the entry to `mcpServers` via the Config Editor UI or Admin API
8. Validate	The platform runs an endpoint reachability check on config submission
9. Monitor	Watch improvement signals in the first weeks for tool invocation failures or misuse

Servers may be added as `enabled: false` stubs before their endpoint is ready. This allows the description to be drafted and reviewed before the server becomes active.

Versioning

Tool registry changes are versioned as part of the overall application config (see 01-host-application-config.md — Config versioning):

Change	Config version increment
New MCP server added	Minor
Existing server description or name updated	Patch
Server removed or `enabled` changed to `false`	Minor
`authType`, `endpoint`, or `accessTier` changed	Minor

Relationship to the MCP Repository

The **MCP Repository** is a complementary ecosystem service (see 17-complementary-mcp-services.md) that provides a discoverable catalogue of available MCP servers — both from the platform ecosystem and from external contributors.

The relationship to the per-tenant registry is:

```
MCP Repository              Per-tenant registry
(discover & browse)    →    (configure & activate)
```

Host teams use the MCP Repository to find servers that already exist and are ready to integrate. Once a suitable server is found, its details (endpoint, suggested description, auth type) are used to populate the `mcpServers` entry in the application config.

The MCP Repository is a read-only browsing surface at config time — it is not invoked at session runtime. The per-tenant registry (derived from the config) is the authoritative source for what is available in a session.

Relationship to the MCP Resources Service

The **MCP Resources Service** is a complementary ecosystem service that provides centralised skills, static resources, and reusable prompt artefacts for use across the MCP ecosystem (see 17-complementary-mcp-services.md).

Host applications may register the MCP Resources Service as an MCP server in their tenant registry (typically as an `opt-in` server). When registered and activated in a session: - The model can invoke the MCP Resources Service to retrieve platform-standard skills, guidance documents, and reusable prompt patterns - Resources returned appear as MCP tool results in the conversation thread with the standard tool call disclosure card - Resources are added to the session artefact tray for download

The MCP Resources Service endpoint is published in the MCP Repository.

System prompt injection

When a server is active in a session (always-on or opt-in enabled), its `description` is injected into the system prompt at session start. This informs the model of the server's purpose and when to invoke it.

Always-on server descriptions are always injected. Opt-in server descriptions are injected only when the user has enabled the server for the session. This keeps the system prompt lean when opt-in tools are not in use.

Combined description injection is included in the prompt cache — the cache hit rate metric accounts for the combined length of all active tool descriptions.

14 — Success Metrics

Governing principle

Metrics are captured from day one at both the platform level (across all tenants) and at the application level (per tenant). Platform-level targets reflect the health of the infrastructure and ecosystem. Application-level targets reflect the value each host application is delivering to its users. Baseline targets for new tenants are set at the end of month 1 based on observed behaviour.

Platform-level metrics

These metrics are tracked across all tenants by the platform team.

Metric	Definition	Target
Active tenants	Tenants with at least one active session in the 7-day window	Growth metric — tracked weekly
Platform uptime	API availability (p99 latency < 2s; error rate < 0.1%)	99.9%
MCP tool error rate (platform)	Tool invocations with non-2xx response across all tenants	< 3%
Cache hit rate (platform)	`cache_read_input_tokens` ÷ total input tokens across all tenants	≥ 40% by month 2
Tenant onboarding time	Time from tenant registration to first active session	Baseline month 1
Improvement signal volume	Total signals across all tenants per week	Baseline month 1; monitor trend
Config validation failure rate	Config submissions that fail validation ÷ total submissions	< 10%

Application-level metrics

These metrics are tracked per tenant and reported to the Application Admin via the tenant analytics dashboard.

Metric	Definition	Target
Weekly active users (WAU)	Distinct users opening a session in a 7-day window	40% of the tenant's eligible user base by day 90

Metric	Definition	Target
Queries per session	Mean user messages per conversation	≥ 4
Workflow invocation rate	Sessions invoking at least one guided workflow	≥ 25% where workflows are configured
Binding click-through	Sessions where user clicks a binding chip (fires `binding-click` event)	≥ 30%
MCP tool error rate (application)	Tool invocations with error in the tenant	< 3%
User satisfaction (CSAT)	Post-session rating (1–5) where offered	Mean ≥ 4.0
Improvement signal rate	Improvement signals per 100 turns	Baseline month 1; reduction month-over-month
Cache hit rate (application)	Cache hit rate for this tenant's sessions	≥ 40% by month 2
Document attachment rate	Sessions with at least one document attachment	Baseline month 1
Artefact download rate	Sessions with at least one artefact download	≥ 20%
Mobile session share	Sessions on mobile/tablet viewport	Baseline month 1
Shared conversation rate	Conversations with at least one invitation sent	Baseline month 1
Participant acceptance rate	Invitations accepted ÷ total invitations sent	≥ 70%

Metric definitions

Weekly active users

A user is counted once per 7-day window regardless of session count. "Active" means at least one user message submitted — opening the component without sending a message does not count.

Queries per session

Total user messages ÷ total conversation sessions in the period. Excludes sessions with zero user messages.

Workflow invocation rate

A session counts if at least one guided workflow was invoked via any method: Workflow Library click, `@`-binding, or natural-language trigger.

Binding click-through

A session counts if at least one binding chip in an assistant response was clicked (firing the `binding-click` event to the host application). Tracks whether users are using the assistant as a gateway into the host application, not just a standalone Q&A surface.

MCP tool error rate

Tool invocations with a non-2xx response or an empty result set (where results were expected) ÷ total tool invocations. The "expected results" qualifier requires a confidence classifier — baseline tracks simple error rate only until the classifier is calibrated.

User satisfaction (CSAT)

Post-session rating prompt (1–5 stars) shown to a random sample configured via `features.csatSampleRate` (default: 20% of sessions).

Improvement signal rate

Total improvement signals (all types) per 100 conversation turns, tracked by signal type. Month-over-month reduction in signal rate per type indicates the improvement pipeline is working.

Cache hit rate

`cache_read_input_tokens` ÷ (`input_tokens` + `cache_read_input_tokens`) per the AI provider API response. Tracked at session level and rolling daily average. A target of ≥ 40% by month 2 reflects the expectation that the system prompt and tool descriptions will be consistently cached after the first turn in each session.

Document attachment rate

Sessions where the user attached at least one document (PDF, Excel, Word, or image). Baseline metric — no target until month 1 data is available.

Artefact download rate

Sessions where the user downloaded at least one artefact from the artefact tray. Tracks whether users find rendered outputs useful enough to take away.

Mobile session share

Sessions where the viewport width is < 768px (mobile) or 768px–1023px (tablet) at any point during the session. Baseline metric to validate mobile-first investment.

Shared conversation rate

Conversations where at least one invitation was sent (regardless of acceptance). Tracks collaborative usage.

Participant acceptance rate

Invitations accepted ÷ total invitations sent. A low rate may indicate users are inviting the wrong people, or that the invitation UX needs improvement.

Measurement responsibilities

Metric	Source	Owner
WAU, queries per session, CSAT, attachment rate, artefact download, mobile share, shared rate, acceptance rate	`assistant` schema + session analytics	Platform Engineering
Workflow invocation rate, binding click-through	`assistant.tool_calls` + binding click events	Platform Engineering
MCP tool error rate	`assistant.tool_calls.status`	Platform Engineering
Improvement signal rate	`assistant.improvement_signals`	Platform team + Application Admin
Cache hit rate	AI provider API response metadata stored in `assistant.turns`	Platform Engineering
Platform uptime, config validation failure rate	Platform infrastructure monitoring	Platform Engineering

---

Review cadence

Cadence	Activity	Owner
Weekly	WAU, tool error rate, CSAT per tenant — operational health	Platform Engineering
Weekly	Application Admin reviews improvement signal issues for their tenant	Application Admin
Monthly	Full metric review; signal rate analysis; cache hit trend	Platform team + Application Admins
Day 90 (per tenant)	Activation target assessment (40% WAU); decision on next feature priorities	Application Admin + Platform team

---

Tenant analytics dashboard

Each tenant's Application Admin has access to a **read-only analytics dashboard** showing their tenant's application-level metrics over time: - WAU trend (30-day rolling) - Queries per session trend - CSAT distribution - Top improvement signal types - MCP tool error rate by server - Cache hit rate trend

Platform-level metrics (cross-tenant) are visible to the Platform team only.

15 — Memory and Recall

Overview

Memory and recall governs how the assistant retains standing context across sessions and how that context is scoped to the right users. Two memory controls work together:

Control	Purpose	Scope
Personal memory	Standing context about the individual user — role, focus areas, preferences, and corrections	That user only
Application context	Standing context about the organisation/application — terminology, structure, governance decisions, standing guidance	All users within the same tenant

A third control — **recall scope** — determines who a user can share a conversation with: only users within the same tenant.

All three controls are **transparent** — users always see exactly what context the assistant is working with. Memory is never injected silently (P1 — transparency first).

Personal memory

Purpose

Personal memory lets users tell the assistant things that are true about them specifically, so they don't have to repeat context at the start of every conversation. Examples:

- *"I am the owner of the Finance domain"*
- *"I primarily work with Customer and Transaction data"*
- *"When I ask for a quality summary, scope it to Gold-tier only"*
- *"In our team, 'account' means a customer record, not a financial account"*

Management UI

Personal memory is managed on the **My Memory** tab within the assistant's profile settings area (accessible from within the `<ai-chat>` component).

Section	Contents
Memory toggle	On/Off switch for personal memory injection. Default: **off** until the user explicitly enables it. Toggle state is always visible.

Section	Contents
Memory items list	All memory items (Active and Inactive) — title, category tag, source (Manual / Extracted), date added, token count, status badge, edit and delete controls.
Search / filter	Full-text search across item titles and content; filter by category and status.
Add item	Free-text input — user types the memory item in plain language, optionally assigns a category, confirms, and it is saved as Active.
Extract from conversation	User-initiated only. See Extraction process below.
Token budget indicator	Shows current usage against the tenant-configured token budget (e.g. `640 / 2,000 tokens`) with a colour-coded bar.
Clear all	Archives all items (moves to Inactive) after confirmation. Does not permanently delete — archived items remain in the audit trail.

Personal memory item categories

Host applications configure memory categories in `memory.personalMemory.categories` in the application config. Default categories:

Category	Examples
Role	Job title, domain ownership, team membership
Preference	Preferred response style, scope defaults, output format preferences
Correction	Organisation-specific term overrides; corrections to the assistant's default assumptions
Context	Current focus areas, active projects, known constraints

Personal memory item lifecycle

```
Active → Inactive (budget exceeded or user toggle) → Archived (deleted by user)
```

Status	Injected?	Editable?	Visible in UI?
Active	Yes	Yes	Yes
Inactive	No	Yes — user can reactivate	Yes (greyed out)
Archived	No	No	No (retained in audit trail only)

Items are never permanently deleted — the record is retained with `status = archived` for the audit trail.

Extraction process

"Extract from conversation" is a structured, user-initiated workflow — nothing is extracted automatically.

1. User opens the conversation picker (full-text search across their conversation history)
2. User selects a past conversation
3. A dedicated **fast-tier model** call analyses the selected conversation using a structured extraction prompt — it identifies discrete, self-contained statements of user context worth persisting (role declarations, expressed preferences, corrections made, constraints stated)
4. The assistant presents a **review panel** showing each proposed item with: suggested text, suggested category, estimated token cost, and source conversation reference
5. User accepts, edits, or discards each proposal **individually** — there is no bulk accept
6. Accepted items are saved as Active; discarded proposals are not stored

The extraction call is a separate API call — it does not affect the conversation context window. The extraction prompt explicitly excludes host application object references (stale risk) and system prompt override attempts before they reach the review panel.

Token budget

Personal memory is injected into the system prompt as a labelled `[Your context]` block. The budget is configurable per tenant via `memory.personalMemory.tokenBudget` (default: 2,000 tokens). When the budget is exceeded, the user is warned and the oldest items are marked Inactive — the user decides which items to reactivate or remove.

What personal memory cannot contain

- Host application object IDs or live data references — these may become stale. The assistant resolves live data via MCP at query time.
- Other users' personal information
- Instructions that would override the platform system prompt — the platform prompt takes precedence

Injection behaviour

When personal memory is enabled, the `[Your context]` block is injected into the system prompt at session start. It appears in the conversation header as a collapsed indicator: **"Your context active [2 items · 640 tokens] ↗"** — click to expand and inspect the full content.

Application context

Purpose

Application context gives the assistant organisation-wide context that applies to all users and sessions within the tenant. It provides ground truth for terminology, structure, and standing decisions that the assistant would otherwise have to infer or ask about. Examples:

- *"Our organisation uses 'account' to refer to customer records. 'Account' does not mean a financial account."*
- *"The Customer domain is the master domain. All other domains reference it for shared definitions."*
- *"Data owner assignments are reviewed quarterly — an assignment may be up to 90 days out of date."*
- *"The Finance domain is under an active regulatory review. Flag any Finance domain queries before providing assessments."*

Management UI

Application context is managed in the **Tenant Admin area** (accessible to users with the Application Admin role).

Section	Contents
Context items list	All items with title, category, approval status, effective date, and approver
Add item	Application Admin proposes a new item with a title, category, and content
Approval workflow	If `memory.applicationContext.approvalRequired: true` (default), new items require two-step approval: Propose → Approve. Proposer and approver must be different users.
Edit item	Any edit to an approved item creates a new draft — the existing approved version continues to be injected until the edit is approved
Retire item	Removes an item from injection without deleting it
Version history	Full version history per item: original text, all edits, approval actions, effective dates
Token budget indicator	Current usage against the tenant-configured budget

Categories

Host applications configure application context categories in `memory.applicationContext.categories` (default):

Category	Examples
Terminology	Organisation-specific term definitions; overrides to common terms
Structure	Organisational hierarchy, domain ownership, team structure
Policy	Standing governance decisions, active reviews, classification conventions
Domain context	Known data quality issues, known gaps, recency caveats

Token budget

Application context is injected as a labelled `[Application context]` block in the system prompt. The budget is configurable per tenant via `memory.applicationContext.tokenBudget` (default: 4,000 tokens). Items are injected in category order and by effective date (newest first) within category.

Visibility for all users

Application context is visible (read-only) to all authenticated users within the tenant from the conversation header indicator. Users can see exactly what organisation-wide context the assistant is working with in any session. They cannot edit it.

Approval pipeline

```
Draft → Proposed → Approved (active) → Retired
```

Transition	Who can trigger	Effect
Draft → Proposed	Application Admin	Item visible in admin UI; not yet injected
Proposed → Approved	Application Admin (different user from proposer)	Item becomes active; injected into all subsequent sessions
Approved → Retired	Application Admin	Item removed from injection; retained in audit trail
Approved → Draft (edit)	Application Admin	New draft created; approved version continues injecting until new draft approved

When `memory.applicationContext.approvalRequired: false`, the two-step approval is replaced by a single-step publish action (any Application Admin can publish immediately).

Combined injection — conversation header

In every conversation, the header shows the combined active memory state:

```
┌──────────────────────────────────────────────────┐
│     Your context       [2 items · 640 tokens] ↗  │
│     Application context [4 items · 1,240 tokens] ↗ │
└──────────────────────────────────────────────────┘
```

Both indicators are collapsed by default. Clicking either opens a two-level panel:

Level 1 — Compact list (default on open) Each active memory item shown as a single line: category tag + title + token count.

Level 2 — Full text (expand per item) Click any item row to expand to its full content — exactly as injected.

A user may **disable personal memory for a single session** by clicking the indicator and toggling off before sending their first message. This does not affect the global personal memory setting.

Application context cannot be disabled by individual users.

Token budget summary

Memory type	Default budget	Injected as
Personal memory	2,000 tokens	`[Your context]` block in system prompt
Application context	4,000 tokens	`[Application context]` block in system prompt
Combined maximum	6,000 tokens	Injected before the first turn; cached on subsequent turns

The combined 6,000-token memory budget is included in the prompt cache — the cache hit rate metric accounts for it.

Recall and access scope

Tenant boundary

A user can only share a conversation with — and recall shared context alongside — other users **within the same tenant**. This is a hard boundary enforced at the data layer.

Boundary	Behaviour
Conversation sharing	Invitation search returns users within the authenticated user's tenant only. Cross-tenant results are excluded.
Application context	The `[Application context]` block is scoped to the authenticated user's tenant. Context from one tenant is never injected into another tenant's sessions.

Boundary	Behaviour
Personal memory	Personal memory items are owned by `user_id` and are never visible to other users.

Why this boundary exists

Conversation threads accumulate context about an organisation's operations, decisions, and data. Permitting cross-tenant access to this context would violate data confidentiality. The tenant boundary is a design constraint, not a configurable option.

Audit trail

Element	Where stored
Personal memory items	`assistant.user_memory` — `tenant_id`, `user_id`, `content`, `category`, `source`, `status`, timestamps
Application context items	`assistant.app_context` — `tenant_id`, `title`, `category`, `content`, `status`, `proposed_by`, `approved_by`, `effective_from`, `retired_at`; version history in `assistant.app_context_versions`
Memory injected per session	`assistant.turns` — the resolved system prompt including both memory blocks is stored on the first turn of every conversation

Memory items themselves are retained indefinitely (they are governance configuration, not conversation data). Only their injection into turns is subject to the tenant's configured retention period.

What memory cannot replace

Memory provides standing context. It does not replace live data. The assistant always resolves live object data, status, and metrics via host MCP servers at query time. Memory items that describe application objects are advisory framing, not authoritative data.

Correct use: *"The Finance domain is under an active regulatory review — flag Finance queries."* This is framing that helps the assistant contextualise queries.

Incorrect use (rejected by validation): *"The Finance domain has 42 entities and Jane Smith is the owner."* This would become stale. The assistant queries the host MCP server for live entity data.

16 — Embedding and Web Component (Mode 2 — Inline Page)

Scope of this document

This document covers the `<ai-chat>` **component** — the inline page embedding mode (Mode 2). It is the complete, full-featured conversation interface for use on dedicated assistant pages or embedded content sections.

The AI Chat Platform has three distinct embedding modes. For the full picture, see 18-entry-points-and-embedding-modes.md:

Mode	Component	Description
1 — Floating Widget	`<ai-chat-widget>`	FAB + collapsible mini/full panel, persists across pages
2 — Inline Page	`<ai-chat>`	This document — full three-zone layout embedded in a page
3 — Form Field Assist	`<ai-chat-field>`	Ephemeral contextual popover scoped to a form field

Modes 1 and 2 share conversation history. Mode 3 is ephemeral and fully independent.

Overview

The `<ai-chat>` component is a **custom HTML element** that host applications embed within their own UI. The component is self-contained — it manages its own state, handles streaming, renders content, and communicates with the platform API independently of the host application's framework.

Host applications include the component script once and configure it via HTML attributes and a JavaScript API. No framework-specific adapters are required; the component works in any modern web environment.

Quick start

```
<!-- 1. Include the component script -->
<script type="module" src="https://chat-platform.io/sdk/v1/ai-chat.js"></script>

<!-- 2. Mount the component -->
<ai-chat
  tenant-id="acme-corp"
  user-token="eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
  theme="light"
></ai-chat>
```

The component renders at 100% of its container's width and height. Host applications control sizing via the container element.

HTML attributes

Attribute	Required	Type	Description
`tenant-id`	Yes	string	The tenant's registered `tenantId` from the application config
`user-token`	Yes	string	The host-authenticated user's JWT. Must contain the required claims (see Authentication bridge). Refreshed by calling `updateToken()`.
`theme`	No	`"light"` | `"dark"`	Initial colour scheme. Defaults to `"light"`. Dark mode is not supported in v1 — this attribute is reserved.
`locale`	No	string	BCP 47 locale tag (e.g. `"en-GB"`). Overrides the tenant config `scope.language` for UI strings.
`initial-conversation-id`	No	string	Opens a specific conversation on mount instead of the default new/recent conversation. Useful for deep-linking from the host application.

Authentication bridge

The platform does not authenticate users independently. The host application authenticates its users and passes the result to the component as a **signed JWT** via the `user-token` attribute.

Required JWT claims

Claim	Type	Description
`sub`	string	User's unique identifier within the tenant
`email`	string	User's email address (used for invitation search)
`name`	string	User's display name (shown in shared conversation attribution)
`tenant_id`	string	Must match the `tenant-id` attribute — enforced server-side
`exp`	number	JWT expiry time — the component checks this and fires `token-expired` when within 60 seconds of expiry

Optional JWT claims

Claim	Type	Description
Any field in `userProfile.styleField`	string	Communication style value (e.g. `"business"`)
Any field in `userProfile.verbosityField`	string	Response verbosity value (e.g. `"concise"`)
Any role identifiers listed in MCP server `roles`	string[]	Role claims used to enforce MCP server access restrictions
`avatar_url`	string	URL to the user's avatar image (used in shared conversation thread)

JWT signature verification

The platform verifies the JWT signature against the JWKS endpoint registered by the host application at tenant registration time. Tokens signed with keys not in the registered JWKS are rejected.

Token refresh

```
const chat = document.querySelector('ai-chat');

// Listen for token expiry warning
chat.addEventListener('token-expired', async () => {
  const newToken = await yourAuthService.refreshToken();
  chat.updateToken(newToken);
});
```

`updateToken(token)` updates the active token without remounting the component. The user's session and conversation state are preserved.

JavaScript API

Access the component's API via a reference to the element:

```
const chat = document.querySelector('ai-chat');
```

Methods

Method	Parameters	Description
`updateToken(token)`	`token: string`	Refresh the user's JWT without remounting
`openConversation(id)`	`id: string`	Navigate to a specific conversation by ID
`startNewConversation()`	—	Open a new conversation, clearing the current one
`setLocale(locale)`	`locale: string`	Change the UI locale at runtime

Properties

Property	Type	Description
`currentConversationId`	`string | null`	The ID of the currently open conversation
`isStreaming`	`boolean`	Whether the assistant is currently generating a response

---

Events

The component fires custom events on the element. Listen with `addEventListener`.

Event	Detail payload	Description
`binding-click`	`{ typeId, objectId, displayId, name }`	Fired when a user clicks a binding chip. The host application should handle navigation to the object's detail page.
`conversation-created`	`{ conversationId, title }`	Fired when a new conversation is created
`conversation-opened`	`{ conversationId, title }`	Fired when an existing conversation is opened
`turn-complete`	`{ conversationId, turnId, model, tokenCounts }`	Fired when an assistant response completes (not on partial/stopped)
`tool-invoked`	`{ conversationId, turnId, serverId, toolName, status }`	Fired on each MCP tool invocation completion

Event	Detail payload	Description
`token-expired`	—	Fired when the user's JWT is within 60 seconds of expiry
`error`	`{ code, message }`	Fired on component-level errors (auth failure, network error, config error)
`csat-submitted`	`{ conversationId, score, comment }`	Fired when a user submits a CSAT rating

Handling `binding-click`

```
chat.addEventListener('binding-click', (event) => {
  const { typeId, objectId, displayId, name } = event.detail;

  switch (typeId) {
    case 'data_domain':
      router.navigate(`/domains/${objectId}`);
      break;
    case 'policy':
      router.navigate(`/policies/${objectId}`);
      break;
    default:
      console.log('Unhandled binding type:', typeId);
  }
});
```

Sizing and layout

The `<ai-chat>` component renders at **100% width and 100% height of its containing element**. The host application controls the component's footprint by sizing the container.

Full-page embedding

```
/* Host application styles */
.chat-page {
  width: 100%;
  height: 100vh;
  overflow: hidden;
}
```

```
<div class="chat-page">
  <ai-chat tenant-id="..." user-token="..."></ai-chat>
</div>
```

Side panel embedding

```
.chat-panel {
  width: 480px;
  height: 100%;
  border-left: 1px solid var(--border-color);
}
```

```
<aside class="chat-panel">
  <ai-chat tenant-id="..." user-token="..."></ai-chat>
</aside>
```

Minimum dimensions

Dimension	Minimum
Width	320px
Height	400px

Below these minimums the component renders a fallback state: *"This panel is too small to display the assistant. Please expand the window."*

Breakpoint behaviour in containers

When the component is mounted in a container narrower than the desktop breakpoint (< 1280px), it uses the container width rather than the viewport width to determine which layout to apply. A 480px side panel will use the tablet layout regardless of the overall page width.

Content Security Policy (CSP)

The `<ai-chat>` component requires the following CSP directives on the host page:

```
script-src    https://chat-platform.io [renderer module origins];
connect-src   https://api.chat-platform.io wss://api.chat-platform.io;
frame-src     'none';
img-src       https://cdn.chat-platform.io [host CDN origins];
style-src     https://chat-platform.io 'nonce-{page-nonce}';
```

Additionally: - Any URLs provided in the `branding` config (`logoUrl` , `faviconUrl`) must be included in `img-src` . - Each origin hosting a custom renderer module (from `renderers[].moduleUrl` in the application config) must be included in `script-src` . Renderer modules are loaded via dynamic `import()` at runtime and are subject to the host page's CSP. A module blocked by CSP will cause that renderer to fall back to syntax-highlighted code — the platform will not bypass CSP.

The component itself does not use `eval`, `inline-script`, or `blob:` URLs. All platform assets are loaded from `https://chat-platform.io` or `https://cdn.chat-platform.io`. Renderer modules are loaded from host-provided origins only.

React / Vue / Angular integration

The web component works without framework adapters. Framework-specific wrappers are provided for convenience:

React:

```
import { AiChat } from '@chat-platform/react';

<AiChat
  tenantId="acme-corp"
  userToken={jwt}
  onBindingClick={({ typeId, objectId }) => navigate(`/${typeId}/${objectId}`)}
/>
```

Vue:

```
<AiChat
  tenant-id="acme-corp"
  :user-token="jwt"
  @binding-click="handleBindingClick"
/>
```

Angular:

```
<ai-chat
  tenant-id="acme-corp"
  [attr.user-token]="jwt"
  (binding-click)="handleBindingClick($event)"
></ai-chat>
```

Security considerations

JWT forwarding

The `user-token` attribute value is passed as a bearer token to the platform API on every request. It must: - Be issued by the host application's authenticated session (never a long-lived API key) - Have a short expiry (recommended: 15 minutes) - Be refreshed via `updateToken()` before expiry

Preventing attribute injection

Do not interpolate untrusted user input into the `tenant-id` or `user-token` attributes. Both are validated server-side, but attribute injection could result in failed sessions or user confusion.

CORS

The platform API enforces CORS. Only origins registered at tenant registration time are permitted to make requests. The host application must register all deployment origins (development, staging, production) during tenant setup.

Iframe considerations

The component must not be embedded within an `<iframe>` — this prevents the platform from accessing the host page's cookies and local storage required for session management. The component is designed for direct DOM embedding only.

17 — Complementary MCP Services

The AI Chat Platform assumes the availability of three ecosystem-level MCP services that operate alongside the platform and host application MCP servers. These services are not owned or operated by the AI Chat Platform itself, nor by individual host applications — they are shared infrastructure within the broader MCP ecosystem. The platform is designed to work with them, and host applications are expected to benefit from them, but none is a hard dependency for platform operation.

MCP Repository

Overview

The **MCP Repository** is a centralised, discoverable catalogue of MCP servers available within the ecosystem. It serves as the primary discovery mechanism for host teams when they are configuring their tenant's MCP tool registry — providing a searchable directory of tools that have already been built, tested, and published, rather than requiring every host team to build their own MCP servers from scratch.

The MCP Repository is a **config-time service**: host teams use it when setting up or updating their application config, not as a runtime tool invoked during user conversations. Its value is in accelerating the time between "we want to add this capability to our assistant" and "this capability is live for our users."

What the MCP Repository provides

The MCP Repository holds metadata records for each published MCP server, including:

- **Server identity:** Name, publisher, version, and a plain-language description of what the server provides
- **Capability catalogue:** The specific tools exposed by the server, with descriptions and example invocations
- **Integration metadata:** The server's MCP endpoint URL, supported authentication types (`bearer`, `api-key`, `none`), and any pre-conditions for integration (e.g. required host-side configuration)
- **Quality indicators:** Verification status, uptime history, average latency, and known compatibility notes with the AI Chat Platform
- **Suggested descriptions:** Pre-written `description` field text optimised for injection into the AI Chat Platform's system prompt — host teams can use these as-is or customise them

How host teams use the MCP Repository

When a host team wants to add a new capability to their assistant, the typical flow is:

1. Search the MCP Repository for a server that provides the needed capability
2. Review the server's capability catalogue and quality indicators

3. Copy the server's endpoint URL, suggested `description`, and recommended `authType` into the `mcpServers` entry in the application config
4. Submit the updated config via the Config Editor UI or Admin API — the platform validates endpoint reachability as part of config submission
5. Monitor tool invocation quality via improvement signals in the first weeks of use

Publishing to the MCP Repository

MCP server publishers — whether internal platform teams, host application developers, or third parties — submit their server records via the MCP Repository's submission API. Submitted records are reviewed for: - Correctness and completeness of metadata - MCP protocol compliance - Endpoint stability and availability - Security posture (auth requirements, data exposure scope)

Approved records are published and immediately searchable in the Repository. Publishers are responsible for keeping their records current as their server's endpoint or capability changes.

Relationship to the per-tenant tool registry

The MCP Repository and the per-tenant tool registry (13-mcp-tool-registry.md) are distinct:

MCP Repository	Per-tenant registry
Ecosystem-wide catalogue of all available servers	Tenant-specific list of servers active for that host application
Read at config time by host teams	Resolved at session runtime by the platform
Covers all publishers and server types	Covers only what the host application has chosen to enable
Not invoked during user conversations	The source of truth for what tools a session can use

---

MCP Resources Service

Overview

The **MCP Resources Service** is a centralised MCP server that provides shared, reusable assets across the MCP ecosystem — skills, guidance documents, prompt templates, and other static artefacts that are useful across many different host applications and conversation types.

Unlike the MCP Repository (which helps teams find and configure tools at setup time), the MCP Resources Service is a **runtime service** — it can be registered as an MCP server in a tenant's tool registry and invoked during user conversations to retrieve resources on demand.

What the MCP Resources Service provides

The MCP Resources Service organises its content into three categories:

Skills Pre-built structured reasoning patterns that the assistant can invoke to approach complex tasks consistently. Skills are prompt fragments with defined input parameters and expected output structures. Examples include: - Structured root-cause analysis prompts - Stakeholder communication drafting frameworks - Risk assessment scoring rubrics - Data quality evaluation methodologies

Skills are not model-specific — they are designed to work across Anthropic, OpenAI, and Gemini models. When the assistant retrieves a skill from the MCP Resources Service, it uses the skill's prompt structure to guide its reasoning for that turn.

Guidance documents Static reference documents that apply across many host applications and are impractical for each host to maintain independently. Examples include: - AI assistant usage guidelines and responsible use principles - Data privacy handling guidance for AI-assisted workflows - Model capability and limitation reference cards - Prompt engineering best practices for domain-specific applications - **Uncertainty handling guidance** — instructions for how the assistant should communicate the limits of its knowledge, signal when data may be outdated, and offer next steps (such as web search) when it cannot answer with confidence. This document is injected as a resource at session start by hosts that want consistent, domain-appropriate uncertainty behaviour beyond the platform's non-overridable baseline.

Guidance documents are versioned by the MCP Resources Service. When retrieved in a conversation, the document version is recorded in the tool call log.

Prompt templates Reusable prompt structures that host teams can adopt as starting points for guided workflows in their application config. These are published as MCP resource objects and can be retrieved, reviewed, and adapted before being embedded in the `workflows` section of a tenant's config.

How host applications integrate the MCP Resources Service

Host applications that want their end users to benefit from MCP Resources Service content register the service as an MCP server in their tenant's tool registry — typically as an `opt-in` server, since not every conversation will require it:

```
{
  "id":          "mcp-resources",
  "name":        "Platform Resources",
  "description": "Provides shared skills, guidance documents, and prompt templates from the
platform ecosystem. Use when you need a structured reasoning approach, usage guidance, or
reference material.",
  "endpoint":    "https://resources.mcp-ecosystem.io/mcp",
  "authType":    "bearer",
  "accessTier":  "opt-in",
  "roles":       []
}
```

The MCP Resources Service endpoint is published in the MCP Repository. The suggested `description` field text above is available in the Repository record and optimised for AI Chat Platform injection.

Content governance

The MCP Resources Service is governed by the platform ecosystem team responsible for its operation. Content goes through an editorial review before publication: - Skills and templates are tested against the platform's AI provider models before publication - Guidance documents are reviewed for accuracy and applicability across a range of host application types - All content is versioned; breaking changes to existing resources trigger a deprecation period before removal - Tenants that have retrieved a specific resource version continue to receive that version until they explicitly update to a newer version

Host applications and their users cannot modify MCP Resources Service content. Feedback and content requests are submitted via the ecosystem's standard issue process.

Relationship to the platform

The MCP Resources Service is independent of the AI Chat Platform — the platform does not own or operate it. However, the platform is designed to work with it cleanly: - Resources returned by the service are rendered using the platform's standard content rendering rules (prose, code blocks, data tables as appropriate) - Resource retrievals appear in the tool call disclosure card with the server name and tool name - Retrieved resources are added to the session artefact tray for download - Resource retrievals are included in the tenant's MCP tool error rate metric and improvement signal pipeline

If the MCP Resources Service is unavailable, it behaves like any other unavailable opt-in MCP server — an error is shown in the tool call disclosure card, and the session continues without it.

Web Search Service

Overview

The **Web Search Service** is an ecosystem-level MCP server that provides the assistant with real-time access to web search results. It is registered as a host application MCP server like any other — the distinction is that it is a shared, ecosystem-managed service rather than a server built and operated by the host team.

Web search complements the platform's primary data access pattern (structured MCP tool calls against the host application's own data) in situations where current information is needed that the host's tools and the model's training knowledge cannot provide — recent news, current pricing, up-to-date documentation, or any query that requires information beyond the host application's data scope.

What the Web Search Service provides

Tool	Description
`web_search`	Executes a web search query and returns a ranked list of results: title, URL, snippet, and publication date

Tool	Description
`fetch_page`	Retrieves and extracts the readable text content of a specific URL from search results or user-provided links

Results are returned as structured MCP tool output — they appear in a tool call disclosure card and are cited inline using the platform's standard source citation mechanism (superscript numerals linking to the disclosure card). The model does not present web results as its own knowledge.

How host applications integrate the Web Search Service

Host applications register the Web Search Service as an MCP server in their tenant tool registry. It is typically configured as `opt-in` — the end user enables it per session when real-time information is needed — though hosts may configure it `always-on` if their use case depends on current information in every session:

```
{
  "id":          "web-search",
  "name":        "Web Search",
  "description": "Searches the web for current information. Use when the user needs up-to-date
facts, recent news, current documentation, or any information beyond the host application's data
and the model's training knowledge. Always cite the source URL.",
  "endpoint":    "https://search.mcp-ecosystem.io/mcp",
  "authType":    "api-key",
  "accessTier":  "opt-in",
  "roles":       []
}
```

The Web Search Service endpoint and API key are available from the MCP Repository. The suggested `description` field text above is optimised for system prompt injection and is published in the Repository record.

Transparency and citation

Web search results follow the platform's mandatory tool call transparency rules — every search invocation and every page fetch appears as a collapsible disclosure card in the conversation thread. The model is instructed (via the platform-managed system prompt layer) to: - Cite all web-sourced claims with inline source citations linking to the disclosure card - Acknowledge when search results are recent vs. when the model is supplementing with training knowledge - Not present search result content as the model's own knowledge

Relationship to the platform

The Web Search Service is independent of the AI Chat Platform. When unavailable, it behaves like any other unavailable MCP server — an error in the disclosure card and the session continues without it. Web results rendered as prose are subject to the platform's standard content rendering rules; results rendered as structured data (tables, lists) use the platform's data table rendering.

18 — Entry Points and Embedding Modes

The AI Chat Platform is delivered through three distinct embedding modes, each with its own component, visual presentation, and conversation model. Host applications may deploy one, two, or all three modes simultaneously — they are independent components that share the same platform backend.

Mode overview

Mode	Component	Conversation model	Persistence	Primary use
1 — Floating Widget	`<ai-chat-widget>`	Full — shared with Mode 2	Persistent across pages	Always-available assistant access from anywhere in the host app
2 — Inline Page	`<ai-chat>`	Full — shared with Mode 1	Persistent	Dedicated assistant page or embedded content section
3 — Form Field Assist	`<ai-chat-field>`	Ephemeral — scoped to the field instance	Discarded on close	Contextual help and content generation for individual form fields

Modes 1 and 2 are **persistent conversation modes** — they share the same conversation history for a given user and tenant. A conversation started in the floating widget appears in the history panel of the inline page view, and vice versa. Mode 3 is entirely **ephemeral and transactional** — it does not create persistent conversation threads and does not appear in any history view.

Mode 1 — Floating Widget

Overview

The floating widget provides always-available assistant access without requiring the user to navigate away from their current page. It is anchored to a corner of the viewport and overlays the host application's content. It has three states: **collapsed** (FAB only), **mini** (compact chat panel), and **full** (complete conversation interface).

States

Collapsed state — the FAB

The widget renders as a **floating action button** (FAB) in the configured corner of the viewport (default: bottom-right). The FAB carries:

- The assistant's icon or initial (from `identity.logoUrl` or the first letter of `identity.assistantName`)

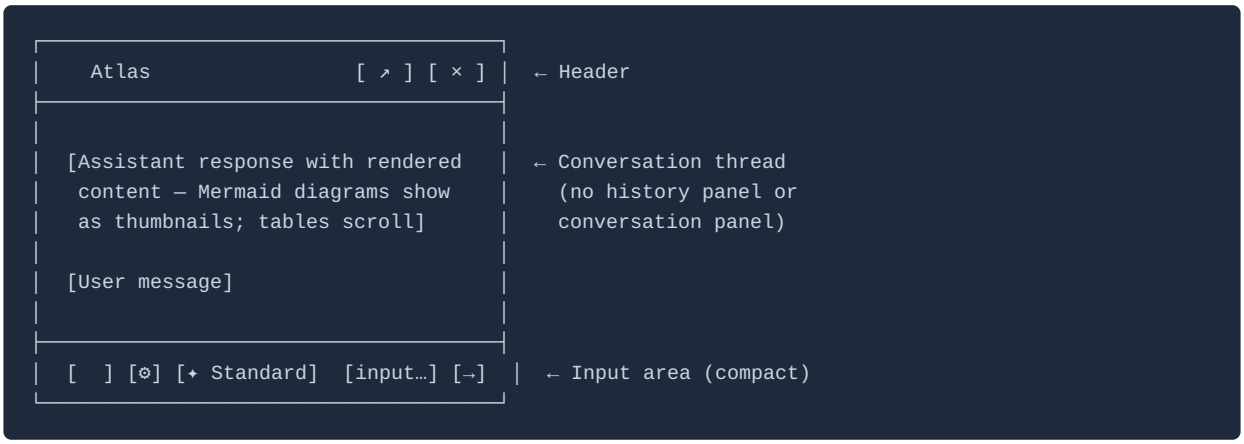
- An unread **badge** count when there are new messages in a shared conversation the user has not viewed
- A subtle pulse animation when the assistant is mid-stream in a minimised conversation

Badge is suppressed when `showBadge: false` is set in the widget config.

Clicking the FAB transitions to the mini state.

Mini state — compact chat panel

The mini state is a compact, non-blocking chat panel anchored above the FAB. Default dimensions: **380px wide × 520px tall**, configurable via the `widget` config section. The panel is elevated above host application content with a shadow; it does not push or reflow page content.



Mini state behaviours:

Element	Specification
Conversation shown	The most recently active conversation. If no prior conversation exists, shows the onboarding welcome state.
History access	No history panel in mini state. A **conversations icon** (☰) in the header opens a compact conversation list overlay within the panel — not a full sidebar.
New conversation	Available via the conversations list overlay; "+ New" button
Input area	Full input capability: text, `@`-binding, attachments, model chip, tool selection. Controls compact but not hidden.

Element	Specification
Content rendering	All content types render (prose, Mermaid, Vega-Lite, tables, code). Diagrams and wide tables are constrained to the panel width and scroll horizontally. Mermaid diagrams collapse to thumbnail by default — tap/click to expand to full-screen overlay.
Tool call disclosures	Rendered as collapsed cards, same as full mode.
Resize	User can drag the top edge to adjust panel height within the range 360px–80vh. Width is fixed.
Draggable	The panel header is draggable — user can reposition the mini panel within the viewport.
Keyboard shortcut	`Escape` collapses mini state back to FAB.

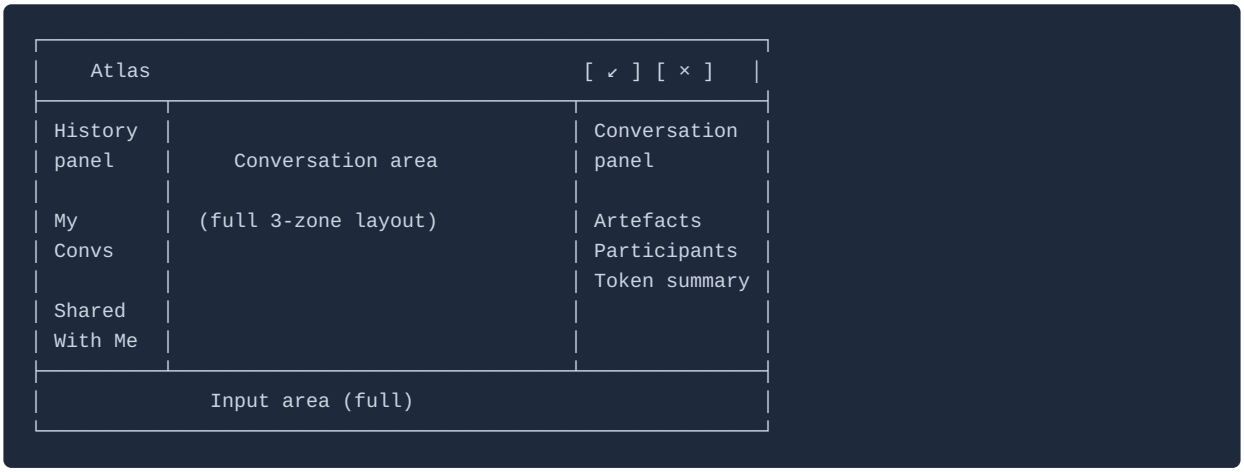
Conversation list overlay (opened via ☰ in the mini-state header):

Header controls in mini state:

Control	Action
↗ Expand icon	Transitions to full state (see below)
× Close icon	Collapses to FAB; does not end the conversation

Full state — complete conversation interface

Clicking the expand icon (↗) from the mini state transitions to the full conversation interface. The full state renders as a **large modal overlay** occupying most of the viewport:



The full state renders the complete three-zone layout from `09-interaction-design.md`. The conversation active in mini state is the conversation shown on transition — no context is lost.

Full state behaviours:

Behaviour	Specification
Layout	Complete three-zone layout: history panel + conversation area + conversation panel
Modal dimensions	Default: `min(90vw, 1200px)` wide × `min(90vh, 860px)` tall, centred in viewport
Background	Host page dimmed (overlay) but not blocked — user can click outside to collapse to mini state
Transition	Smooth expand animation from mini panel position to modal centre
Collapse	↙ icon collapses back to mini state; × closes to FAB
Page navigation	If the user navigates while full state is open, full state collapses to mini state automatically

Conversation continuity between mini and full

There is no transition cost when moving between mini and full states. The same conversation is active in both views, and they share state in real time — a response streaming in mini state continues streaming when the user expands to full state.

Floating widget across page navigation

The floating widget persists across client-side page navigation (SPA routing). It retains the current conversation and streaming state as the user moves between pages. On hard page reload, the widget remounts and restores the most recently active conversation.

Web component

```
<!-- Place once in the host application shell / layout -->
<ai-chat-widget
  tenant-id="acme-corp"
  user-token="..."
  position="bottom-right"
></ai-chat-widget>
```

Widget-specific HTML attributes

Attribute	Default	Description
`position`	`bottom-right`	FAB and panel position. Options: `bottom-right`, `bottom-left`
`mini-width`	`380`	Width of the mini panel in pixels (range: 320–600)
`mini-height`	`520`	Default height of the mini panel in pixels (range: 360–800)
`z-index`	`9000`	CSS z-index for the widget stack
`offset-x`	`24`	Horizontal margin from the viewport edge in pixels
`offset-y`	`24`	Vertical margin from the viewport edge in pixels

Widget-specific events

Event	Detail	Description
`widget-expanded`	`{ state: 'mini' | 'full' }`	Widget transitioned from collapsed to mini or full
`widget-collapsed`	`{ previousState: 'mini' | 'full' }`	Widget collapsed to FAB

All other events from `16-embedding-and-web-component.md` (`binding-click`, `turn-complete`, `token-expired`, etc.) fire identically from `<ai-chat-widget>`.

Browser notifications

When `showBadge` is enabled, the widget can optionally request the **Web Notifications API** permission to surface a browser notification when a shared conversation receives a new message and the widget is collapsed. This is relevant when the user is on a different browser tab or has the host application in the background.

Behaviour	Specification
Permission request	The widget never requests notification permission proactively. Permission is only requested when the user explicitly opts in — via a *"Notify me of shared conversation activity"* control in the conversations panel.
Notification payload	*"[Participant name] replied in [conversation title]"* — no message content in the notification for privacy
Click action	Clicking the notification opens the host application and expands the widget to the mini state, navigating to the relevant conversation
Denial handling	If the user declines or revokes permission, the badge count remains the only unread signal. No fallback prompt is shown.
Host control	`showBadge: false` in the widget config suppresses both badge and browser notifications entirely.

Config

The `widget` config section in the application config controls widget-level defaults:

```
{
  "widget": {
    "defaultState":           "collapsed",
    "fabIcon":                "https://cdn.acme.com/atlas-icon.svg",
    "showBadge":              true,
    "returningUserThreshold": 4
  }
}
```

Field	Default	Description
`defaultState`	`"collapsed"`	State on first page load: `"collapsed"` or `"mini"`
`fabIcon`	Assistant logo from `branding.logoUrl`	Custom icon for the FAB. 40×40px SVG or PNG.
`showBadge`	`true`	Show unread badge count on FAB for shared conversation activity
`returningUserThreshold`	`4`	Hours of inactivity after which the widget opens to the returning-user card instead of resuming mid-conversation. Set to `0` to always resume mid-conversation.

Mode 2 — Inline Page

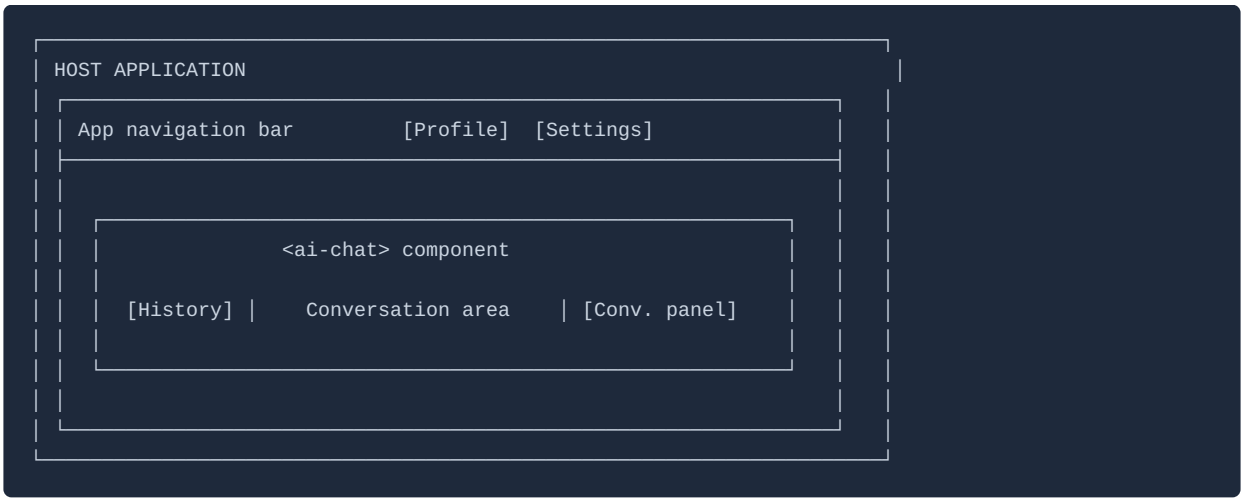
Overview

The inline page mode embeds the assistant as a first-class content block within a host application page. It uses the `<ai-chat>` component described in full in `16-embedding-and-web-component.md`. It is the **richest and most capable** embedding mode — the full three-zone layout with no constraints on features or content rendering.

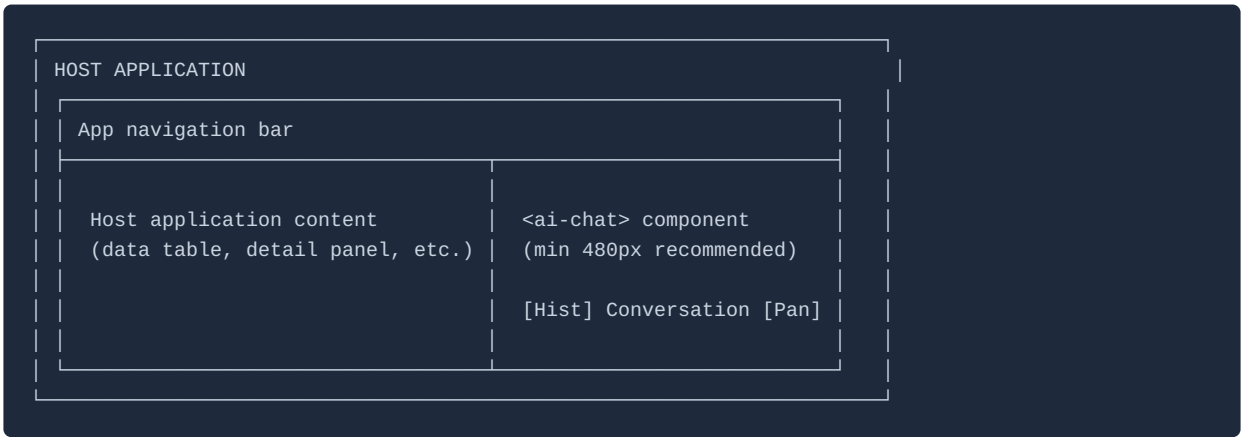
Mode 2 in host application context

The `<ai-chat>` component occupies its mounting point only — the host application retains full ownership of its navigation, header, and page shell.

Dedicated assistant page (full content area):



Split-panel page (assistant alongside host content):



Conversation model

Mode 2 shares conversation history fully with Mode 1. The same history panel, the same conversation list, the same threads — a user moving between a dedicated assistant page (Mode 2) and the floating widget (Mode 1) sees the same conversations in both. There is no concept of "page-scoped" conversations in Mode 2.

Typical placements

Placement	Description
Dedicated assistant page	A route in the host application dedicated to the assistant (e.g. `/assistant`). The `<ai-chat>` component fills the main content area. Most capable experience.
Split-panel page	The host application page is split: content on the left, assistant on the right. The component is mounted in a fixed-width right panel (minimum 480px recommended).
Tab within a page	The assistant occupies one tab of a tabbed page. The component mounts/unmounts on tab switch — component state is restored on remount via the platform API.

Relationship to Mode 1

A host application that deploys both `<ai-chat>` and `<ai-chat-widget>` on the same page will have two simultaneous conversation views. This is an unusual configuration but is supported. Both components connect to the same platform backend with the same `tenant_id` and `user_id` — changes in one are reflected in the other in real time.

For most host applications, Mode 1 (floating widget) is deployed site-wide in the layout shell, and Mode 2 (inline) is deployed on a dedicated assistant page. When the user navigates to the dedicated page, the floating widget collapses automatically (the host application controls this by calling `widget.collapse()` on the `<ai-chat-widget>` element).

Mode 3 — Form Field Assist

Overview

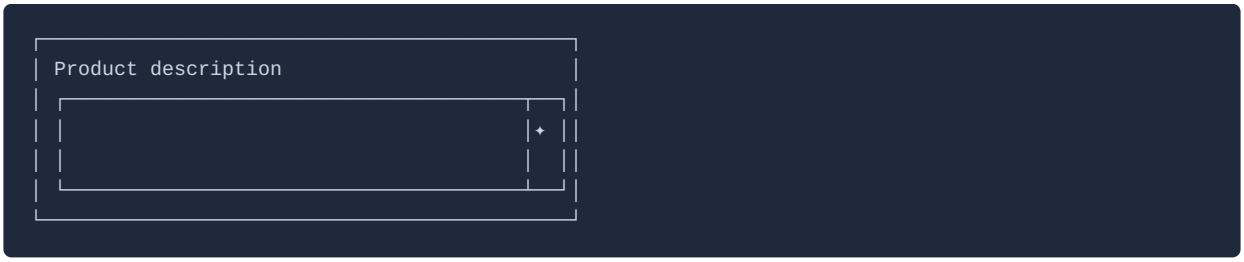
The Form Field Assist provides contextual, ephemeral AI assistance scoped to an individual form field. It appears as a small icon or button adjacent to the field and opens a **compact contextual popover** when activated. The interaction is entirely transactional — there is no conversation history, no history panel, and no persistent thread. When the popover closes, the session is discarded.

Form Field Assist is designed for tasks like: - *"Write a product description for this item"* - *"Suggest a title based on these bullet points"* - *"Explain what this field is asking for"* - *"Translate this text into French"* - *"Summarise this paragraph more concisely"*

Trigger presentation

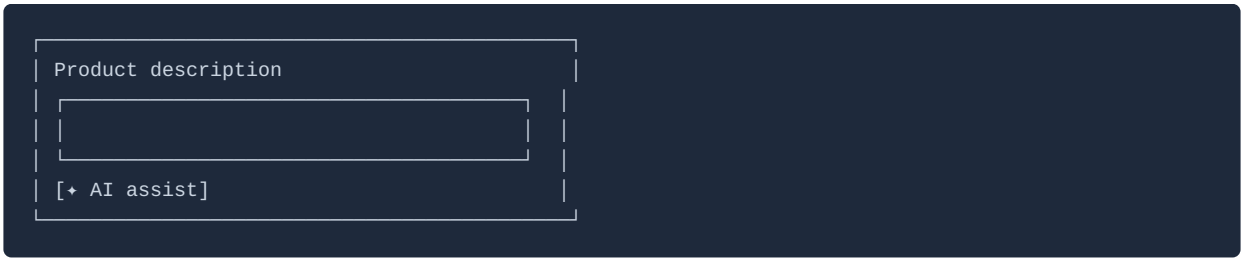
The trigger can be presented in two ways, depending on the host application's UI:

Inline icon trigger (inside the field)



A small ✦ (or assistant icon) appears inside the field at the trailing edge. It does not overlay field content — the field's right padding is adjusted by the component to accommodate it.

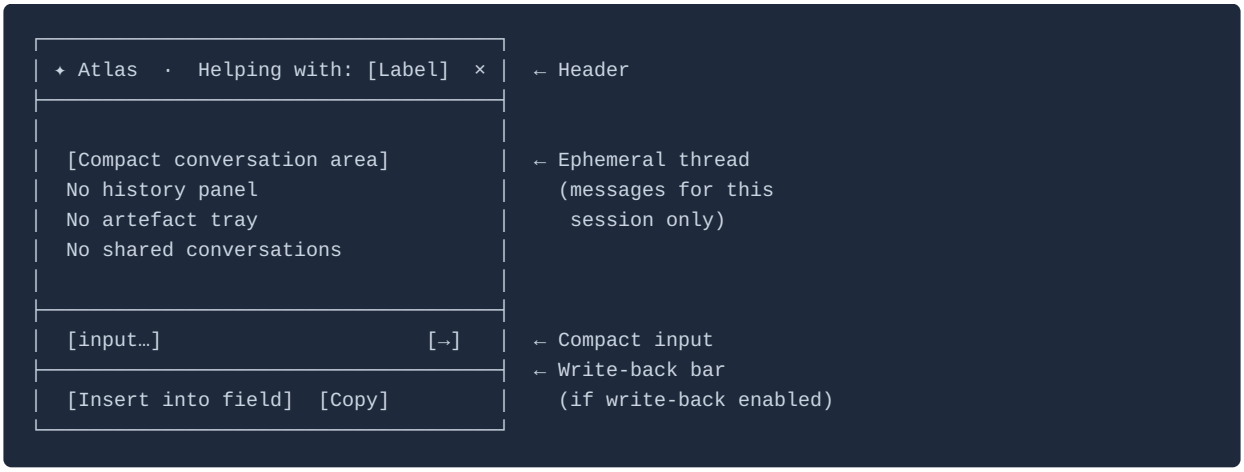
External button trigger (beside or below the field)



A button or link appears below or beside the field. The host application styles the button; the `<ai-chat-field>` component fires a `field-trigger-click` event the host handles. This variant gives the host full control over the trigger's visual appearance.

The trigger variant is configured via the `trigger` attribute on the component (see Web component below).

Popover layout



Default popover dimensions: **320px wide × flexible height** (grows with conversation, max 480px, then scrolls). The popover anchors to the trigger element and repositions automatically to stay within the viewport.

Ephemeral session model

Property	Specification
Conversation persistence	None — the session is discarded when the popover closes
History access	No history panel; no access to past conversations
Shared conversations	Not available
Model switching	Not available — uses the tenant's `defaultModel`
Workflow Library	Not available
Artefact tray	Not available (individual outputs can be inserted or copied)
`@`-binding typeahead	Available — useful for referencing application objects in the field context
MCP tools	Always-on tools are available by default; can be disabled per-field via `tools-disabled` attribute
Personal memory	Not injected by default; optionally enabled via `inject-personal-memory` attribute
Application context	Always injected — org terminology and standing context is relevant to field composition
Improvement signals	Captured and attributed to the tenant; no turn-level audit trail (ephemeral session not persisted)

Context injection

When the popover opens, the platform assembles a field-specific system prompt from:

1. **Tenant base prompt** — from `scope.systemPrompt` in the application config
2. **Application context** — `[Application context]` block, always injected
3. **Field context block** — assembled from the component attributes:

```
[Field context]
Field: {{field-label}}
Current value: {{field-value}} (if provided and non-empty)
Form context: {{field-context}} (if provided)
Task: Help the user compose or refine the content for this field.
```

1. **Platform-managed instructions** — write-back guidance (if enabled), tool transparency, injection mitigation

The field context block keeps the model tightly focused on the field's purpose. The model does not reference other conversations or memory items not explicitly injected.

Write-back behaviour

When `write-back` is configured for a field, a **write-back bar** appears at the bottom of the popover after each assistant response that contains renderable text content.

Control	Behaviour
Insert into field	Fires the `field-insert` event with the response content. The host application handles this event to update the field value.
Copy	Copies the response text to the clipboard. No host event required.

For multi-turn field sessions, the write-back bar always shows the most recent response. Earlier responses can be copied by selecting text directly in the popover.

Write-back format is configurable per field:

Format	Use case
`plain` (default)	Standard text fields, input fields
`markdown`	Rich text editors that accept markdown
`html`	WYSIWYG editors
`json`	Structured data fields

Web component

Two component variants are provided:

Variant A — Wrapper (component wraps the field element)

```
<ai-chat-field
  tenant-id="acme-corp"
  user-token="..."
  field-label="Product Description"
  field-context="Help the user write a compelling product description for an e-commerce listing.
Focus on benefits, not features."
  write-back="true"
  write-back-format="plain"
  trigger="inline"
>
  <textarea name="description" rows="4"></textarea>
</ai-chat-field>
```

Variant B — Standalone trigger (component placed near the field)

```
<textarea id="desc-field" name="description" rows="4"></textarea>
<ai-chat-field
  tenant-id="acme-corp"
  user-token="..."
  field-label="Product Description"
  field-context="Help the user write a compelling product description."
  field-target="#desc-field"
  write-back="true"
  trigger="button"
  trigger-label="AI assist"
></ai-chat-field>
```

`<ai-chat-field>` attributes

Attribute	Required	Description
`tenant-id`	Yes	Tenant identifier
`user-token`	Yes	Host-authenticated user JWT
`field-label`	Yes	The field's label — shown in the popover header and injected as field context
`field-context`	No	A plain-language description of what this field is for and how the assistant should help. Injected into the field context block.
`field-value`	No	The field's current value, passed by the host for editing assistance. Injected into the field context block. If omitted, the component reads the value from the wrapped field element (Variant A only).
`write-back`	No	`"true"` enables the write-back bar. Default: `"false"`.
`write-back-format`	No	`"plain"` | `"markdown"` | `"html"` | `"json"`. Default: `"plain"`.
`trigger`	No	`"inline"` — icon inside the field; `"button"` — external button. Default: `"inline"`.
`trigger-label`	No	Button label text when `trigger="button"`. Default: `"AI assist"`.
`field-target`	No	CSS selector for the target field element (Variant B).
`tools-disabled`	No	`"true"` disables always-on MCP tools for this field session. Useful for simple text-generation fields that don't need live data. Default: `"false"`.
`inject-personal-memory`	No	`"true"` injects the user's personal memory into the field session. Useful for fields where personal preferences or role context are relevant. Default: `"false"`.
`placeholder-prompt`	No	Pre-filled text shown in the input field on first open (e.g. `"Describe this product in 2–3 sentences"`). Helps orient the user.

`<ai-chat-field>` events

Event	Detail payload	Description
`field-insert`	`{ fieldLabel, content, format }`	Fired when user clicks "Insert into field". Host updates the field value using `content`.
`field-copy`	`{ fieldLabel, content }`	Fired when user clicks "Copy"
`field-opened`	`{ fieldLabel }`	Fired when the popover opens
`field-closed`	`{ fieldLabel, messageCount }`	Fired when the popover closes
`token-expired`	—	Identical to other component variants — host refreshes via `updateToken()`

`<ai-chat-field>` JavaScript API

Method	Description
`open()`	Programmatically open the popover
`close()`	Programmatically close and discard the session
`updateToken(token)`	Refresh the user JWT — same as other components
`setFieldValue(value)`	Update the `field-value` context without reopening the popover

Handling `field-insert`

```
document.querySelectorAll('ai-chat-field').forEach((field) => {
  field.addEventListener('field-insert', (event) => {
    const { fieldLabel, content, format } = event.detail;
    const textarea = document.querySelector(`[data-label="${fieldLabel}"]`);
    if (textarea) {
      textarea.value = content;
      textarea.dispatchEvent(new Event('input', { bubbles: true }));
    }
  });
});
```

Cross-mode design decisions

Decision	Rationale
Modes 1 and 2 share conversation history	Both are persistent conversation modes accessing the same user session and tenant. There is no "page-specific" conversation context in these modes — the user's conversations belong to them, not to a page.
Mode 3 is entirely ephemeral	Form field interactions are transactional. The user wants help completing a specific field right now. Persisting these as conversations would pollute the history with low-value threads and create privacy/audit implications for draft field values.
Mode 3 does not escalate to Modes 1 or 2	The form field context is scoped to a specific field instance at a specific moment. It would not be useful as a persistent thread — the field context would be stale, and the user's goal (filling the field) would already be complete.
Application context is injected in Mode 3	Org terminology, standing guidance, and governance decisions are just as relevant when composing field content as in a full conversation. Omitting them would cause inconsistent terminology in AI-generated field values.
Personal memory is opt-in in Mode 3	Personal memory may contain preferences relevant to some fields (e.g. writing style, role context for approvals). But for most fields it adds unnecessary token overhead. The host decides per field.
Always-on MCP tools available in Mode 3 by default	Some fields benefit from live data access (e.g. a "domain owner" field might benefit from the governance MCP). Host can disable per field if not needed.
Write-back requires host event handling	The platform does not manipulate the DOM directly — it fires events and the host application updates its own fields. This keeps the component framework-agnostic and respects the host's form state management.

Summary: which component to use

Scenario	Component
Provide assistant access from anywhere in the app without navigation	`<ai-chat-widget>` (Mode 1)
Build a dedicated assistant page or embed a full assistant panel	`<ai-chat>` (Mode 2)
Add contextual AI assistance to individual form fields	`<ai-chat-field>` (Mode 3)
All three simultaneously	All three components can coexist — they share the same auth and platform backend

AI Chat Platform — Roadmap

This file captures planned enhancements beyond the current release. Items here are named and scoped at a level sufficient for future prioritisation. Each requires a Product Design Document and platform team approval before build begins.

Current release	v1 — multi-tenant embed, host config, MCP tool integration, full audit trail
Date	May 2026

Near-term — Semantic search (RAG)

Objective: Allow the assistant to retrieve semantically relevant content from host application data using vector search, as a complement to structured MCP tool calls.

Item	Description
Per-tenant vector search	Platform provides a vector index per tenant. Host applications push embeddings via the Platform Admin API. The assistant can retrieve semantically similar content when structured MCP tool calls return insufficient results.
Conversation memory RAG	Embed past conversation content per tenant. The assistant retrieves semantically relevant past turns at the start of a new session, surfaced as an optional `[Related past conversations]` context block.
Embedding refresh pipeline	Host applications trigger re-embedding via webhook when their source data changes. The platform maintains freshness indicators per embedding.

Pre-requisites: pgvector infrastructure provisioned in platform storage; Platform Admin API embedding endpoints designed; per-tenant embedding budget model defined.

Near-term — Improvement webhook

Objective: Allow host applications to receive improvement signals in their own tooling rather than relying solely on the platform's improvement dashboard.

Item	Description
Webhook registration	Host teams register a webhook endpoint via the Platform Admin API. The platform sends structured JSON payloads for qualifying improvement signals.

Item	Description
Payload schema	Signal type, turn summary, remediation recommendation, tenant context. Full turn content available on request via the Platform Admin API (not in webhook payload for security).
Retry and delivery guarantees	Platform retries failed webhook deliveries with exponential backoff. Delivery status tracked in the platform admin dashboard.

Phase 2 — Context memory

Objective: Allow the assistant to recall relevant context from past conversations at the start of a new session, without the user having to re-establish it manually.

Item	Description
Session summary injection	At session start, optionally inject a compressed summary of the user's recent conversations (last 30 days) as a labelled `[Recent context]` block. Opt-in per session. Transparent — block visible and dismissible. Generated by the fast-tier model.
Context globbing	User-initiated: pull selected context from one or more past conversations into a new session. Requires a conversation selection UI, a context budget allocator, and provenance labelling.

Pre-requisites: Auto-summarisation infrastructure (already in v1); per-tenant conversation memory RAG (above).

Phase 2 — Extended config capabilities

Objective: Give host applications more control over conversation behaviour and user experience.

Item	Description
Conversation folders	Host applications can enable user-created conversation folders/projects (grouping by topic, project, or review cycle). User-created and personal — not shared. Requires history panel redesign.
Real-time presence	Typing indicators and active-viewer presence for shared conversations. Requires WebSocket infrastructure upgrade.
Extended thinking mode	Expose the powerful-tier model's extended reasoning budget as an optional session setting (where supported by the configured provider). Displayed as a collapsible `<thinking>` block in the thread. Host configures whether this option is available.
Participant limit increase	Raise the shared conversation participant cap above 10. Requires load and permission testing at higher counts. Platform-level config change.

Item	Description
Conversation export	Export a conversation as PDF or markdown. Requires host applications to declare their data classification handling before enabling. Off by default.

Phase 2 — Analytics and operational tooling

Objective: Give Application Admins richer insight into their tenant's assistant usage and quality.

Item	Description
Enhanced analytics dashboard	Expand the tenant analytics dashboard with: query topic clustering, peak usage patterns, workflow completion rates, binding click-through by type, improvement signal resolution time.
Signal dashboard	Dedicated improvement signal view: volume by type, confidence distribution, issue lifecycle status, resolution time by remediation target.
Application config change log	Auditable log of all config changes within the tenant, with before/after diff view and the Application Admin who made the change.
MCP server health panel	Per-server uptime, error rate, and latency trends over time, visible to Application Admins within the tenant analytics dashboard.

Phase 2 — Multi-provider AI models

Objective: Allow host applications to configure alternative AI providers as the model backend.

Item	Description
Provider abstraction layer	Formalise provider configuration in the application config. Test OpenAI and Gemini models against the platform's system prompt injection, MCP tool call patterns, and content rendering.
Model capability parity	Validate that alternative models correctly invoke MCP tools, respect system prompt constraints, and produce rendering-compatible outputs (Mermaid, Vega-Lite, math).
Provider selection UI	Admin-level provider selection per tenant. End users may be able to select provider per session subject to host policy.

Phase 2 — SDK and programmatic config

Objective: Give host engineering teams a first-class programmatic interface for managing their tenant configuration.

Item	Description
Platform Admin SDK	TypeScript/Python SDK wrapping the Platform Admin API — typed config objects, config diff preview, validation helpers, rollback support.
Config-as-code	CI/CD integration that treats the application config as a versioned artefact. Config changes trigger validation and deployment via the SDK rather than the Config Editor UI.
Config diff and preview	Before applying a config update, show a structured diff of what changes and a preview of how the system prompt will assemble.

Phase 3 — MCP ecosystem expansion

Objective: Grow the MCP ecosystem to accelerate host team onboarding.

Item	Description
MCP Repository growth	Increase the number of published, verified MCP servers in the Repository across more domains (HR, finance, engineering, legal, customer success).
MCP Resources Service expansion	Add more skills, guidance documents, and prompt templates to the MCP Resources Service, with category-level browsing and search.
MCP server scaffolding tool	CLI tool that generates a boilerplate MCP server codebase for a given set of capabilities, reducing the time for host teams to publish their own server to the MCP Repository.
Cross-tenant MCP benchmarking	Anonymised, aggregated MCP tool error rate and latency benchmarks across tenants using the same server — helps host teams identify whether their server performance is typical or an outlier.

Not in roadmap

Item	Rationale
Voice or multimodal input	Not aligned with current platform direction
General-purpose AI assistant mode	Outside product vision — every deployment is a specialist scoped by the host application
Public shareable conversation links	Conflicts with tenant-scoped privacy model
Cross-tenant sharing	Data confidentiality constraint — tenant boundary is non-configurable
AI model training or fine-tuning	Platform uses foundation AI provider models; no training infrastructure

---

For current product specification, see README.md and the numbered specification documents.